

ROGEAP

Environmental and Social Management System (ESMS) Development Guidelines

**For Companies in the Off-Grid Solar Products
and Services Sector**

Table of contents

Table of Contents

1. Introduction to Environmental and Social Management System	5
2. Developing an Environmental and Social Management System.....	10
3. Environmental & Social (E&S) Policy.....	15
4. Identification of Environmental and Social Risks and Impacts	19
5. Compliance	32
6. Organisational Capacity and Competency.....	40
7. Environmental and Social Management Plans	47
8. Health and Safety Plan	49
Key Elements of an Occupational Health and Safety (OHS) Plan	49
Community Health and Safety.....	57
Consumer Protection	61
9. Human Resources Policy and Plan.....	64
Code of Conduct	66
10. Waste Management Plan	72
11. Emergency Preparedness and Response Plan.....	80
12. Stakeholder Engagement Plan (SEP)	86
13. Grievance Mechanism.....	90
14. Monitoring and Review	95
15. Environmental and Social Action Plan (ESAP)	98
16. References	102

Table of Figures

Figure 1: Depiction of the benefits to a business of implementing an environmental and social management system (ESMS).....	5
Figure 2: Environmental and social risks and impacts for key stakeholders in ROGEAP	6
Figure 3: Key elements of an environmental and social management system (ESMS).....	7
Figure 4: Tasks and responsibilities of the ROGEAP Granter/Lender (PIU/PFM/CFI) and Applicant/Recipient (OGS Business)	9
Figure 5: Phases and steps for designing and developing an Environmental and Social Management System (ESMS)	10
Figure 6: Components of the Environmental and Social Management System (ESMS) required of Off Grid Solar Businesses applying for ROGEAP grants and loans.....	11
Figure 7: Policy development and implementation process.....	15
Figure 8: Environmental and Social Risks and Impacts faced by Businesses and the Financial Institutions financing their projects and operations.....	19
Figure 9: Illustration of the steps in an E&S risk assessment process for a financial institution financing a business (Client/Investee)	19
Figure 10: Health and safety risks and hazards associated with stand-alone solar power systems	22
Figure 11: A general root cause tree diagram	31
Figure 12: Key elements to be considered towards achieving compliance.....	32
Figure 13: Environmental and social planning, management and review in the context of Plan-Do-Check-Act framework	47
Figure 14: Waste management hierarchy.....	72
Figure 15: E-waste management value chain	78
Figure 16: General emergency preparedness and response management process	80
Figure 17: Stakeholder map depicting the level engagement with stakeholders depending on their level of interest and influence.....	86
Figure 18: Grievance mechanism development steps.....	90
Figure 19: Grievance mechanism process suggested by the World Bank's Compliance Advisory Ombudsman (CAO)	91
Figure 20: Planning, monitoring, and evaluation process for results-based management.....	95
Figure 21: Environmental and social risk monitoring form for solar companies (source: ROGEAP Operational Manual)	
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Figure 22: Components of an Environmental and Social Action Plan (ESAP).....	98
Figure 23: IFC ESMS Handbook and Supplemental Resources	99

Table of Tables

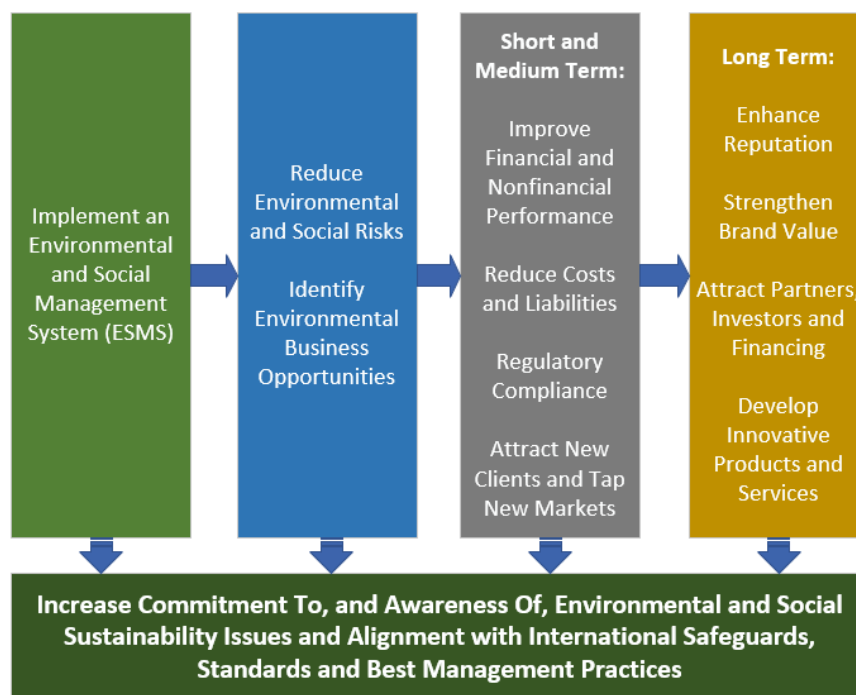
Table 1: World Bank Group (WB and IFC) E&S Standards and Equator Principles their applicability to OGS Businesses	6
Table 2: ESMS components, responsibilities and requirements for solar companies.....	7
Table 3: Description of the level of development of ESMS components expected of businesses depending on their stage of growth and grant window pursued.....	11
Table 4: Environmental and social (E&S) requirements and actions required at each step in the ROGEAP grant process	13
Table 5: Key Environmental and Social Risks and Mitigation Measures Often Associated with Off-Grid Solar Products and Services and the International Standards of Relevance.....	23
Table 6: ROGEAP Environmental and Social Screening Questionnaire for Solar Businesses	27
Table 7: Example of a Compliance Tracking Checklist Template.....	38
Table 8: Legal and Regulatory Register Template	39
Table 9: Permit and License Tracking Template	39
Table 10: Sample responsibility matrix. The tasks and positions listed in the matrix are simply examples and should be replaced with tasks applicable to the company's own ESMS tasks and staffing positions of relevance to the ESMS.....	41
Table 11: Sample Incident / Accident Report Form	52
Table 12: Sample OHS Plan Table of Risks and Management Measures	53
Table 13: Examples of community health and safety concerns and management measures	59

Table 14: Waste disposal management template.....	73
Table 15: Waste management options for common off-grid solar products.....	74
Table 16: Examples of common types of emergencies, preparedness measures and response procedures	82
Table 17: Examples of typical stakeholders and engagement methods	88
Table 18: Grievance mechanism process suggested in ROGEAP Operational Manual.....	92
Table 19: Sample grievance form to be completed by a Complainant. Similar form can be made accessible on the company's website for online submissions.....	93
Table 20: Example of a Grievance Mechanism Log where received grievances are documented along with the response process undertaken.....	94
Table 21: Template for Environmental and Social Action Plan (ESAP)	99

1. Introduction to Environmental and Social Management System

An Environmental & Social Management System (ESMS) helps companies proactively identify and address environmental and social risks in their operations and supply chain, while also demonstrating a commitment to sustainability and responsible business practices. Developing and maintaining an ESMS enables a company to proactively identify and minimize potential environmental, social, health and safety risks faced by the company, staff, subcontractors, suppliers, customers, and the local communities where the company operates. Through development and implementation of an ESMS a company is able to demonstrate its commitment to:

- Promote Environmental and Social (E&S) sustainability;
- Avoid, minimize, or mitigate against adverse E&S impacts through implementing E&S risk management policies, processes, and capacity building at solar companies
- Promote decision-making informed by E&S risks.
- Address cross-cutting issues (e.g., hazardous waste/ batteries, gender impacts, and citizen engagement) at the market level through strategic multi-stakeholder solutions.



Source: Adapted from FIRST for Sustainability Website¹

Figure 1: Depiction of the benefits to a business of implementing an environmental and social management system (ESMS)

An ESMS provides a framework of principles, requirements, processes, tools, roles and responsibilities that should complement and integrate with the company's existing risk management procedures. An ESMS should be structured to reflect the Plan, Do, Check, Act (PDCA) management cycle approach, which characterizes voluntary management systems such as ISO 9001 (Quality Management Systems), ISO 14001 (Environmental Management Systems), and ISO 45001 (Occupational Health and Safety Management Systems)²:

- **Plan:** Define policies and objectives, identify risks and opportunities, develop action plans and allocate financial and human resources for implementation.
- **Do:** Implement action plans with allocated resources.
- **Check:** Monitor and evaluate the adequacy of action plans and resources by measuring results against policies and objectives.
- **Act:** Implement corrective actions to improve performance.



¹ Managing E&S Risks. (n.d.). FIRST for Sustainability. Retrieved April 12, 2024, from <https://firstforsustainability.org/managing-environmental-and-social-risk>

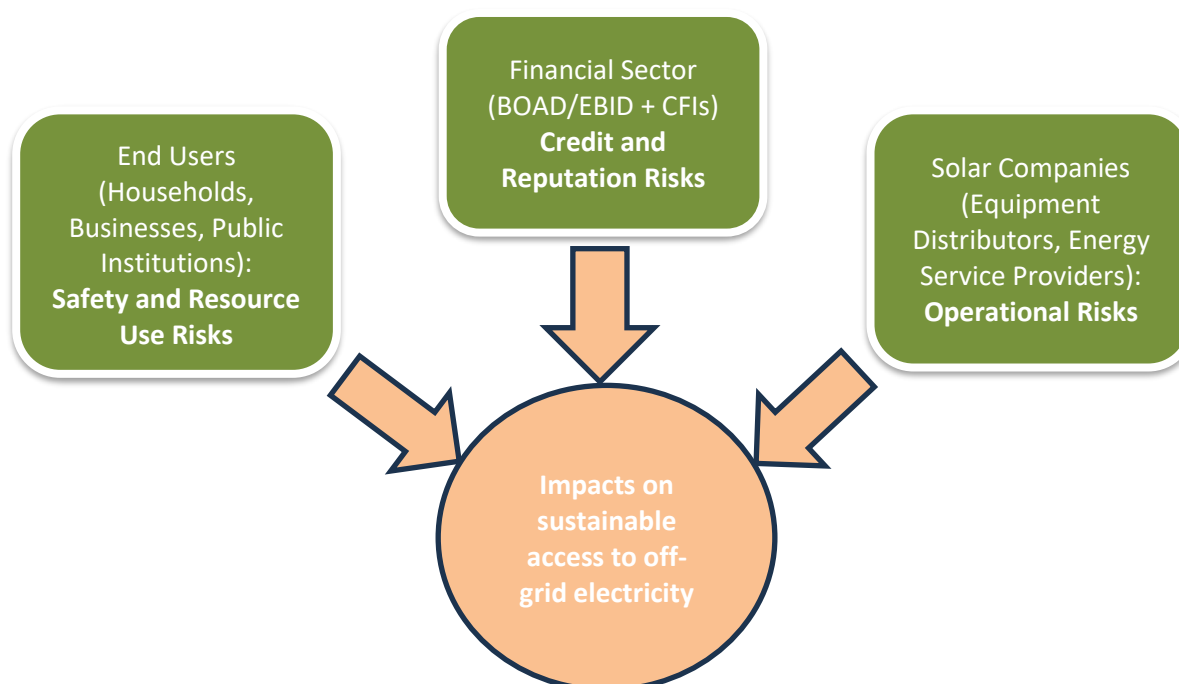
² Cleantech Solar. 2021. Environmental and Social Management System (ESMS) Manual: Rooftop Solar India.

International financial institutions generally require projects they finance to meet specified E&S standards. Below are examples of the World Bank and International Finance Corporation Standards and Equator Principle (used by other financial institutions) applicable to OGS businesses wishing to qualify for financing from such sources. The key mechanism for applying these standards is through development of an ESMS.

Table 1: World Bank Group (WB and IFC) E&S Standards and Equator Principles their applicability to OGS Businesses

International Finance Corporation (IFC) Performance Standards (PS) / World Bank (WB) Environmental and Social Standards (ESS)	Applicability to OGS Business
IFC PS 1 / WB ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Applicable
IFC PS 2 / WB ESS 2: Labor and Working Conditions	Applicable
IFC PS 3 / WB ESS 3: Resource Efficiency and Pollution Prevention and Management	Applicable
IFC PS 4 / WB ESS 4: Community Health Safety [and Security]	Applicable
IFC PS 5 / WB ESS 5: Land Acquisition, [Restrictions on Land Use] and Involuntary Resettlement	Rarely Applicable
IFC PS 6 / WB ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Rarely Applicable
IFC PS 7 / WB ESS 7: Indigenous Peoples / Sub-Saharan African Historically Underserved Traditional Local Communities	Rarely Applicable
IFC PS 8 / WB ESS 8: Cultural Heritage	Rarely Applicable
WB ESS 9: Financial Intermediaries	Not Applicable
WB ESS 10: Stakeholder Engagement and Information Disclosure	Applicable
Equator Principles	
Principle 1: Categorization of Projects - [using IFC E&S Categorization Process]	Applicable
Principle 2: Borrower required to conduct an Environmental and Social Impact Assessment (ESIA) for Category A and B - [a more basic E&S Risk and Impact Assessment is applicable in OGS context]	Applicable [Category B]
Principle 3: Applicable Social and Environmental Standards – [Compliance with national E&S requirements]	Applicable
Principle 4: Action Plan [i.e. ESAP] and Management System [i.e. ESMS]	Applicable
Principle 5: Consultation and Disclosure - [Stakeholder Engagement]	Applicable
Principle 6: Grievance Mechanism	Applicable
Principle 7: Independent Review [Required mostly of Category A projects and thus not applicable to OGS]	Not Applicable
Principle 8: Covenants [Environmental Performance Compliance Covenants included in Financial Documentation]	Applicable
Principle 9: Independent Monitoring and Reporting	Applicable
Principle 10: Reporting and Transparency	Applicable

In ROGEAP, risks and impacts are identified as arising from and affected the main sets of stakeholders including Financial Sector (e.g. BOAD/EBID and CFIs), Solar Companies, and End-Users, and the impact on sustainable access to electricity (see figure below). This manual is focused on managing risks and impacts associated with Solar Companies and their operations, which in turn also mitigates negative risks and impacts, while also enhancing positive impacts (benefits), for the financial institutions and end users.

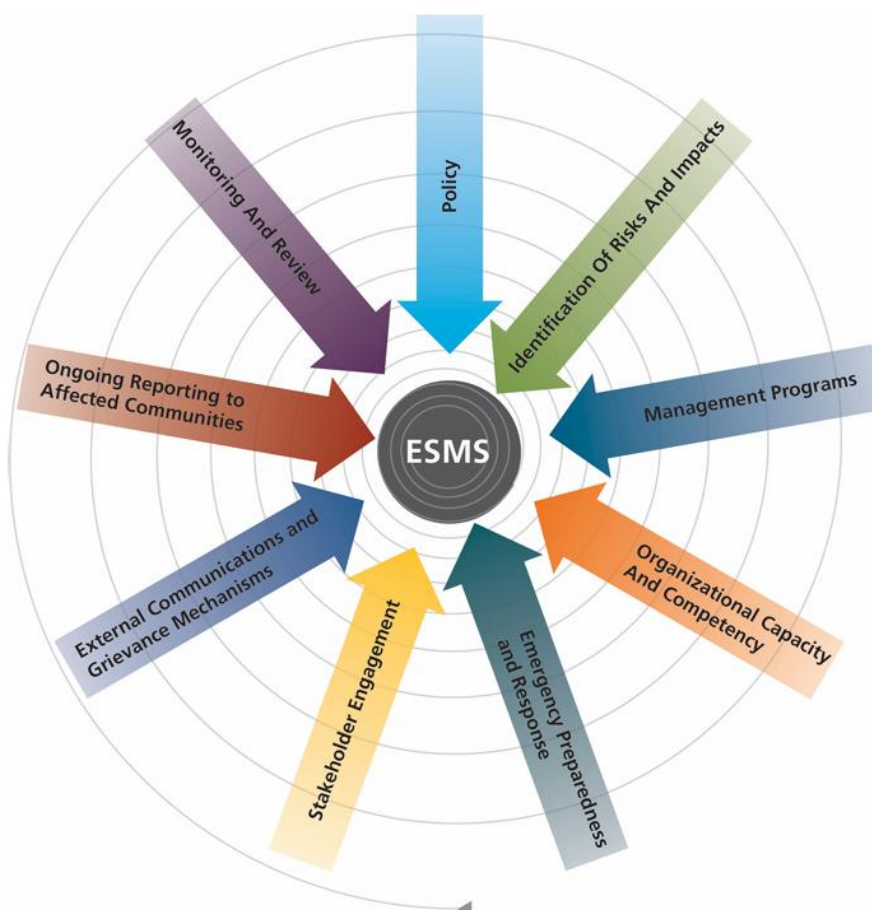


Source: ROGEAP Operational Manual 2024.

Figure 2: Environmental and social risks and impacts for key stakeholders in ROGEAP

In the context of ROGEAP, as stated in the Environmental and Social Risk Management (ESRM) Strategic Paper (ECOWAS,2020), “environmental and social management operational processes are needed to (i) promote E&S sustainability in lending operations under ROGEAP; (ii) avoid, minimize, or mitigate against adverse impacts, mainly through implementing E&S risk management policies, process, and capacity at the level of wholesale lenders (BOAD), participating commercial financial institutions (CFIs), and solar companies; and (c) promote informed decision-making relating to E&S effects.”

IFC has issued guidelines describing nine interrelated elements that should be included in an effective ESMS (see figure below).



Source: IFC ESMS Implementation Handbook 2014

Figure 3: Key elements of an environmental and social management system (ESMS)

The ROGEAP Project requires qualifying grant [or loan] applicants to have an Environmental and Social Management System (ESMS) in place to ensure [business] company operations are aligned with national and international environmental and social safeguards towards fostering sustainable development of the off-grid solar products sector. World Bank Environmental and Social Standards (ESS) and IFC Performance Standards require establishment of an ESMS for all projects funded by World Bank, with other international finance institutions (IFI) and development finance institutions (DFI) having similar requirements and expectations³.

Table 2: ESMS components, responsibilities and requirements for solar companies

Component	Responsibilities	Requirements
E&S Policy	Summarize the company's commitment to manage E&S risks and impacts.	Policy statement that, according to the activity, focuses mainly on waste management (with an emphasis on e-waste), health and safety, and labor risks
Identification of E&S risks and impacts	Identification of the potential E&S risks and impacts	Risk identification and assessment process in place

³ The International Finance Corporate (IFC) has published an Environmental and Social Management System Implementation Handbook to help guide businesses on how to integrate an environmental and social management system (ESMS) into their management and operations.

Component	Responsibilities	Requirements
Ensure legal and regulatory compliance	Identify and adhere to all applicable laws and regulations of countries of business operations	- Description of legal and regulatory requirements and procedure for compliance - Documentation of required permits/licenses
Management program	Environmental and Social Management Plan (ESMP) to avoid, minimize or compensate for the E&S risks and impacts identified	- Documented plan that must necessarily include according to the activity: - Policy/ procedure and records on occupational health and safety - Human Resource policy (including code of conduct for workers and grievance mechanism for workers) - Waste management policy or procedures - Stakeholder engagement plan and grievance mechanism
Organisational capacity and competency	Definition of the roles and responsibilities to implement the ESMS	Internal procedures documents on the ESMS implementation, including internal training if necessary.
Emergency preparedness and response	To respond effectively to emergencies	- Internal training on first AID to mitigate occupational accidents. - Operation procedures and practices must be prepared to respond to emergencies such as fire and flooding.
Stakeholder engagement	Identification and engagement with key stakeholders at the strategic business and operating level	- Develop a stakeholder engagement plan that involves disclosures and consultation with stakeholders on E&S risks and mitigation practices (e.g., e-waste management). - Stakeholder engagement should be accessible, inclusive, and in a format that is appropriate to ensure the inclusion of marginalized and vulnerable groups (such as illiterate persons, elderly, persons with disabilities, etc.).
Grievance mechanisms	To establish a way for individuals, groups or communities to contact if they have inquiries, concerns or complaints	- Grievance mechanisms should be accessible, inclusive, and in a format that is appropriate to ensure the inclusion of marginalized and vulnerable groups (such as illiterate persons, older adults, persons with disabilities, etc.). - Establishment of a grievance mechanism adapted to handle SEA/SH complaints in a confidential, ethical, and survivor-centered manner.
Monitoring and review	To monitor compliance with the contractual E&S obligations made with financial institutions	E&S Monitoring plan

This ESMS Guidelines document has been developed to provides guidelines to facilitate preparation and implementation of an ESMS that aligns with the requirements of the ROGEAP project.

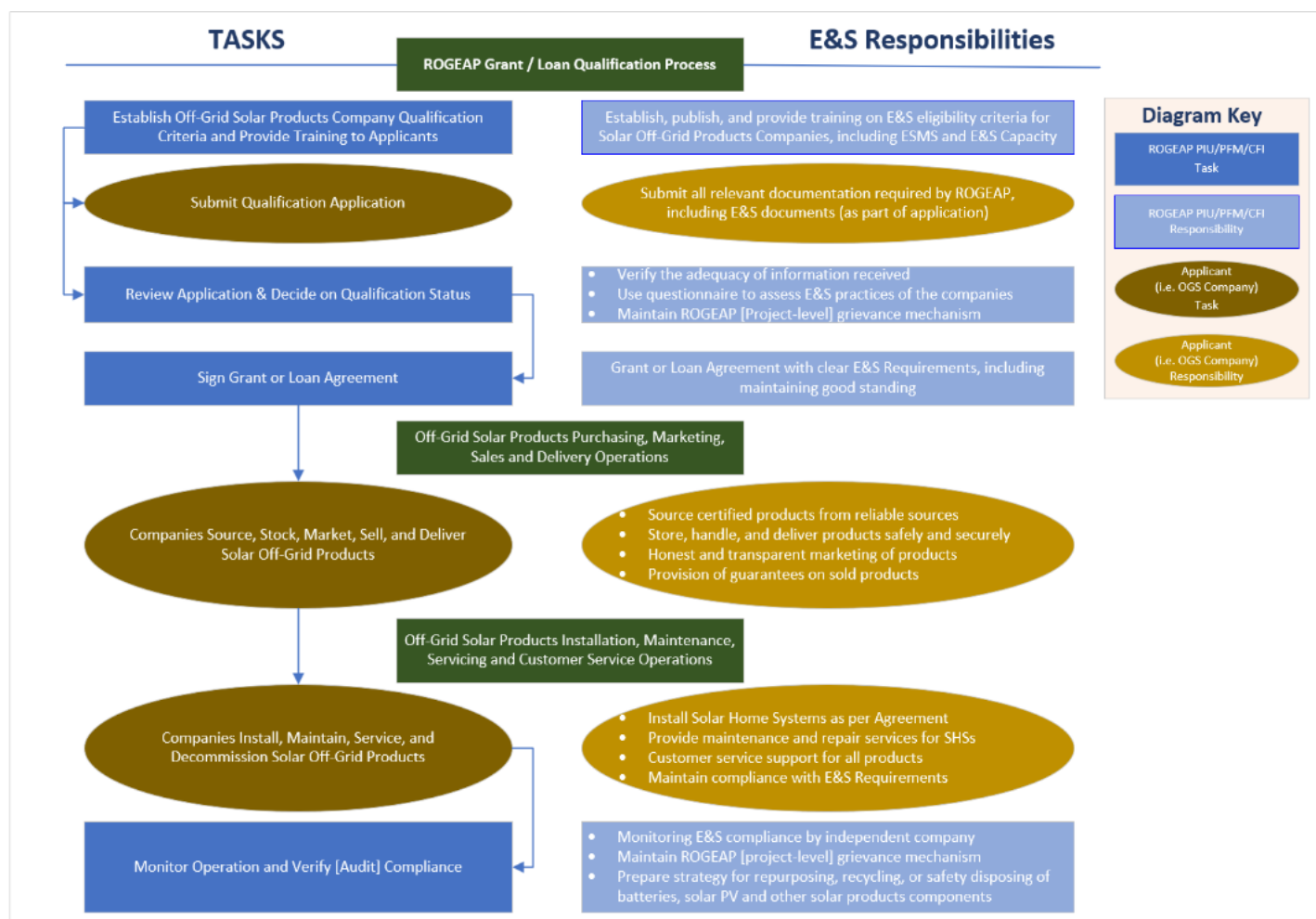
An Environmental and Social Risk Management (ESRM) Sector Guide on Off-Grid Standalone Solar for Financial Service Providers (a.k.a. E&S Due Diligence Guideline) has already been prepared and published by ECOWAS and BOAD for CFIs, Fund Managers, and Entities acting in similar capacity *“to assess the potential E&S risks and opportunities associated with providing debt or grant financing to solar off-grid standalone businesses and activities”*. That guideline provides *“information on what is needed for FIs to assess the E&S risks, communicate requirements to solar businesses, as well as monitor and report on implementation.”*⁴

The present ESMS Guidelines document is intended to complement the E&S Due Diligence Guideline by providing ESMS development guidelines for companies applying for and receiving grants and loans from the CFIs and Fund Managers under the ROGEAP program. These guidelines will hopefully prove useful for any solar off-grid products

⁴ (N.d.). Ecowas.Int. Retrieved April 12, 2024, from https://ecowas.int/wp-content/uploads/2022/08/ROGEAP-Annex7-ESRM-Section-Guide-on-Off-Grid-Solar-for-CFI_19Dec2020-final.pdf

companies seeking to develop or strengthen their ESMS, regardless of whether or not they are pursuing or awarded a ROGEAP associated grant or loan.

The following diagram depicts the main tasks and E&S responsibilities required as part of ROGEAP.

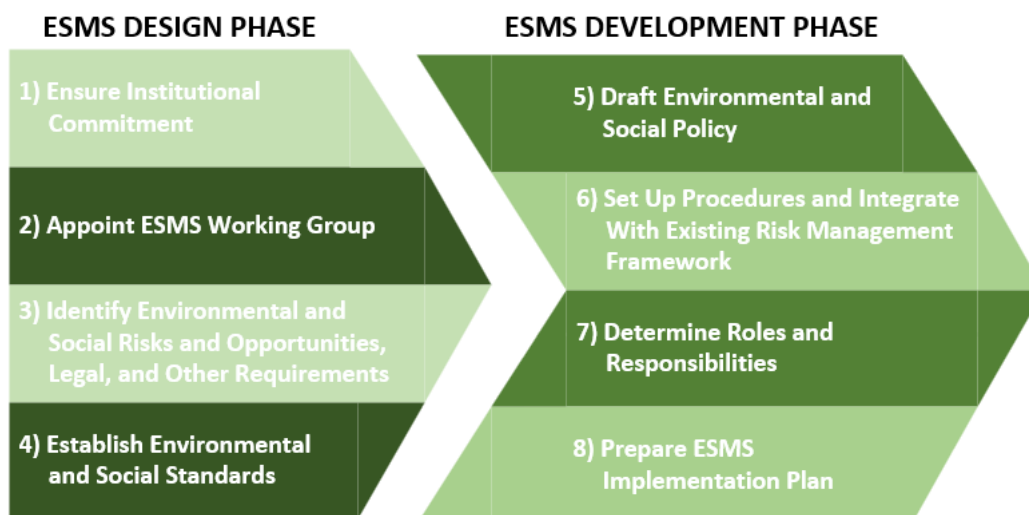


Source: Adapted from NEP ESMP 2018

Figure 4: Tasks and responsibilities of the ROGEAP Granter/Lender (PIU/PFM/CFI) and Applicant/Recipient (OGS Business)

2. Developing an Environmental and Social Management System

Environmental and Social Management Systems (ESMS) are a requirement as per **IFC PS 1 / WB ESS: Assessment and Management of Environmental and Social Risks and Impacts**. To develop an effective Environmental and Social Management System (ESMS), it's essential to have strong support from senior management. The ESMS should align with existing risk management practices. Begin the development process by forming a dedicated working group to develop the ESMS. A well-designed ESMS should include a clear E&S policy, step-by-step procedures for identifying and assessing E&S risks, plans for managing E&S risks, a framework for monitoring and evaluating and reporting performance, and capacity building of internal staff to undertake assigned roles and responsibilities associated with the ESMS. The following diagram shows the main steps in developing an ESMS:



Source: Adapted from FIRST for Sustainability Website⁵

Figure 5: Phases and steps for designing and developing an Environmental and Social Management System (ESMS)

Businesses is required to develop an Environmental and Social Management System (ESMS) comprised of the following components, at a minimum, to qualify for a grant or loan under ROGEAP⁶:

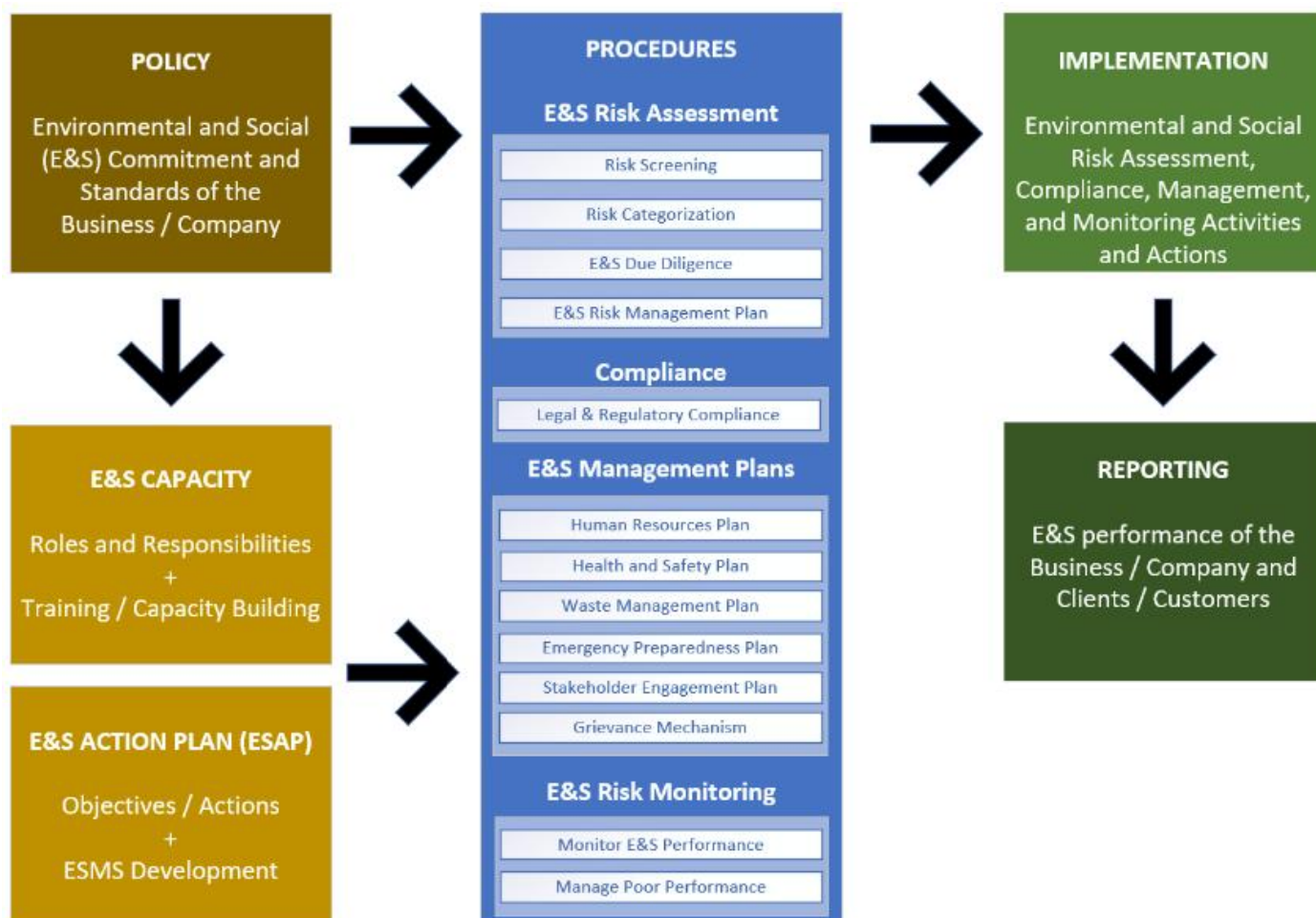
- 1) Environmental and social **(E&S) policy**
- 2) Environmental and social **risk identification and assessment** process or procedure
- 3) **Compliance** with the laws and regulations of the countries of operations
- 4) Policy, procedure and records on occupational [and community] **health and safety**
- 5) **Human resource** policy, including a code of conduct for workers and grievance mechanism for workers
- 6) **Waste management** policy or procedures
- 7) **Stakeholder engagement** plan
- 8) **Grievance mechanism**, including a process for grievances related to Sexual Exploitation and Abuse (SEA) and Sexual Harassment (SH)

Businesses are encouraged to also pursue voluntary management systems (e.g. ISO 14001, ISO 45001, OHSAS 19001), but these are not a requirement for ROGEAP funding.

The following diagram shows the required components of the Environmental and Social Management System (ESMS) required for businesses applying for grants/loans through ROGEAP:

⁵ Managing E&S Risks. (n.d.). FIRST for Sustainability. Retrieved April 12, 2024, from <https://firstforsustainability.org/managing-environmental-and-social-risk>

⁶ Solar businesses classified as Tier 1 and 2 (typically businesses that are less than 3 years old, with few employees, and that may have sold few units of Standalone Home Solar systems) may not yet have all of the basic ESMS components (e.g. policies, procedures and tools) in place. These small-scale early-stage businesses will be required to develop an E&S Action Plan (ESAP) committing to implement the ESMS requirements within a year. The ESAP is to be included in the financing agreement for the ROGEAP grant or loan.



Source: Adapted from FIRST for Sustainability Website⁷

Figure 6: Components of the Environmental and Social Management System (ESMS) required of Off Grid Solar Businesses applying for ROGEAP grants and loans

The level of complexity of an ESMS and the components thereof are expected to be commensurate with the stage of growth of the business, with start-ups and early-stage businesses requiring a basic ESMS, while mid-stage businesses would be expected to develop a more comprehensive ESMS suited to the growth in complexity and scale operations, increased number of staff and customers, increased volume of OGS products, and corresponding increase in potential environmental and social risks and impacts.

Table 3: Description of the level of development of ESMS components expected of businesses depending on their stage of growth and grant window pursued.

ESMS Component	Stage 1: Start-Up (Matching Grant Applicants)	Stage 2: Early-Stage (Market Entry Grant & Performance-Based Grant Applicants)	Stage 3: Growth-Maturity (Market Entry Grant & Performance-Based Grant Applicants)
E&S Policy	Basic, high-level commitment to sustainability (in company documentation) aligned with the organization's size, nature, and environmental and social risks and impacts.	Documented policy, outlining key E&S principles and goals/commitments reflecting the company's values.	Comprehensive E&S policy, aligned with international standards (e.g., ISO 14001), including quantitative targets and commitments, and integrated into the organization's business strategy
E&S Risk Assessment	Basic risk identification of key E&S risks relevant to business operations.	Risk identification and assessment procedure, including some quantification of risks and impacts, and development of risk mitigation actions.	Comprehensive risk identification and assessment procedure aligned with international standards, with quantification and prioritization of risks, risk registers, and development of risk mitigation plans.
Legal and Regulatory	Basic compliance with core	Initiative-taking compliance	Robust compliance program with

⁷ Managing E&S Risks. (n.d.). FIRST for Sustainability. Retrieved April 12, 2024, from <https://firstforsustainability.org/managing-environmental-and-social-risk>

ESMS Component	Stage 1: Start-Up (Matching Grant Applicants)	Stage 2: Early-Stage (Market Entry Grant & Performance-Based Grant Applicants)	Stage 3: Growth-Maturity (Market Entry Grant & Performance-Based Grant Applicants)
Compliance	laws and regulations relevant to the organization's activities, with support from partners or consultants.	approach, including tracking of new regulations and standards relevant to the industry, and ensuring permits and licenses are obtained and up-to-date.	regular audits, legal reviews, and compliance training for staff, and integration into the organization's management system.
HR Policy / Plan	Basic employee handbook and/or guidelines with a Code of Conduct addressing non-discrimination and [SEAH] harassment.	Formal HR policies addressing fair labor practices, diversity, and inclusion, with a Code of Conduct, including non-discrimination and [SEAH] harassment clauses	Comprehensive HR policies aligned with international labor standards, including training on the ESMS and code of conduct for all employees, and a formal Grievance Mechanism for staff.
Organizational Capacity and Competency	ESMS responsibilities shared among existing staff, with basic awareness raising on E&S issues.	Designated personnel with clear roles/responsibilities for ESMS, with some training and external support for ESMS implementation.	ESMS team or function emerging with dedicated resources and expertise in E&S management, regular training, and integration of E&S into job descriptions and performance reviews.
Health and Safety Plan (including Occupational Health and Safety, Community Health and Safety, and Consumer Protection)	Basic safety measures in place, with focus on common workplace hazards. Consumer Protection Principles mentioned in business documentation.	OHS plan under development with risk assessments and basic safety procedures. Community Health and Safety measures considered. Consumer Protection Principles and Code documented.	Formalized OHS plan (aligned with ISO 45001) with regular inspections, incident reporting, and continuous improvement of safety performance. Community Health and Safety Plan under development. Consumer Protection Principles and Code documented and implemented.
Waste Management Plan	Basic waste segregation, disposal and recycling practices.	Waste management plan with waste reduction targets, segregation, recycling, proper disposal, and hazardous waste management.	Comprehensive waste management plan with waste minimization strategies (e.g. 5 R strategies), resource recovery, and compliance with waste regulations.
Emergency Preparedness Plan	Basic emergency procedures documented and communicated to staff.	Emergency plan covering potential scenarios, roles/responsibilities, and communication.	Formalized emergency preparedness and response plan with regular drills, community engagement, and continuous improvement.
Stakeholder Engagement Plan	Informal engagement with immediate stakeholders, such as employees and local communities.	Stakeholder engagement plan under development, identifying key stakeholders and engagement methods.	Formalized stakeholder engagement plan with regular communication, consultation, information disclosure, and a grievance mechanism accessible to all stakeholders, and alignment with relevant international standards (e.g., IFC Performance Standards).
Grievance Redress Mechanism	Basic/informal process for receiving and addressing complaints.	Documented grievance mechanism with clear steps for submitting and resolving complaints.	Formal grievance mechanism accessible to all stakeholders, including external parties, and including capacity to handle grievances related to Sexual Exploitation and Abuse (SEA) and Sexual Harassment (SH).
Monitoring Reporting &	Basic monitoring of key E&S parameters and ad-hoc reporting.	Regular monitoring with some key performance indicators (KPIs), internal and some external reporting, and an improvement process.	Comprehensive monitoring system with detailed KPIs, regular, transparent internal and external reporting, and continuous improvement cycle with management review.

Companies should identify and review all the existing components of their ESMS and prepare an Environmental and Social Action Plan (ESAP) specifying how and when any underdeveloped or missing ESMS components will be developed and applied within a year. The following table describes what ESMS development and documentation is expected of applicants at each stage of ROGEAP grant program.

Table 4: Environmental and social (E&S) requirements and actions required at each step in the ROGAP grant process

Grant Program Stage	ESMS Requirements	ESMS Development
EOI Application Step 1: Applicants submit baseline business information per the EOI form. Step 2: Technical committee evaluates and shortlists the applicants based on eligibility assessment criteria. Eligible businesses that obtain the minimum threshold will be invited to submit a full grant proposal.	A company characterized by any of the following environmental and social (E&S) exclusion criteria is disqualified from grants or loans under ROGAP: 1) Production or activities involving forced labor 2) Production or activities involving child labor 3) Cross-border trade in waste and waste products, unless compliant to the Basel Convention and the underlying regulations 4) Confirmed cases of occupational, health, and safety incidents or accidents 5) Confirmed cases of Gender Based Violence/Sexual Exploitation and Abuse 6) Confirmed cases of discrimination of vulnerable groups, including gender and disability Companies must comply with national environmental, social, health and safety (ESHS) laws and regulations of countries where they operate. Businesses with minimal or no adverse E&S impact risk require simple E&S risk mitigation policies, procedures, and tools, collectively referred to as an Environmental and Social Management System (ESMS).	In addition to ESMS core components, businesses identified as medium risk businesses are required to develop and implement an environmental and social action plan (ESAP) with monitoring and reporting, to mitigate for potential adverse E&S impacts. Factors that may trigger the ESAP requirement include: size and/or characteristics of the business, and/or identified risks, such as occupational accidents or labor and GBV incidents, irregularities with regard to fair working conditions and terms of employment, or installers operating without applying safe installation measures. The ESAP should itemize specific actions, indicators of completion, timeline for completion, and verification methods. The ESAP and/or E&S covenants will be included in grant/loan (legal) agreements.
Training Training and coaching provided to shortlisted businesses by the PFM. Training covers ESHS risks and requirements as well as technical aspects, business and financial, management, and business plans (Training is optional for Stage 3 companies).	Participate in training offered by ROGAP to learn more about ROGAP E&S requirements, ESHS risks in OGS sector, E&S management, ESMS components, and Grievance Mechanisms.	A copy of the ESHS Training Module is shared with training participants, along with this ESMS Guideline document to help applicants with their efforts to build upon their business's existing E&S safeguards and develop missing components of their ESMS.
Proposal (if qualified from EOI stage) Pre-qualified businesses submit full proposals, including business plan, financial plan, market entry plan, growth plan, statutory documents, and proof of matching contributions. The PFM provides coaching and application support services, incl. business plan finalisation, to eligible applicants to ensure the documentation is complete and consistent.	ESMS Components required or expected to be developed: 1) Environmental and social (E&S) policy 2) Environmental and social risk identification and assessment process or procedure 3) Compliance with the laws and regulations of the countries of operations 4) Policy, procedure and records on occupational [and community] health and safety 5) Human resource policy, including a code of conduct for workers and grievance mechanism for workers 6) Waste management policy or procedures 7) Stakeholder engagement plan 8) Grievance mechanism, including a process for grievances related to Sexual Exploitation and Abuse (SEA) and Sexual Harassment (SH)	Solar businesses classified as Tier 1 and 2 (typically businesses that are less than 3 years old, with few employees, and that may have sold few units of Standalone Home Solar systems) may not yet have all of the basic ESMS components (e.g. policies, procedures and tools) in place. Businesses will be required to develop an E&S Action Plan (ESAP) committing to implement the ESMS requirements within a year.
Due Diligence (if advanced to next stage of Proposal evaluation process) The PFM conducts on-site due diligence on all applicants who obtained the minimum threshold points to check and validate their physical business presence, verify submitted documents and assess risks.	Documentation/evidence of ESMS existence and implementation (see list of required components in row above).	Preparation of an E&S Action Plan (ESAP) committing to implement any missing ESMS requirements within a year.
Financing Agreement (if awarded a grant) The financing agreement is duly executed upon the signature of the grant	E&S Action Plan (ESAP) to be included in the financing agreement for the ROGAP grant or loan.	Sign the ROGAP financing agreement and commit to implementing the E&S Action Plan

Grant Program Stage	ESMS Requirements	ESMS Development
awardee and the PFM.		
Post-Disbursement / Performance Reporting and Monitoring For performance-based grants, the disbursement schedule will depend on the milestones/results achieved as set by the SME	E&S Action Plan (ESAP) implementation including development and implementation of all missing ESMS components within a year.	Progress Reports with updates on implementation of the ESAP and ESMS.

The following sections of this document describe and provide guidelines for developing each of the required ESMS components, followed by a section with guidance on preparing an ESAP.

3. Environmental & Social (E&S) Policy

The Company E&S Policy is considered the cornerstone of the ESMS, establishing expectations for conduct and actions in all aspects of company operations aligned with the policy statement⁸. Care should be taken in developing the policy to ensure it addresses the issues of greatest concern and aligns with the key values of the company. It is helpful to have an implementation-centered policy model⁹ that prioritizes the successful policy execution from the early stages of its development. Among the recommendations of such an approach are to consider the following:

- **Policy Leadership:** Establishing clear leadership for each policy issue, ensuring accountability and coordination throughout the implementation process.
- **Planning:** Developing a detailed work plan and accountable structure early in the process to track progress and identify potential obstacles.
- **Coordination, Consultation, and Communication:** Engaging with internal and external stakeholders early and often to ensure buy-in, promote collaboration, and address any concerns or obstacles that may arise during implementation.
- **Policy Options:** Analyzing various policy options and instruments, considering their feasibility, potential impact, and ethical implications.
- **Innovation:** Encouraging innovative solutions and exploring new approaches to policy challenges, leveraging technology and cross-sector collaboration.
- **Policy Evaluation:** Conducting ongoing evaluations throughout the policy process, not just at the end, to assess effectiveness, identify areas for improvement, and ensure that the policy is achieving its intended goals.

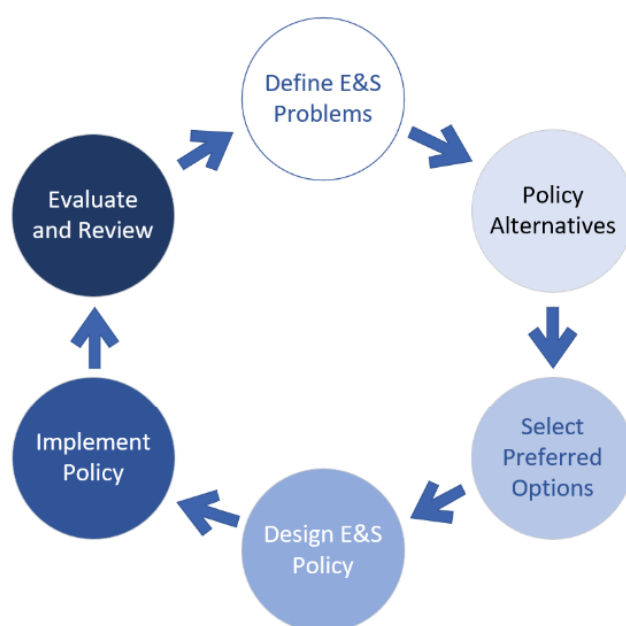


Figure 7: Policy development and implementation process

The E&S Policy should, in clear and simple terms, summarize the commitment of the company to manage E&S risks and impacts. The policy statement should be tailored to focus on issues of relevance to the company's operations, which in the context of off-grid solar may be waste management (with an emphasis on e-waste) and OHS and labor risks.

Here are some elements to consider including in the E&S policy statement¹⁰:

⁸ IFC. 2014. Environmental and Social Management System Implementation Handbook: Construction. [esms-handbook-construction-v7.pdf](https://www.ifc.org/esms-handbook-construction-v7.pdf) (ifc.org)

⁹ Salar K. 2022. Policy Process: An Implementation-Centered Model. LinkedIn.Com. Retrieved May 23, 2024, from <https://www.linkedin.com/pulse/policy-process-implementation-centered-model-salar-khudadad/>

¹⁰ Adapted from Nigeria Electrification Project (NEP): Environmental and Social Management System Template for Off-grid solar products business

1. Declaration of Compliance

- **Laws and Regulations:** Emphasize your company's strict adherence to all applicable environmental and social (E&S) laws and regulations within its areas of operation. Clearly state a zero-tolerance policy for Gender-Based Violence (GBV) and Sexual Exploitation and Abuse (SEA).
- **Exclusion List:** Mention a commitment to avoid activities you consider unethical (e.g. forced labor), such as those listed in the Exclusion List of international finance institutions.
- **Stakeholder Standards:** Acknowledge your responsibility to meet the E&S standards set by lenders, investors, and shareholders.

2. Environmental Stewardship and Social Responsibility

- **Environmental Stewardship:** Describe your commitment to implementing environmentally sustainable practices and resource conservation, including addressing issues such as sustainable procurement, waste management, resource efficiency, and environmental education.
- **Social Responsibility:** Describe your commitment to social responsibility including issues such as community engagement, fair labor practices, health and safety, gender equity and empowerment, access and affordability, and transparency and accountability.

3. Capacity and Resources

- **Dedicated Oversight:** Outline plans to establish an environmental department/unit or the designation of an E&S Manager/Coordinator. This signals your commitment to effective implementation of environmental and social safeguards.
- **Leadership Commitment:** Stress the role of senior management in championing E&S compliance and modeling responsible practices.
- **Staff Training:** Highlight comprehensive training programs for relevant personnel on E&S issues. This includes investment screening, public consultation, environmental mitigation, monitoring processes, codes of conduct, and grievance redress mechanisms (with particular attention to safe and confidential reporting of GBV cases).

4. Monitoring & Reporting

- **Grievance Redress:** Describe the implementation of a transparent Grievance Redress Mechanism with meticulous record-keeping to enable the prompt resolution of any complaints.
- **Accountability:** Emphasize the importance of maintaining detailed records of E&S non-compliance incidents, fines, code of conduct violations, and their resolution.
- **Transparency:** State your commitment to regular reporting to government agencies, lenders, and investors. This includes full disclosure of any incidents, accidents, or GBV/SEA cases.

Box 1: The following template demonstrates an E&S Policy statement that a company can adapt to its context. The sample policy statement integrates the elements recommended above.¹¹

[Your Institution's Name] Environmental & Social Policy Statement

We, at [your institution's name], are committed to environmental sustainability, social responsibility, and ethical conduct across all our operations.

Our Commitment

[Company Name], is committed to providing sustainable and accessible off-grid solar (OGS) solutions. We recognize our responsibility to ensure that our operations, from product procurement to installation and customer service, contribute positively to the environmental and social well-being of the communities we serve and the planet we share. We are dedicated to upholding the highest standards of environmental and social responsibility in all aspects of our business.

Environmental Stewardship

1. **Sustainable Procurement:** We carefully select our OGS product suppliers based on their commitment to environmental and social responsibility. We prioritize products that are designed for durability, longevity, and minimal environmental impact, utilizing sustainable materials and energy-efficient technologies.
2. **Waste Management:** We actively manage waste generated during installation and end-of-life disposal of OGS products. We prioritize recycling and responsible disposal of solar components, particularly batteries, to minimize

¹¹ Sample Environmental and Social Policy Statement was generated iteratively with assistance of AI tools based on input of sample policy statements of multiple companies as well as ROGEAP PFM's sample E&S Policy Statement.

environmental harm. We educate our customers on proper disposal methods and work with local partners to establish recycling programs.

3. **Resource Efficiency:** We strive to minimize our resource consumption, including water and energy, in our offices, warehouses, and during installations. We promote energy-efficient practices and encourage responsible energy use among our customers.
4. **Environmental Education:** We empower our customers and communities with knowledge about sustainable energy practices and the environmental benefits of OGS. We provide guidance on proper system maintenance and energy conservation to maximize the lifespan of our products and minimize their environmental impact.

Social Responsibility

1. **Community Engagement:** We actively engage with the communities we serve, seeking their input and participation in our projects. We prioritize local hiring and capacity building, supporting local businesses, and contributing to community development initiatives.
2. **Fair Labor Practices:** We uphold fair labor practices throughout our operations, ensuring safe and healthy working conditions, fair wages, and respect for workers' rights. We have zero tolerance for forced or child labor.
3. **Health and Safety:** We provide a safe and healthy work environment, including occupational safety measures, emergency preparedness, injury and illness prevention, proper hygiene, and physically demanding work safeguards.
4. **Gender Equity and Empowerment:** We promote gender equity and empowerment in the OGS sector. We strive to create an inclusive workplace and actively encourage women's participation in the industry, from sales and installation to entrepreneurship. We have zero-tolerance for Gender-Based Violence (GBV) and Sexual Exploitation, Abuse and Harassment (SEAH).
5. **Access and Affordability:** We are committed to making OGS products affordable and accessible to all, particularly those living in remote and underserved areas. We offer flexible payment options and work with local partners to develop financial solutions that expand access to clean energy.
6. **Transparency and Accountability:** We are committed to conducting our business with transparency and accountability. We regularly assess our environmental and social performance, disclose relevant information to stakeholders, and continuously seek ways to improve.

Alignment with National and International Standards

We align our operations with and adhere to relevant national and international standards, including the ECOWAS Directives, ECREEE Standards, and the World Bank's Environmental and Social Framework (ESF).

Continuous Improvement

We are committed to developing and implementing an environmental and social management system (ESMS) to ensure compliance with laws, regulations, and customer requirements, as well as continuous improvement of our social and environmental performance. We regularly review and update our policies and practices to ensure they remain relevant and effective in meeting the evolving needs of our stakeholders and the planet.

Contact Us

If you have any questions or concerns regarding our environmental and social responsibility commitment, please do not hesitate to contact us at [contact information].

This policy statement embodies our pledge to operate in a manner that prioritizes the environment, the wellbeing of communities, and the highest ethical standards.

In order for an E&S Policy to be effective the following recommended conditions should be considered following the development of the E&S Policy Statement¹²:

- **Build Awareness** amongst company management, staff and stakeholders about the E&S Policy and the environmental and social issues of relevance to the company.
- **Build Commitment** or “buy-in” amongst company management and staff to operate in accordance with the E&S Policy.
- **Drive Implementation** of the E&S Policy through development and implementation of an ESMS (policies, procedures and plans), assignment of responsibilities, provision of training, and allocation of required resources.

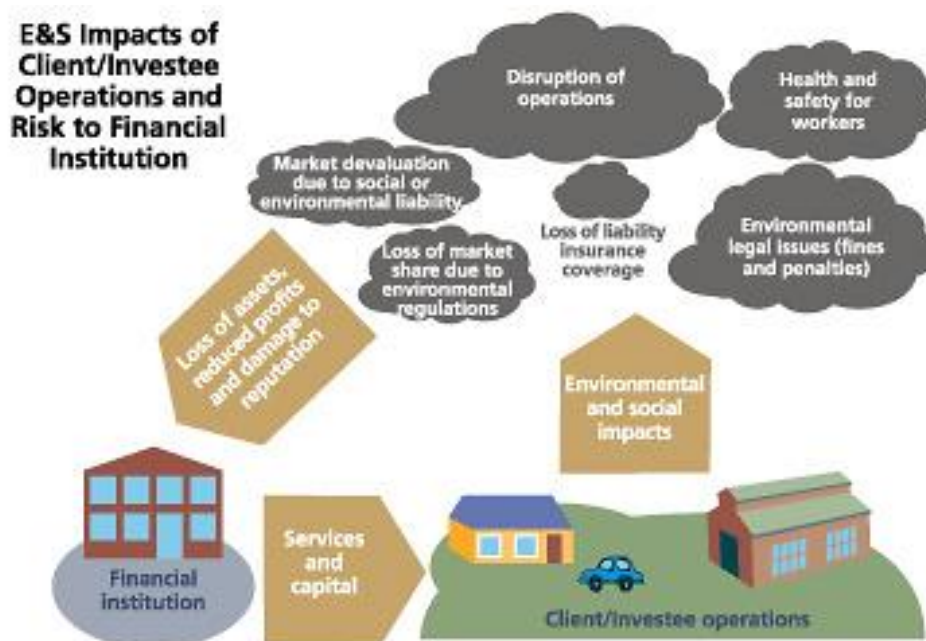
By prioritizing implementation and incorporating these key factors, your company can increase the likelihood of successful and impactful policy outcomes.

¹² IFC. 2014. Environmental and Social Management System Implementation Handbook: Construction. esms-handbook-construction-v7.pdf (ifc.org)

4. Identification of Environmental and Social Risks and Impacts

Environmental and Social (E&S) Risk and Impacts Assessment is a basic ESMS component as per **IFC PS 1 / WB ESS: Assessment and Management of Environmental and Social Risks and Impacts**. Screening and risk categorization are often used in environmental and social (E&S) risk assessment before a more detailed due diligence process is implemented to further study and analyze the likelihood and impacts of the identified risks and determine the mitigation measures required to manage the risk and impacts. The extent of E&S due diligence required is determined by the regulatory requirements of the country and the nature of the business (e.g. size, location, products, services and activities) and the risk categorization identified through the screening process.

Different stakeholders in OGS face different risks and impacts as illustrated in the following diagram.



Source: FIRST for Sustainability Website¹³

Figure 8: Environmental and Social Risks and Impacts faced by Businesses and the Financial Institutions financing their projects and operations.

The following diagram demonstrates steps of an E&S Risk Assessment process used by financial institutions to limit their exposure to E&S risks. Similar steps can and should be taken by OGS businesses financed by these institutions.



Source: FIRST for Sustainability Website¹⁴

Figure 9: Illustration of the steps in an E&S risk assessment process for a financial institution financing a business (Client/Investee)

¹³ Managing E&S Risks. (n.d.). FIRST for Sustainability. Retrieved April 12, 2024, from <https://firstforsustainability.org/managing-environmental-and-social-risk>

¹⁴ Managing E&S Risks. (n.d.). FIRST for Sustainability. Retrieved April 12, 2024, from <https://firstforsustainability.org/managing-environmental-and-social-risk>

The current document is directed primarily at companies involved in the procurement, stocking (temporary storage), sale, transport, installation, maintenance, and decommissioning of OGS, which encompasses business activities that would be classified as involving no high-risk activities and overall medium to low environmental and social impacts (i.e. Category II / B or lower)¹⁵. *“Potential adverse environmental and social risks or impacts [of such OGS activities] will be few in number, generally site-specific, largely reversible, and readily addressed through mitigation measures.”*¹⁶ These OGS business activities would generally not require an extensive environmental and social due diligence process, but rather a more simplified check-list type assessment and basic environmental social management system with plans to address and mitigate the key risks with negative impacts, while enhancing positive impacts.

The environmental screening adopted by ROGEAP categorizes businesses as either:

- **Not Eligible for Funding:** Businesses that fail to meet exclusion criteria and/or not meeting minimum E&S risk mitigation policies, procedures and tools required by ROGEAP.
- **Category B1 (Low risk):** Businesses with minimal or no adverse E&S impact requiring only simple ESMS with core components to address key risks and mitigation measures such as OHS procedures, HR manual, grievance redress mechanism, and a waste management policy
- **Category B2 (Medium risk):** Businesses that require – in addition to the above – an environmental and social action plan (ESAP) and/or E&S covenants in legal/ loan agreements, as well as close monitoring, due to the severity of their potential adverse E&S impacts as a result of the size and/or characteristics of the solar business, and/or identified situations, such as occupational accidents or labor and GBV incidents, irregularities with regard to fair working conditions and terms of employment, or installers without applying safe installation measures. The ESAP should itemize specific actions, indicators of completion, timeline for completion, and verification methods (to be included in the loan/ grant agreement).

Box 2: Exclusion Criteria applicable to applicants for ROGEAP financing

Companies seeking to qualify for a grant or loan under the ROGEAP program will need to meet the following environmental and social (E&S) exclusion criteria. A company characterized by any of the following E&S exclusion criteria is disqualified from receiving a grant or loan under ROGEAP:

- 1) Organizations promoting religious or ideological doctrines
- 2) Activities involving child labour and/or forced labour
- 3) Production or trade of tobacco
- 4) Production and sale of weapons
- 5) Gambling, casinos, and related activities
- 6) Businesses that discriminate based on race, religion, age, gender, or sexual orientation
- 7) Businesses involved in the production or trade of any product or activity deemed illegal under the country's laws or regulations or international conventions and agreements.
- 8) Cultivation, production, or business practices that degrade the environment (e.g., forest products from unmanaged forests, production methods/businesses that endanger wildlife).
- 9) Companies blacklisted by World Bank, CTF, Kingdom of the Netherlands government or any other notable entities.
- 10) Companies involved in the production or trade of products subject to international bans, such as certain pesticides/herbicides and ozone-depleting substances.
- 11) Production or activities that encroach on land belonging to indigenous peoples without the prior, informed, and documented consent of these people.
- 12) Production or trade of alcoholic beverages (excluding beer and wine).
- 13) Production or trade of drift-net fishing in the marine environment using nets longer than 2.5 km.
- 14) Projects involving dam construction or rehabilitation.
- 15) Companies classified in the 'High' risk category according to the World Bank's environmental and social policy and procedures.
- 16) Cross-border trade in waste and waste products, unless compliant with the Basel Convention and the underlying regulations
- 17) Companies with confirmed cases of occupational, health, and safety incidents or accidents (If the company can demonstrate with reports and documentation that it has been able to adequately be addressed the OHS issues and this can be verified, the business can be considered eligible for ROGEAP financing).
- 18) Companies with confirmed cases of Gender Based Violence/Sexual Exploitation and Abuse
- 19) Companies with confirmed cases of child or forced labor

¹⁵ Companies undertaking construction and operation of large manufacturing or recycling facilities for OGS products or components thereof, may however, be categorized as high risk and high impact (Category I), which would require much more detailed and extensive risk assessment procedures (e.g. Environmental and Social Impact Assessment) and comprehensive environmental and social management and monitoring plans (ESMP).

¹⁶ The World Bank. 2020. Project Information Document / Integrated Safeguards Data Sheet (PID/ISDS). Regional Off Grid Electricity Access Project Additional Financing (P174885).

The following criteria ("warning criteria") are not automatic exclusion criteria but require specific analysis by the PFM on the basis of corrective measures put in place:

- Involvement of members of the company's management in another company that has gone bankrupt.
- Violations of the company's labour laws in the last five years.
- Violations of the company's tax laws in the last five years.
- Environmental and social violations by the company in the last five years.
- Involvement of politically exposed persons in the company.
- Non-compliance by a company on environmental and social performance criteria, mitigation and improvement plans must be analysed and highlighted as per ROGEAP's Environmental and Social Management Framework; an environmental and social action plan (ESAP) is required of companies classified as medium/high E&S risk (B2) and companies that do not have all ESMS elements in place to plan for the development and implementation of a full ESMS.

Companies must comply with national environmental, social, health and safety (ESHS) laws and regulations of the countries where they operate. ESHS requirements should be assessed for each business activity.

OGS standalone system installation, maintenance, and decommissioning projects undertaken by OGS solar businesses would involve minimal civil works and comprise mainly installation, servicing or removal of solar generation equipment and associated electro-mechanical works. The World Bank's integrated safeguards datasheet for ROGEAP states that *"...the nature of business activities in this sector that will be supported by the project, key environment and social risks are expected to include waste management (disposal and recycling of solar panels and used solar equipment, and especially lead acid and lithium ion batteries, which are considered hazardous waste), water quantity and quality for panel washing, Sexual Exploitation and Abuse (SEA)/Sexual Harassment (SH) safety of OHS practices for solar companies' workers, and labor issues (potential instances of child or forced labor, absence of proper grievance mechanism including a GM to address SEA/SH complaints, fair terms of employment, inclusive access to women in employment... Land-related issues are not expected to be significant. In general, land acquisition leading to economic and/or physical displacement is not expected to occur in the project and, in case of commercial solar equipment installations would be done on willing-seller-willing-buyer basis."*¹⁷

Most ESHS risks for private sector solar companies involved with OGS are associated with the occupational, health and safety (OHS) risks arising from the activities of installation, maintenance and decommissioning of the standalone solar equipment and the environmental and health risks from the e-waste generated from the products. The Figure below lists common ESHS risks associated with each component of OGS, while a more extensive list of risks and potential mitigation measures are listed in the table below.

¹⁷ The World Bank. 2020. Project Information Document / Integrated Safeguards Data Sheet (PID/ISDS). Regional Off Grid Electricity Access Project Additional Financing (P174885).

Off-Grid Solar Product and Project Environmental Health and Safety (ESHS) Risks	OGS Component	ESHS Risks and Impact
	Photovoltaics	Electric Shock, Fire, Smoke, Fall of PV Module, Head Injury, Cuts, Bumps, Waste, Environmental Pollution
	Inverter, Charge Controller	Electric Shock, Fire, Smoke, Noise, Waste
	Batteries	Electric Shock, Fire, Smoke, Chemical Fumes and Burns, Explosion, Waste, Chemical Spill and Environmental Contamination
	Cables	Electric Shock, Fire, Smoke, Punctures, Cuts, Waste
	Appliances	Electric Shock, Fire, Smoke, Burns, Cuts, Waste, Noise, Greenhouse Gas (GHG) and Ozone Depleting Emissions, Water Extraction
	Transport	Accidents, Fire, Smoke, Burns, Chemical Burns, Back Injuries, Muscle Strains, Cuts, Bumps, Waste, GHG/Air Pollution Emissions
	On-Site Work	Electric Shock, Fire, Smoke, Burns, Fall From Ladder or Rooftop, Chemical Burns, Cuts, Bumps, Insect and Snake Bites, Gender Based Violence, Sexual Exploitation and Abuse, Sexual Harassment, Labor Issues, Exclusive Employment

Source: Adapted from SNV 2015 Solar PV Standardized Training Manual

Figure 10: Health and safety risks and hazards associated with stand-alone solar power systems

The probability of each identified risk occurring and the severity of the impact can be assessed and mapped to determine which risks are critical, medium-level, or low-level, which in turn helps inform the level of prioritization of mitigation measures and resources. Each risk can be assigned a probability rating of low (1), medium (2), high (3), or extreme (4) and an impact severity rating of low (1), medium (2), high (3), or extreme (4). The risks can then be assessed and mapped out using the diagram to the right, to determine the risk rating.

The table below provides an overview of common risks associated with OGS. For each risk a possible risk and impact rating is provided, though this will vary depending on the context and thus should be assessed in each case. And finally the table provides suggested mitigation measures as well as applicable standards from World Bank ESS and IFC PS.

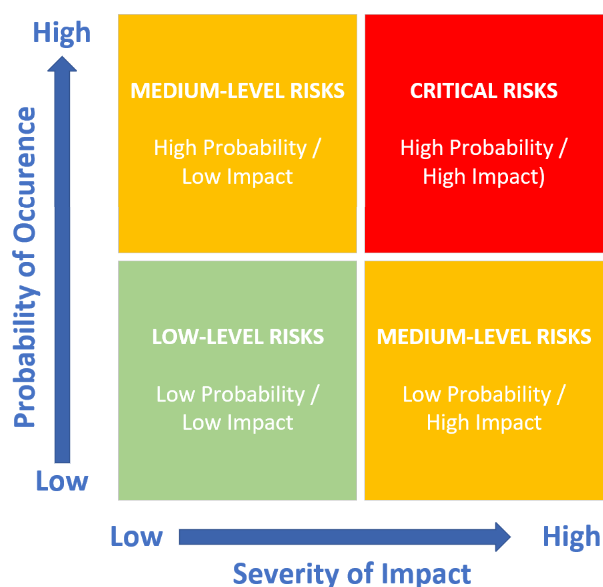


Table 5: Key Environmental and Social Risks and Mitigation Measures Often Associated with Off-Grid Solar Products and Services and the International Standards of Relevance

Risks and challenges	Description	Probability	Impact	Suggested mitigation measures	International Standards
Waste management (e-waste, used batteries / chemical waste) from the OGS and equipment and devices powered or charged by solar energy.	- Used panels, used batteries and units (both lead-acid and lithium-ion) are hazardous waste - Improper recycling of lead acid batteries causes wide-scale lead pollution/poisoning, including air, soil, and water contamination; lead entry into the food chain resulting in diseases and fatalities - Management of used batteries will be a significant risk - Additional waste issues are related to plastic material, polystyrene residues, aluminium, copper, steel	High	High	- Encourage incorporating the cost of the responsible management of waste into the business budgeting and financial prospects. Responsibly choose a waste management partner - Ensure compliance with country regulations for recycling batteries - Engagement with other businesses and stakeholders to explore practical waste management solutions - Proper management of waste systems and channel - Establishment of a reparation network and replacement parts channel - Communication on e-waste management for end users and communities - Internal training on waste management processes - Promote reuse, recycling or energy recovery plastic treatment - Reuse, recycling or energy recovery in an equipped unit for polystyrene treatment - Implementation of voluntary management systems such as ISO 14001¹	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 3 / WB ESS 3: Resource efficiency and pollution prevention
Occupational Health and Safety (OHS) for Workers	- Slips and trips, falls - Manual handling issues - Hazards of musculoskeletal disorders, Injuries, lack of protective equipment, etc.	High	High	- Solar equipment needs to be installed safely - Assessment of workplace hazards. Consult and involve workers in the workplace risk assessment as well as in the choice of prevention measures - Internal training on the type of risks and suppression measures (safe working procedures) - Supply appropriate Personal Protective Equipment (PPE) and train staff and subcontractors on its use and ensure that it is properly maintained - Maintain a fully stocked and accessible first aid kit	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 2 / WB ESS 2: Labour and working conditions
Labour Issues	- Child or forced labour - Improper grievance redress for workers - Unfair terms of employment	Medium	High	- No employment or use of child or forced labour - Put in place HR policies that articulate transparent and fair terms of employment and provide for no discrimination and equal opportunity - Fair and inclusive employment practices (including employing women) can lead to better business and better workers - Develop and implement a proper grievance redress mechanism - Proper training and record on the GRM system or working procedure	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 2 / WB ESS 2: Labour and working conditions - IFC PS 4 / WB ESS 4: Community health safety and security - WB ESS 10: Stakeholder Engagement and Information Disclosure

Risks and challenges	Description	Probability	Impact	Suggested mitigation measures	International Standards
Gender actions	- Women are disproportionately affected by lack of reliable access to energy (exclusion risks); this includes underserved female-headed households and women with disabilities, illiterate women and landless women - Gender-based violence/Sexual - Exploitation and Abuse/Sexual - Harassment may occur in the context of OGS business and service delivery activities or the context of changes in OGS end user lifestyle and activities - Social tension and decreased social cohesion may occur in the context of OGS access and the resulting changes in end-user lifestyle and activities	Medium	High	- Employment of women as part of fair and inclusive labour practices - Gender-sensitive stakeholder engagement and capacity building; - Promote active participation of women entrepreneurs, women's organisations, civil society and non-governmental organisations working on gender and energy issues - Increase information and awareness of women that will allow them to enter the renewable energy market - Capacity building and internal training on Gender-based violence (GBV)/Sexual Exploitation and Abuse (SEA)/ and Sexual Harassment (SH) prevention and mitigation measures; - GBV, SEA, and SH to be reported and dealt with as per the law - Establish a code of conduct for beneficiary companies to prevent GBV/SEA, including termination of employees/contractors engaging in GBV/SEA - Stakeholder engagement measures to identify and take into consideration possible social tensions and conflicts within communities - The GRMs handling local complaints should include a process for handling GBV/SEA/SH complaints	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 2 / WB ESS 2: Labour and working conditions - IFC PS 4 / WB ESS 4: Community health safety and security - WB ESS 10: Stakeholder Engagement and Information Disclosure
Inclusion of vulnerable groups	Prejudice or discrimination toward individuals or groups who, due to their peculiar circumstances, may be at a disadvantage in getting access to energy	Medium	Medium	- Engage with all staff, customers, suppliers and other stakeholders without prejudice or discrimination toward vulnerable groups - Produce and services communication accessible to illiterate and semi-literate customers/end users - Issue company policy statement of non-discrimination in employment and business practices	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 2 / WB ESS 2: Labour and working - IFC PS 4 / WB ESS 4: Community health safety and security - WB ESS 10: Indigenous peoples [if present] - Stakeholder Engagement and Information Disclosure
Sustainability of the supply chains	Production of solar panels or other components of solar equipment may be associated with several unsustainable practices such as labour rights violations, poor working conditions, absence of clean production standards, etc.	Low	Low	- Awareness of supply chain E&S risks and including E&S risks in procurement decision-making process; - Promote products produced sustainably.	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 2 / WB ESS 2: Labour and working conditions

Risks and challenges	Description	Probability	Impact	Suggested mitigation measures	International Standards
					• IFC PS 3 / WB ESS 3: Resource efficiency and pollution prevention
End user (consumer) health and safety	• Risks to end-user health and safety from falling objects (e.g. roof-top solar panels), leaked hazardous chemicals, electrical shocks, fires or explosion of batteries.	Low	Low	• Awareness raising of end users on key E&S issues that affect them (e.g. water consumption and water quality for panel washing) • Educate customers / end users about proper and safe practices for use of equipment • Ensure safe installation • Proper isolation of equipment • Proper signalisation of the solar power system • Awareness raising on e-waste generation and management	• IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts • IFC PS 4 / WB ESS 4: Community health safety and security • WB ESS 10: Stakeholder Engagement and Information Disclosure
Stakeholder Engagement and Grievance Redress Mechanism (GRM)	• Stakeholders must be engaged (informed and consulted) during project implementation per WB requirements. Furthermore, GRM must be in place to enable channels for stakeholders to communicate complaints and comments for resolution and response.	Low	Low	• Awareness raising of customers/end users on key E&S issues that affect them (e.g. water consumption and water quality for panel washing) • Prepare and implement a stakeholder engagement plan for disclosures and consultation • Development and implementation of a GRM, including a process for handling GBV/SEA/SH complaints confidentially, quickly (within 72 hours), and under a survivor-centred approach, with referral pathways established to ensure that all survivors are referred to essential medical, psychosocial, and if possible and available legal services at a minimum	• WB ESS 10: Stakeholder Engagement and Information Disclosure
Land issues	• If the installations need some on-the-ground space (as opposed to rooftops), ensure the areas are suitable for installation • Key risks may be related to voluntary land donation in the case of public/community buildings	Low	Medium	• Photovoltaic installations on the ground must take into account the protection of existing agricultural and forestry areas • Prioritize “degraded sites” (brownfield sites). Avoid areas subject to natural hazards • Obtain written consent from the building users/owners. Similar procedures would be adopted for ground-mounted systems.	• IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts • IFC PS 3 / WB ESS 3: Resource efficiency and pollution prevention • IFC PS 5 / WB ESS 5: Land acquisition and involuntary resettlement [unlikely] • IFC PS 8 / WB ESS 8: Cultural heritage [unlikely]
Water Resource Risk	• Increased extraction and use of water using solar-powered pumps and for cleaning solar panels.	Low	Medium	• Creating awareness among the customers / end users of the consequences of the increase in water usage as their operations increase • Education of end users in the productive use of the technology	• IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts

Risks and challenges	Description	Probability	Impact	Suggested mitigation measures	International Standards
	Water scarcity may lead to social conflicts between community users				IFC PS 3 / WB ESS 3: Resource efficiency and pollution prevention
Air Pollution Risk	Increased use of halocarbon-containing appliances such as refrigerators and air conditions	Low	Low	- Creating awareness among the end users of the consequences of usage of appliances using halocarbon gas and encouraging the use of more climate-friendly systems - Prioritize the import and sale of climate-friendly air conditioners and dehumidifiers.	- IFC PS 1 / WB ESS 1: Assessment and management of E&S risks and impacts - IFC PS 3 / WB ESS 3: Resource efficiency and pollution prevention

Source: ROGEAP Operation Manual 2020

The following Environmental and Social Screening Questionnaire was developed by ROGEAP for determining which solar businesses may qualify for financing under the project:

Table 6: ROGEAP Environmental and Social Screening Questionnaire for Solar Businesses

Section A. Company details		
Company name:		
Company type:		
Year of establishment		
Short description of the company core business:		
Company Address:		
<i>(Name of the company / Street, Avenue, etc. name and number /Village / City / State/ Country/ Postal Code)</i>		
Contact person for E&S issues:		
<i>(Name and title / Phone /E-mail)</i>		
Name and title of the person filling the questionnaire		
Website <i>(if any)</i> :		
Section B. Business activities (solar sector)		
Total number and value of business transactions over the last year in the solar photovoltaic systems sector):	Number	Value (specify currency)
Solar Home Systems		
Stand-alone Solar Systems		
Pico PV products		
Solar Irrigation Power Systems		
Other <i>(specify)</i>		
Section C. Environmental and Social Risk Management		
Has the company been found to be involved in activities meeting the E&S exclusion criteria:	Yes	No
Production or activities involving forced labor?		
Production or activities involving child labor?		
Cross-border trade in waste and waste products, unless compliant to the Basel Convention and the underlying regulations?		
Confirmed cases of occupational, health, and safety incidents or accidents? (Note: If the company can demonstrate with reports and documentation that it has been able to adequately be addressed the OHS issues and this can be verified, the business can be considered eligible for financing among other requirements)		
Confirmed cases of Gender Based Violence/Sexual Exploitation and Abuse?		
Confirmed cases of discrimination of vulnerable groups, including gender and disability?		
If the financial institution has a cause to believe that these issues have been duly dealt with according to their satisfaction then the business can be considered for financing	Yes	No
Labour-related risks (terms of employment, labor practices)?		

Occupational health and safety of workers?		
Waste management risks (e-waste, used batteries, other waste)?		
Gender and vulnerable groups issues?		
Land-related issues (customers need to purchase land, need land donation from communities, resettlement of households for installations on the ground for public buildings etc.?)		
Interactions with communities where the company works?		
Water/ resource efficiency issues?		
End users' health and safety?		
Environmental or social issues in supply chains (e.g. reputational risks due to well-known poor labor conditions of panel or other equipment manufacturers, non-responsive or weak grievance mechanism?)		
Other (please describe)?		
Environmental and / or Social Policy	Yes	No
Does the company have an Environmental and / or Social Policy?		
<i>(If yes, please attach any documents that can serve as evidence of such policy)</i>		
Does the company conduct any environmental, social, gender awareness and SEA/SH training internally for its staff?		
Does the company conduct any E&S monitoring visits with its business customers?		
Human Resources Policy	Yes	No
Does the company have a Human Resources Policy?		
<i>If yes, please provide a brief description of key provisions of the HR policy and attach any documents that describe HR policy</i>		
Does the company's Human Resources Policy explicitly provide for the following:	Yes	No
Compliance with country's labor laws and regulations?		
Clear terms and conditions of employment and worker's rights related to hours of work, wages, overtime, compensation, benefits, etc.		
Non-discrimination and equal employment opportunities for women and vulnerable groups?		
Prohibition of child labor?		
Prohibition of forced labor?		
Is there a workers' grievance mechanism in place?		
Occupational Health and Safety (OHS)	Yes	No
Does the company have a policy or guidelines on occupational health and safety?		
<i>If yes, please provide a brief description of key provisions of such policy/guidelines and attach any documents that describe this</i>		
Does the company have qualified internal designated coordinators, officers, or other staff responsible to oversee OHS issues?		
Does the company provide Personal Protective Equipment (PPE) to its workers?		
Does the company conduct regular OHS training for its workers and employees?		
Does the company have a clear, documented workplace incident and accident tracking system?		

Does the company have a Code of Conduct for workers? <i>If yes, attach copy</i>		
Does the company provide internal training on Gender Based Violence?		
Does the company have a monitoring system for workplace conditions and safety (e.g. regular internal audits, field visits by company OHS staff/coordinators etc.?)		
E-waste and other environmental issues	Yes	No
Does the company have any policy or process for collecting used batteries (both lead acid and lithium ion), as well as used units and equipment from customers?		
<i>If yes, please describe the approach and process currently on place. What challenges does the company encounter with disposal and recycling of these products? Is the company contributing to proper reparation channels?</i>		
Does the company have any policy or process for collecting, sorting, recycling and disposal of used lead acid and lithium-ion batteries (or, units containing such batteries) or any other used material resulting from solar systems installation process and subsequent use?		
<i>If yes, please describe the approach and process currently on place</i>		
Does the company have any buy-back agreements with equipment manufacturers as part of its waste management approach?		
Does the company systematically collect use batteries and/ or units from its consumers?		
Does the company inform end users on the e-waste issue and provide them with information on proper e-waste management?		
Does the company inform end users on other environmental issues? i.e. overexploitation of water resources*? <i>*Only in consideration in case of solar water pumping installation activities</i>		
Stakeholder engagement and grievance mechanisms	Yes	No
Does the company have a plan for engaging with stakeholders, communities, customers (including a mechanism to receive and address complaints)?		
<i>If yes, please provide a brief description of key provisions of such policy/guidelines and attach any documents that describe this</i>		
Does the company engage in consumer and public education about environmental and social aspects of solar energy?		
Did the company identify key external stakeholders for its business?		
Does the company hold events or similar actions to engage with stakeholders relevant to its business?		
Is there a written mechanism to receive and address complaints? If yes, please attach a copy of the procedure, website link etc.		
Any other relevant information		

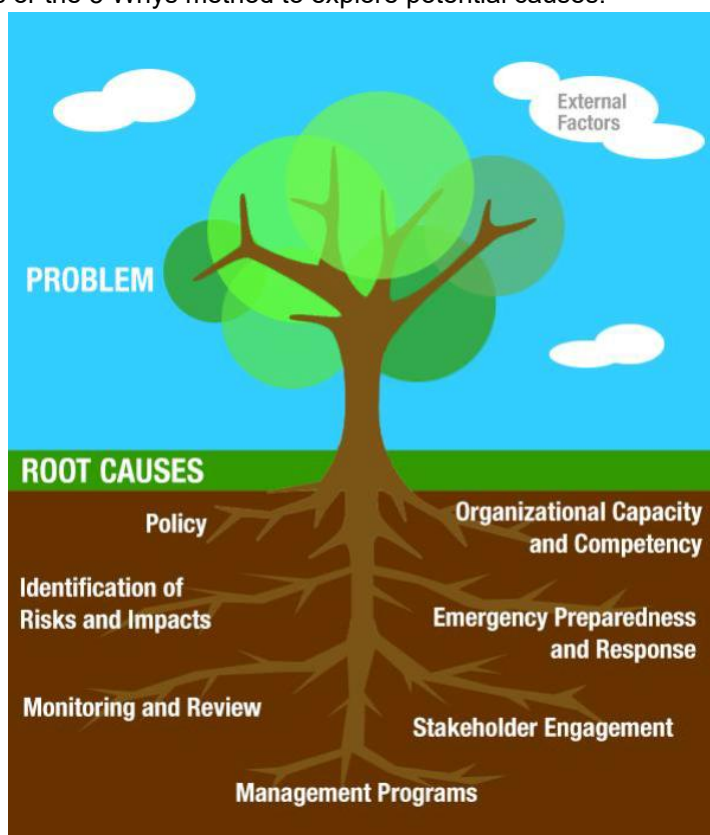
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Root Cause Analysis

After assessing and rating environmental and social risks and impacts, the next step is identifying the underlying causes of the risks through root cause analysis (RCA). This process aims to uncover the fundamental reasons behind issues, rather than just addressing the surface-level symptoms.

Steps of root cause analysis:

1. **Define the Problem:** Clearly articulate the specific environmental or social risk/impact you're investigating. Be precise and consider the unique context of the off-grid solar sector. For example, instead of "waste management issues," specify "improper disposal of lead-acid batteries leading to soil and water contamination in rural communities."
2. **Gather Data:** Collect relevant information related to the problem, considering both environmental and social factors. This may include data on carbon emissions, waste generation, labor conditions, community engagement, customer feedback, and incident reports. Partner with local organizations or stakeholders for additional insights.
3. **Identify Potential Causes:** Brainstorm and list all possible factors contributing to the problem, keeping in mind the interconnected nature of environmental and social issues. Consider factors like raw material sourcing, manufacturing processes, distribution networks, product design, consumer behavior, government policies, and cultural norms. Use tools like fishbone diagrams or the 5 Whys method to explore potential causes.
4. **Construct the Root Cause Tree Diagram:**
 - Place the main environmental or social risk/impact at the top of the tree.
 - Branch out to identify direct causes, considering both environmental and social factors. Ask "Why does this happen?" for each cause.
 - Continue branching out to identify deeper causes, delving into systemic issues, social inequalities, environmental degradation, or failures in supply chain management, product design, or community engagement.
 - Categorize similar root causes under relevant ESMS elements (Policy, Risk Identification and Impact, Management Programs, etc.).
 - Note any external factors outside your control, such as climate change impacts, political instability, or economic fluctuations.
5. **Analyze and Prioritize:** Evaluate each root cause based on its severity, likelihood, and potential impact on both the environment and society. Prioritize the most significant and actionable root causes, considering their interconnectedness and potential for long-term positive impact.
6. **Develop Corrective Actions:** Design and implement solutions that address the identified root causes, integrating environmental and social considerations. This may involve diversifying suppliers, improving product design for recyclability, establishing fair labor practices, partnering with local communities, or advocating for supportive policies. Ensure the solutions are measurable and aligned with your company's sustainability goals.
7. **Monitor and Adapt:** Continuously track the effectiveness of the corrective actions, collecting data on environmental and social outcomes. Adapt your strategies as needed based on the evolving context and feedback from stakeholders.

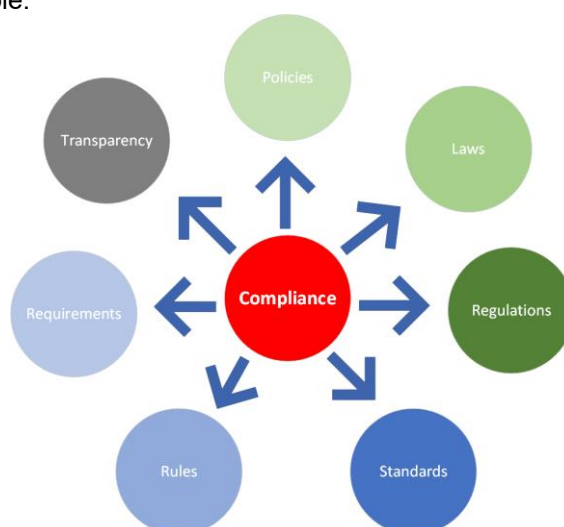


Source: IFC 2015

Figure 11: A general root cause tree diagram

5. Compliance

Compliance, within the context of an Environmental and Social Management System (ESMS), refers to the systematic adherence to all applicable legal, regulatory, tax, contractual, and other internal and external requirements related to environmental performance, social responsibility, occupational health and safety (OHS), and ethical business conduct. Given the dynamic regulatory landscape and diverse operational environments across ROGEAP countries, a proactive and country specific approach to compliance is indispensable.



Source: Adapted from integritybackgrounds.com 2025

Figure 12: Key elements to be considered towards achieving compliance

The rationale for managing compliance extends beyond avoiding penalties and is fundamental to operating a resilient and responsible business, with the following potential benefits:

- Legal Protection and Avoidance of Penalties
- Enhanced Reputation and Stakeholder Trust
- Access to Finance and Investment
- Operational Continuity and Stability
- Proactive Risk Mitigation
- Promotion of Ethical Business Practices
- Expanded Market Access and Competitive Advantage

Legal Compliance

Legal compliance refers to adherence to national, state, and local laws and decrees.

Environmental Laws:

- **National Environmental Agencies:** Compliance with acts and regulations enforced by national bodies.
- **Specific Regulations:** This includes regulations on **Environmental Impact Assessment (EIA)** (required for projects above a certain threshold, though this is rarely triggered by OGS projects), **Waste Management** (e.g., proper segregation, storage, transportation, and disposal of general waste, hazardous waste like spent batteries, and e-waste), **Pollution Control** (e.g., limits on noise, air emissions from generators used for backup, or wastewater discharge from facilities), and **Chemical and Hazardous Substances Management**.

Social Laws:

- **Labor Laws:** Strict adherence to national labor acts. This covers minimum wage, maximum working hours, overtime pay, holidays, leave entitlements, prohibition of child labor and forced labor, non-discrimination in employment (gender, religion, ethnicity), freedom of association (right to join trade unions), and termination procedures. Compliance also extends to mandatory social security contributions.
- **Community Relations:** Laws related to **land tenure and acquisition** (if relevant to larger installations or company premises), requirements for **community consultation** for projects impacting local livelihoods or resources, and the provision for **local content development**.

(e.g., local employment, local sourcing of materials).

- **Consumer Protection:** Laws ensuring product safety, fair trade practices, transparent pricing, warranty obligations, and consumer data privacy.

Occupational Health & Safety (OHS) Laws:

- **Workplace Safety Acts:** Compliance with national OHS acts and specific industry regulations.
- **Key Requirements:** Mandates for conducting **workplace risk assessments**, providing **Personal Protective Equipment (PPE)**, implementing **first aid provisions**, establishing **safety committees** (for larger operations), providing **OHS training**, and ensuring **mandatory accident reporting** to regulatory bodies.

Business Registration & Licensing:

- Compliance with laws governing company incorporation, annual company filings, and obtaining general business operating licenses from state or local governments.
- Specific professional body registrations for technical roles (e.g., electrical engineers).

Regulatory Compliance

Regulatory compliance covers rules issued by specific governmental or quasi-governmental agencies often with delegated authority.

- **Energy Sector Regulations:** Specific directives from national energy regulators regarding licensing requirements for power generation (even small scale if connected to grid or above certain capacity), product standards, technical specifications for solar installations, and consumer tariffs for larger systems. Rural electrification agencies may have specific compliance requirements for projects they fund or regulate.
- **Product Standards:** Adherence to national product certifications and international product safety and quality standards (e.g., **IEC standards**, **VeraSol Quality Standards**) for solar panels, batteries, inverters, charge controllers, and accessories. This ensures product reliability and safety for consumers.
- **Local Government Regulations:** Requirements from municipal or district authorities for building permits (for larger installations or company premises), environmental permits (e.g., for boreholes, waste disposal in specific areas), signage regulations, and business rates/taxes.
- **Sector-Specific Directives:** Any specific guidelines or directives issued by industry associations, donor organizations, or financing bodies that the company has voluntarily or contractually committed to.

Tax Compliance

Tax compliance refers to meeting all tax obligations fundamental for legal operation and financial stability.

- **Corporate Income Tax:** Annual taxation on the company's net profits, filed with the national tax authority.
- **Value Added Tax (VAT) / Goods and Services Tax (GST):** Collection of VAT on sales of OGS products and services from customers, and timely remittance to the relevant tax authority. Businesses must register for VAT if their turnover exceeds a certain threshold.
- **Pay As You Earn (PAYE):** Income tax deducted directly from employee salaries and remitted to the relevant Internal Revenue institution.
- **Withholding Tax (WHT):** Tax deducted at source on certain types of payments made by the company (e.g., rent, professional service fees, contracts) and remitted to the tax authority.
- **Customs Duties and Levies:** Payment of applicable import duties and levies on solar products, components, and equipment brought into the country.
- **Other Levies:** Potential payment of local government levies, environmental levies, or development contributions.

Other Commitments

Other non-statutory compliance considerations include:

- **Contractual Obligations:** Adherence to all terms and conditions set forth in contracts with clients, suppliers, investors, or financial partners. This often includes specific E&S clauses (e.g., "no child labor in the supply chain," "adherence to IFC Performance Standards" in a loan agreement).

- **Voluntary Standards and Certifications:** Commitments to international best practices (e.g., **UN Guiding Principles on Business and Human Rights**, **ISO 9001 for Quality Management**, **ISO 14001 for Environmental Management**, **ISO 45001 for Occupational Health & Safety**), or specific industry initiatives (e.g., **GOGLA Code of Conduct**). These are often adopted to improve performance, enhance reputation, and attract specific types of investment or partnerships.
- **Anti-Bribery and Corruption (ABC) Policy:** Compliance with national anti-corruption laws and, if operating internationally or engaging with certain foreign entities, international laws like the UK Bribery Act or US Foreign Corrupt Practices Act (FCPA). This involves establishing clear policies, training staff, and maintaining transparent financial records.

Procedures for Ensuring Compliance

For companies, a structured, systematic, and documented approach to compliance is vital to navigate the complexities effectively.

Identifying Applicable Requirements

- 1) **Legal and Regulatory Register Development:**
 - **Process:** Create and maintain a dynamic register (e.g., using a spreadsheet or dedicated software) that lists all pertinent legal acts, regulations, permits, and licenses.
 - **Content:** For each entry, include:
 - Law/Regulation Name and Citation (e.g., *National Environmental (Waste) Regulations, 2009*)
 - Issuing Authority (e.g., Ministry of Environment)
 - Effective Date and any amendments
 - Relevant Sections/Articles pertinent to OGS operations (e.g., "Part IV: Management of Used Electrical and Electronic Equipment")
 - Specific Requirements/Obligations (e.g., "Duty to segregate e-waste from other waste streams")
 - Applicability to Company Operations (e.g., "Applies to product sales and end-of-life management")
 - Frequency of Compliance (e.g., Annually, Per incident, Monthly)
 - Evidence of Compliance (e.g., Waste disposal manifest, Training certificate)
 - Responsible Person/Department
 - Review Date
- 2) **Stakeholder Engagement and Consultation:**
 - **Process:** Actively engage with key government agencies, industry associations, and legal/accounting counsel.
 - **Purpose:** To gain insights into current requirements, anticipate upcoming changes, and clarify ambiguities in existing regulations. Participate in relevant public consultations or workshops where available.
- 3) **Permit and License Matrix:**
 - **Process:** Develop a dedicated matrix that centralizes information on all necessary permits, licenses, and registrations.
 - **Content:** Include permit name, issuing authority, application requirements, validity period, renewal dates, associated fees, and the responsible individual for application/renewal. This serves as a critical tracking tool.

Tracking and Interpreting Changes

- 1) **Designated Responsibility:**
 - **Process:** Clearly assign the responsibility for tracking legislative and regulatory changes to a specific individual or team (e.g., the ESMS Manager, HR Manager, or a dedicated Compliance Officer if the company grows). This person is the "compliance champion."
- 2) **Alert and Monitoring Systems:**
 - **Process:** Subscribe to official government gazettes, national energy/environmental agency newsletters, industry association updates, and reputable legal news services.
 - **Purpose:** To receive timely notifications of new laws, amendments, or policy changes.

3) **Regular Legal and Expert Review:**

- **Process:** Periodically (e.g., annually or bi-annually), engage external legal counsel or specialized E&S consultants to conduct a formal review of the compliance register.
- **Purpose:** To interpret the implications of new legislation for the company's specific operations and advise on necessary adjustments to policies and procedures.

Implementing Controls and Operational Procedures

1) **Integration into ESMS Management Plans:**

- **Process:** Ensure that compliance requirements are not standalone but are directly embedded within all relevant ESMS management plans (e.g., the Waste Management Plan must detail how e-waste is disposed of in compliance with regulations; the OHS Plan must stipulate adherence to the workplace safety laws).

2) **Standard Operating Procedures (SOPs) Development:**

- **Process:** For every task with a significant compliance implication, develop clear, written SOPs. These should be accessible to all relevant employees.
- **Content:** SOPs should detail the purpose, scope, assigned responsibilities, step-by-step instructions, necessary equipment, safety precautions, quality control checks, and relevant forms/records for documenting compliance. Examples include: "SOP for Safe Battery Handling and Storage," "SOP for E-waste Segregation and Disposal," "SOP for Electrical Installation Safety (LOTO)," "SOP for Customer Data Privacy."

3) **Permit Acquisition and Renewal System:**

- **Process:** Establish a centralized system (e.g., a shared calendar, dedicated software) for tracking all permits and licenses. Assign specific individuals responsibility for each permit's application, tracking, and timely renewal. Implement automated reminders.

4) **Financial Controls and Tax Calendars:**

- **Process:** Implement robust internal financial controls and accounting practices to ensure accurate calculation and timely remittance of all taxes (corporate, VAT, PAYE, WHT).
- **Tools:** Utilize tax calendars to track submission deadlines for various tax obligations. Implement segregation of duties for financial transactions.

5) **Contractual Review and Due Diligence:**

- **Process:** Establish a procedure for legal review of all new contracts with clients, suppliers, and partners.
- **Focus:** Ensure that E&S clauses, OHS responsibilities, labor standards, and anti-corruption provisions are clearly defined, especially when dealing with third-party installers or suppliers. Conduct due diligence on key partners to assess their own compliance records.

Monitoring, Measurement, and Reporting

1) **Internal Compliance Audits:**

- **Process:** Conduct regular, scheduled internal audits (e.g., quarterly for key areas, annually for comprehensive review) of compliance status across all operational areas (e.g., OHS compliance at installation sites, waste management practices at the warehouse, labor practices in HR).
- **Scope:** Define the scope of each audit, the criteria (based on the compliance register and SOPs), the qualifications of internal auditors, and the reporting structure for audit findings.

2) **Compliance Checklists:**

- **Process:** Develop and utilize specific checklists during site inspections, facility audits, and process reviews to verify adherence to particular compliance requirements (e.g., "Daily PPE Compliance Checklist," "Monthly Waste Segregation Check," "LOTO Procedure Adherence Check").

3) **Key Performance Indicators (KPIs) for Compliance:**

- **Process:** Establish measurable KPIs to track compliance performance.
 - **Examples:** Percentage of staff receiving mandatory OHS training, number of OHS incidents (Lost Time Injury Frequency Rate - LTIFR), percentage of permits renewed on time, volume of hazardous waste properly disposed, tax payment punctuality rate, number of grievances received/resolved.
- 4) **Internal Reporting System:**
- **Process:** Implement a clear system for internal reporting of compliance status, identified non-conformances, and the progress of corrective actions to relevant management levels.
 - **Frequency:** Regular reports (e.g., monthly ESMS committee meetings, quarterly board reports) ensure that compliance issues are visible and addressed promptly.
- 5) **External Reporting:**
- **Process:** Ensure the timely and accurate preparation and submission of all required reports to regulatory bodies (e.g., environmental monitoring reports to environmental authorities, OHS incident reports to relevant labor authorities, annual tax filings to tax authorities).
 - **Verification:** Implement a review process to verify the accuracy and completeness of all external reports before submission.

Grievance Mechanisms

Effective grievance mechanisms are critical for identifying and addressing issues before they escalate, demonstrating commitment to accountability.

- 1) **Internal (Employee) Grievance Procedure:**
- **Process:** Establish clear, accessible, and confidential procedures for employees to report concerns related to OHS issues, labor practices, discrimination, harassment, or ethical conduct.
 - **Features:** This should include multiple access points (e.g., anonymous suggestion box, direct contact with HR or a designated independent person, email), a clear investigation process, timelines for resolution, and a strong non-retaliation policy to protect reporters.
- 2) **External (Community/Stakeholder) Grievance Mechanism:**
- **Process:** For community-related issues (e.g., noise during installation, dust from operations, perceived land use impacts), implement a transparent, culturally appropriate, and accessible grievance mechanism.
 - **Features:** This may involve designated community liaison officers, accessible phone lines/emails, suggestion boxes in local communities, or regular community meetings. The process should clearly outline how grievances are logged, investigated impartially, responded to (with clear timelines), and followed up. It should be culturally sensitive and easy for non-literate individuals to use.

Best Practices for SMEs in Compliance

Successfully embedding compliance within an OGS SME's operations requires strategic yet practical approaches:

- **Appoint a Dedicated Compliance Champion:** Even if it's not a full-time role initially, designate one individual (e.g., the ESMS Manager, Operations Manager, or a senior administrator) as the primary owner and coordinator of all compliance activities. This person will maintain the compliance register, track permits, coordinate training, and be the first point of contact for compliance questions.
- **Simplify Documentation and Communication:** While procedures must be thorough, they should also be easy to understand and follow. Use clear language, bullet points, flowcharts, and visuals. Translate key safety procedures or community engagement guidelines into relevant local languages where necessary. Avoid unnecessary jargon or bureaucratic complexity.

- **Leverage Simple Technology Tools:** You don't need expensive software. Spreadsheets can be powerful for managing compliance registers, permit trackers, training records, and incident logs. Cloud storage (e.g., Google Drive, Dropbox) can ensure documents are accessible to authorized personnel from different locations. Consider basic project management tools for tracking tasks and deadlines.
- **Conduct Regular Internal Reviews and Self-Assessments:** Implement frequent, even if informal, checks to ensure adherence to key procedures. These might be weekly site safety walks, monthly waste segregation checks, or quarterly reviews of tax payment records. These regular "pulse checks" help catch potential issues early before they become major non-conformances.
- **Prioritize and Invest in Continuous Training:** Compliance is a continuous learning process. Regularly train all staff on relevant compliance aspects, particularly OHS, ethical conduct, product safety, and specific operational procedures. Emphasize why compliance is important and how each employee contributes. Include modules on anti-bribery and corruption for relevant staff.
- **Foster a Strong Culture of Compliance from the Top:** Leadership must visibly commit to compliance. This means setting clear expectations, allocating necessary resources, leading by example, and actively participating in compliance reviews. Reward good compliance behavior and address non-compliance consistently and fairly. Encourage employees to speak up about concerns without fear of retaliation.
- **Proactively Engage with Stakeholders:** Build positive, collaborative relationships with local authorities, regulatory bodies, and community leaders. Proactive communication and engagement can facilitate understanding of requirements, resolve ambiguities, and build goodwill that can be invaluable in times of challenge.
- **Learn from Every Incident and Non-Conformance:** Treat every compliance breach, near-miss, or incident as a critical learning opportunity. Conduct thorough root cause analyses to understand *why* something happened, not just *what* happened. Implement robust corrective and preventive actions (CAPAs) to prevent recurrence and improve the ESMS.
- **Seek External Expertise Strategically:** While SMEs handle much in-house, don't hesitate to engage external legal counsel for complex regulatory interpretations, experienced accountants for tax compliance, or specialized E&S consultants for specific impact assessments or ESMS audits. This can save significant time and costs in the long run.
- **Implement Robust Supply Chain Due Diligence:** For OGS products, ensuring compliance often extends to the supply chain. Develop a process for screening key suppliers and partners for their own E&S and labor performance. Include clear E&S clauses in supplier contracts.

Compliance Tracking Checklist Template

This checklist provides a quick, high-level overview of the company's compliance status across critical areas. It's ideal for regular (e.g., monthly or quarterly) management reviews.

Company Name: [Your Company Name] **Date of Review:** [DD/MM/YYYY] **Reviewed By:** [Name/Position]

Table 7: Example of a Compliance Tracking Checklist Template

Compliance Area	Status (Compliant/ Non-Compliant/ N/A)	Date Last Checked	Responsible Person/ Department	Comments/ Actions Required
BUSINESS & REGULATORY				
Company Registration & Annual Filings				
Business Operating Licenses				
Product Certifications (National Standards, IEC, VeraSol)				
Energy Sector-Specific Permits				
TAXATION				
Corporate Income Tax Filings & Payments				
VAT/GST Collection & Remittance				
PAYE Deductions & Remittance				
Withholding Tax (WHT) Remittance				
ENVIRONMENTAL				
Environmental Permits				
Waste Management & Disposal				
Chemical/Battery Storage & Handling				
SOCIAL & OHS				
Labour Law Adherence (Contracts, Wages, Hours)				
Social Security Contributions				
OHS Registrations/Certifications				
OHS Training & Inductions				
Incident Reporting & Investigation				
Internal Grievance Mechanism Functionality				
Community Engagement Records				
External Grievance Mechanism Functionality				
ETHICAL & OTHER				
Anti-Bribery & Corruption Policy Training				
Data Privacy Compliance				
Contractual E&S Clauses Adherence				

Legal and Regulatory Register Template

This register provides a detailed inventory of all applicable legal and regulatory requirements, forming the core of your compliance framework. This should be a living document, reviewed periodically (e.g., annually or bi-annually) and updated as laws change.

Company Name: [Your Company Name] **Version:** [e.g., 1.0] **Date Last Updated:** [DD/MM/YYYY]

Table 8: Legal and Regulatory Register Template

No.	Legislation/ Regulation Name (Citation)	Issuing Authority	Relevant Section/ Article	Applicability to OGS Operations	Specific Requirement/ Obligation (Summary)	Compliance Status (Yes/ No/ Partial/ N/A)	Evidence of Compliance (Document/ Record)	Responsible Dept./ Person	Last Review Date	Next Review Date	Comments/ Action Required

Permit and License Tracking Template

This matrix helps an OGS SME keep track of all necessary permits and licenses, ensuring timely renewals and avoiding operational disruptions.

Company Name: [Your Company Name] **Date of Review:** [DD/MM/YYYY] **Reviewed By:** [Name/Position]

Table 9: Permit and License Tracking Template

No.	Permit/License Name	Issuing Authority	Date Issued	Expiry Date	Renewal Frequency	Renewal Due Date	Application (Applied/ Approved)	Status Pending/ Approved	Responsible Person/ Department	Cost (Annual Fee)	Notes

6. Organisational Capacity and Competency

It is important to define the roles and responsibilities for implementation the ESMS and ensure company staff have the required competencies for carrying out their assigned tasks and activities

- **Organizational capacity**, in the context of the ESMS, refers to the resources (human, financial, and technical) available within a company to implement and maintain an ESMS effectively.
- **Competency** encompasses the knowledge, skills, and experience of personnel (e.g. management and staff) needed to fulfill their roles in the ESMS successfully.

A well-defined organizational structure with specified roles and responsibilities ensures everyone knows their part in managing environmental and social impacts. Without this clarity, tasks might be overlooked or neglected, leading to inefficiencies or non-compliance. Clear roles and responsibilities foster a sense of ownership. Employees know what's expected of them, making them more likely to take initiative and be accountable for their actions. Organizational diagrams (organograms) are a helpful tool for communicating organizational hierarchy, reporting lines, and areas of responsibility.

It is important to identify and note what skills and knowledge is required to successfully implement the various tasks required by the ESMS and the management plans therein. If employees lack the skills or knowledge to execute their tasks (e.g., conducting audits, assessing risks), the ESMS's effectiveness is compromised. Ensuring competency through training and development is essential as a competent team is better equipped to identify areas for improvement, propose innovative solutions, and adapt to changing environmental and social regulations.

A procedure should be established to verify that third-party employers, including any subcontractors, are reputable businesses with suitable Environmental and Social Management Systems (ESMS). This should include procedures to manage and monitor the performance of these third-party employers and their compliance with relevant environmental and social standards.

It is helpful to develop internal procedures documents for ESMS implementation including a responsibility matrix, standard procedures (aligned with management plans and processes), competency requirements for specific roles and tasks, and training plans, if necessary.

Responsibility Matrix

A responsibility matrix, also known as a RACI matrix (Responsible, Accountable, Consulted, Informed), is an excellent tool for defining and communicating roles and responsibilities within an ESMS. The key roles specified in the matrix include:

- **Responsible:** The person(s) who do the work to complete the task.
- **Accountable:** The person who has the ultimate ownership of the task and its outcome.
- **Consulted:** The person(s) whose input is sought before a decision or action is taken.
- **Informed:** The person(s) who are kept up-to-date on progress but don't have decision-making authority.

The responsibility matrix typically lists ESMS activities along one axis and personnel or roles along the other. By assigning RACI designations to each intersection, it becomes clear who's responsible for what. The matrix provides a quick, easy-to-understand overview of who's involved in each ESMS activity and prevents misunderstandings about roles and responsibilities, reducing the likelihood of duplicated effort or tasks being overlooked. The matrix serves as a reference point for discussions, ensuring everyone has the same vision and understanding of the specific tasks and activities required by the ESMS and the role each team member plays towards successful implementation. It offers a convenient tool for onboarding of new employees, who can quickly grasp their role within the ESMS by referring to the matrix.

Steps to developing a responsibility matrix include the following:

- 1) **Identify Key ESMS Activities:** List all the essential activities involved in implementing and maintaining your ESMS.
- 2) **List Roles and Responsibilities:** Identify all the individuals or groups involved in your ESMS.
- 3) **Create the Matrix:** Draw a table with the ESMS activities listed along one axis and the roles and responsibilities listed along the other. In each cell, assign a RACI designation (R (Responsible), A (Accountable), C (Consulted), I (Informed). Additional role designations where relevant such as S (Support)
- 4) **Fill in the Matrix:** For each activity, determine who plays each RACI role. It's helpful to involve relevant personnel in this process to ensure accuracy and buy-in.
- 5) **Review and Refine:** Once the matrix is filled in, review it carefully for completeness and accuracy. Look for gaps

or overlaps in responsibilities. Ensure that each activity has at least one person accountable (A).

- 6) Communicate and Implement: Share the matrix with all relevant personnel. Explain the purpose of the matrix and how it will be used. Encourage feedback and make any necessary adjustments.

The responsibility matrix should be tailored to your organization's specific structure and the components of your ESMS. Similarly to the overall ESMS, the responsibility matrix should be maintained as a living document that is reviewed and updated regularly to reflect changes in personnel, processes, or regulations. Ensure the matrix is easily accessible and understood by all relevant personnel.

Table 10: Sample responsibility matrix. The tasks and positions listed in the matrix are simply examples and should be replaced with tasks applicable to the company's own ESMS tasks and staffing positions of relevance to the ESMS.

ESMS Task / Activity	ESHS Manager	E&S Team Member	Department Manager	External Consultant
<i>Task / Activity</i>	<i>(specify role)</i>	<i>(specify role)</i>	<i>(specify role)</i>	<i>(specify role)</i>
Policy Development and Review				
Risk Assessment and Management				
Legal and Regulatory Compliance				
Occupational Health and Safety				
Waste Management				
Emergency Planning and Response (Incident Management)				
Grievance Mechanism (grievance management)				
Communications and Stakeholder Engagement				
Human Resources Training and Awareness Building				
Monitoring and Evaluation				
Audit and Inspection				

Standard Procedures

By establishing HR and ESHS procedures and protocols, OGS companies can create a safe, ethical, compliant, and sustainable work environment, fostering employee well-being, environmental responsibility, and long-term business success. The existence of standard procedures allows for communication and development of a shared understanding amongst company staff of how specific tasks and activities are to be implemented by company staff.

Developing a standardized procedure can be a simple yet impactful way to improve efficiency, ensure consistency, and mitigate risks. Procedures turn tasks into routine actions and can be communicated through various formats like text, flowcharts, or even pictures to best suit your employees.

Here's a suggested step-by-step guide for developing procedures:

- 1) **Identify the Need:**
 - **Determine the Process or Task:** Pinpoint the specific process or task that needs standardization. This could be anything from customer onboarding to inventory management or safety procedures.
 - **Define the Goal:** Clearly articulate the desired outcome of the procedure. What do you want to achieve through standardization? Is it to improve efficiency, reduce errors, ensure compliance, or enhance customer satisfaction?
- 2) **Gather Information:**
 - **Research and Best Practices:** Gather information from various sources, including industry guidelines, best practices, existing procedures, and relevant regulations. Consult with experts or experienced employees for their input.
 - **Document Current Practices:** If the process already exists, document the current steps and identify any pain points, bottlenecks, or areas for improvement.
- 3) **Draft the Procedure:**
 - **Format:** Choose a format that is easy to read and follow, such as a step-by-step guide, flowchart, or checklist.
 - **Clear and Concise Language:** Use simple, direct language that avoids jargon or technical terms.
 - **Include All Necessary Details:** Ensure the procedure includes all the steps required to complete the task successfully, including decision points, potential risks, and troubleshooting tips.
- 4) **Review and Refine:**

- **Peer Review:** Have colleagues or subject matter experts review the draft procedure for accuracy, completeness, and clarity.
 - **User Testing:** If possible, have employees test the procedure in a real-world setting to identify any practical issues or potential for improvement.
 - **Incorporate Feedback:** Revise the procedure based on feedback and testing to ensure it is user-friendly and effective.
- 5) Implement and Communicate:**
- **Training:** Provide training to all relevant employees on the new procedure, ensuring they understand the steps, rationale, and importance of following it.
 - **Accessibility:** Make the procedure easily accessible to employees, either through a central repository, online portal, or physical copies.
 - **Communication:** Communicate the implementation of the new procedure to all relevant stakeholders and explain its benefits.
- 6) Monitor and Review:**
- **Regular Monitoring:** Monitor the implementation of the procedure to ensure it is being followed correctly and achieving the desired outcomes.
 - **Feedback:** Gather feedback from employees on the effectiveness of the procedure and any challenges they encounter.
 - **Continuous Improvement:** Review the procedure periodically and update it as needed to reflect changes in processes, technologies, or regulations.

Below are a list of procedures and protocols that may be developed and implemented, if applicable to the business. The list includes typical standard procedures and protocols included in ESMS management plans. The specific procedures and protocols to be developed and included in your ESMS will vary depending on nature, extent and context of your business operations and activities, potential environmental and social risks identified during your risk assessment, and the applicable laws and regulations of the location where you are operating. Begin with developing procedures and protocols for high-impact processes that will yield the most significant benefits, as well as those required for regulatory compliance.

Examples of typical ESMS related procedures and protocols:

- **Sustainable Sourcing:** Protocols for sourcing materials and components from suppliers who adhere to environmental and social standards as well as quality assurance standards.
- **Waste Management:** Procedures for waste segregation, collection, storage, transportation, treatment, and disposal, considering hazardous and non-hazardous waste streams.
- **Water Management:** Protocols for water usage, wastewater treatment and discharge, monitoring water quality, and ensuring compliance with local regulations.
- **Carbon Footprint:** Procedures for assessing and minimizing the carbon footprint of the business.
- **Energy Management:** Strategies for energy conservation, use of renewable energy sources, and monitoring energy consumption.
- **Chemical Management:** Procedures for the safe handling, storage, and disposal of chemicals, including spill response plans.
- **Noise Management:** Measures to mitigate noise pollution, such as using quieter equipment, installing noise barriers, and monitoring noise levels.
- **Community Engagement:** Protocols for engaging with local communities, including consultation, information sharing, grievance mechanisms, and conflict resolution processes.
- **Product Recall:** Procedure for product recall and repair/replacement in case of safety defects or malfunctions.
- **Labor and Working Conditions:** Procedures for ensuring fair labor practices, safe working conditions, and compliance with labor laws and regulations.
- **Staff Onboarding:** Develop a structured onboarding procedure to familiarize new employees with company culture, values, policies, and procedures.
- **Human Rights:** Policies and procedures for respecting and protecting human rights throughout operations, including non-discrimination, freedom of association, and prevention of forced labor or child labor.
- **Gender Equality and Social Inclusion:** Strategies for promoting gender equality and social inclusion in project activities, ensuring equal opportunities and participation for all.
- **Hazard Identification and Risk Assessment (HIRA):** Procedures for identifying and assessing workplace hazards, evaluating risks, and implementing control measures. Develop checklists, templates, and guidelines for conducting HIRAs, considering specific tasks, work environments, and potential hazards (electrical, falls, heat stress, etc.).
- **Personal Protective Equipment (PPE):** Procedure defining required PPE for various tasks and ensure its availability, proper fit, and regular inspection. Create guidelines for selecting, using, maintaining, and replacing PPE, including specific requirements for different hazards (e.g., hard hats, safety glasses, gloves, safety footwear).

- **Working at Heights:** Procedure for working at heights, including the use of fall protection systems (harnesses, lanyards, lifelines), ladders, and scaffolding. Develop detailed guidelines for safe ladder and scaffold use, inspection, and maintenance. Train employees on fall prevention techniques and emergency procedures.
- **Electrical Safety:** Procedures for de-energizing electrical systems before work begins. Ensure proper grounding and use of insulated tools. Develop step-by-step lockout/tagout (LOTO) procedures, electrical safety checklists, and guidelines for handling electrical components and wiring.
- **Manual Handling:** Procedures for proper lifting techniques and the use of mechanical aids (dollies, hoists) to reduce the risk of musculoskeletal injuries. Develop guidelines for safe manual handling practices, weight limits, and team lifting.
- **Heat Stress Management:** Protocols for working in hot weather, including hydration, rest breaks, and protective clothing.
- **Vehicle and Transportation Safety:** Procedures for company vehicles to be well-maintained and equipped with safety features and drivers with proper training and safe driving protocols. Develop driver safety checklists, vehicle inspection procedures, and load securement guidelines.
- **Remote Work Safety:** Procedures to address specific risks associated with working in remote areas, such as limited access to medical care and communication challenges.
- **Security:** Establish security procedures to protect staff from security risks and safeguard equipment and materials from theft or vandalism.
- **Safety Training:** Procedures for training employees on safe work practices, hazard awareness, emergency procedures, and the proper use of personal protective equipment (PPE). Develop training modules, materials, and assessment tools for different roles (sales, installers, technicians) and conduct regular refresher training.
- **Emergency Preparedness and Response:** Protocols for responding to workplace accidents, injuries, or emergencies, including first aid. Develop incident report forms, investigation procedures, and guidelines for documenting findings and recommendations.
- **Monitoring and Evaluation:** Procedures for monitoring and evaluation of the environmental and social aspects of company operations.

Competency Requirements

Developing and specifying competency requirements for company roles is instrumental to organizational success.

Competency requirements allow for:

- **Targeted Recruitment and Selection:** Clear competency requirements help attract and select candidates who possess the necessary skills, knowledge, and attributes to excel in specific roles. This reduces the risk of mismatches and ensures a better fit between individuals and positions.
- **Effective Performance Management:** Defined competencies provide a framework for evaluating employee performance against objective standards. This enables fair and consistent performance reviews, identifies areas for improvement, and supports informed decisions regarding promotions, training, and development.
- **Targeted Training and Development:** By understanding the competencies required for each role, organizations can design tailored training programs to address skill gaps and enhance employee capabilities. This maximizes the return on investment in training and accelerates employee development.
- **Improved Employee Engagement and Retention:** Employees who understand the expectations for their roles and have opportunities to develop their skills are more likely to be engaged and motivated. This can lead to increased job satisfaction, improved productivity, and reduced turnover.
- **Enhanced Organizational Effectiveness:** When employees possess the required competencies, they can perform their tasks more effectively, contribute to team goals, and drive overall organizational success. This leads to improved efficiency, innovation, and customer satisfaction.
- **Risk Mitigation:** Clear competency requirements help mitigate risks associated with incompetent employees. This can include errors, accidents, customer dissatisfaction, and financial losses.
- **Succession Planning:** Identifying and developing competencies enables effective succession planning. Organizations can groom internal talent to fill critical roles, ensuring continuity and minimizing disruptions.
- **Regulatory Compliance:** For certain roles, competency requirements may be mandated by law or regulations. Adhering to these requirements is essential for maintaining compliance and avoiding legal penalties.

Steps for developing competency requirements for a company role:

- 1) **Job Analysis:** Conduct a thorough analysis of each role to identify the specific tasks, responsibilities, and knowledge required.
- 2) **Competency Identification:** Determine the core competencies needed for success in each role, considering both technical skills and behavioral attributes.
- 3) **Competency Framework:** Develop a competency framework that defines and describes each competency, outlining the expected levels of proficiency.
- 4) **Validation:** Validate the competency requirements through input from subject matter experts, managers, and

employees.

- 5) **Communication:** Clearly communicate the competency requirements to employees, ensuring they understand the expectations for their roles and how their performance will be assessed.

Box 3: Sample Competency Requirements for an ESHS Manager Role

Title: ESHS (Environmental, Social, Health, and Safety) Manager

Department: Operations/Sustainability

Reports To: Chief Operating Officer (COO)

Summary: The ESHS Manager is responsible for developing, implementing, and maintaining a comprehensive Environmental and Social, Management System (ESMS) program for the company's operations. This includes ensuring compliance with local and international regulations, mitigating risks, promoting a safe and healthy work environment, and integrating sustainability principles into all aspects of the business.

Key Competency Requirements:

Technical Skills:

- **ESMS Expertise:** In-depth knowledge of ESMS principles, IFC ESMS guidelines, standards (ISO 14001, ISO 45001), and best practices, particularly in the context of off-grid solar products and services.
- **Risk Assessment and Management:** Proven ability to identify and assess environmental, social, and health & safety risks, develop mitigation strategies, and implement effective control measures.
- **Environmental and Social Risks and Impacts Assessment:** Experience in conducting environmental and social risk assessments and developing environmental and social management plans (ESMPs) for business operations and projects.
- **Regulatory Compliance:** Familiarity and understanding of relevant national environmental, social, and labor laws and regulations, as well as international standards.
- **Data Analysis and Reporting:** Proficiency in collecting, analyzing, and reporting environmental and social performance data, including key performance indicators (KPIs).

Interpersonal and Leadership Skills:

- **Communication:** Excellent written and verbal communication skills to effectively engage with diverse stakeholders, including employees, management, investors, regulators, customers and community members.
- **Leadership:** Strong leadership abilities to motivate and inspire teams, foster a culture of safety and sustainability, and drive continuous improvement.
- **Collaboration:** Ability to build effective working relationships with internal and external stakeholders, including government agencies, community leaders, and NGOs.
- **Problem-Solving:** Strong analytical and problem-solving skills to identify and address complex environmental and social challenges.
- **Training and Facilitation:** Ability to develop and deliver training programs on ESMS topics to employees at all levels.

Additional Desirable Competencies:

- **Local Experience:** Familiarity with the local cultural, social, and political context is a significant asset.
- **Language Skills:** Fluency in English [and/or French] and relevant local languages is highly desirable.
- **Project Management:** Experience in managing environmental and social projects from initiation to closure, including budget and timeline management.
- **Technical Knowledge:** Basic understanding of solar energy systems, products, installation, and maintenance is a plus.

Education and Experience:

- **Bachelor's degree** in environmental science, engineering, sustainability, or a related field.
- **5+ years of experience** in ESMS implementation and management, preferably in the renewable energy sector.
- **Professional certifications** (e.g., Certified Environmental Professional, Certified OHS Professional) are a plus.

Key Performance Indicators:

- Effective implementation and maintenance of the ESMS program.

- Compliance with relevant regulations and standards.
- Reduction in environmental and social impacts.
- Improved health and safety performance.
- Positive engagement with stakeholders.

Training Plans

Developing and maintaining environmental, social, health and safety (ESHS) training plans for staff will facilitate implementation of the company's Environmental and Social Management System (ESMS) and will lead to the following benefits:

- 1) **Skill Enhancement:** Training ensures staff skills and knowledge are up-to-date with respect to the latest E&S principles, standards, risk assessment methodologies, and stakeholder engagement techniques.
- 2) **Safety Assurance:** Comprehensive training minimizes accidents, environmental incidents, and social conflicts, reducing liability for the company.
- 3) **Stakeholder Satisfaction:** Well-trained personnel can effectively communicate the company's commitment to environmental and social responsibility and build positive relationships with local communities.
- 4) **Increased Effectiveness:** Training equips employees with the knowledge and tools to identify, assess, and manage environmental and social impacts effectively, leading to improved E&S performance and compliance.
- 5) **Employee Empowerment:** Investing in E&S training demonstrates the company's commitment to sustainability and responsible business practices. This can empower employees to contribute to the company's E&S goals and foster a sense of ownership and accountability.
- 6) **Regulatory Compliance:** Training ensures compliance with relevant environmental and social regulations, minimizing legal risks and penalties for the company.
- 7) **Risk Mitigation:** A well-trained workforce can proactively identify and address potential environmental and social risks, reducing the likelihood of negative impacts and associated costs.
- 8) **Reputation Enhancement:** A company known for its commitment to environmental and social responsibility, demonstrated through well-trained staff, builds a strong reputation and brand image, attracting customers, investors, and partners.
- 9) **Continuous Improvement:** E&S training fosters a culture of continuous learning and improvement, encouraging employees to identify opportunities for enhancing environmental and social performance.
- 10) **Long-Term Sustainability:** E&S training contributes to the long-term sustainability of the company's operations and its positive impact on the environment and society.

Process for Planning and Implementing E&S Training (a consultant may be procured to lead this process):

- 1) **Needs Assessment:**
 - a. **Identify Knowledge Gaps:** Conduct surveys, interviews, and reviews of ESMS documentation to identify areas where employees lack understanding of E&S principles and ESMS procedures.
 - b. **Prioritize Training Needs:** Focus on training that will address the most critical E&S risks and compliance requirements and enable development and implementation of missing ESMS elements.
- 2) **Set Training Objectives:**
 - a. **Clear and Measurable:** Define specific, measurable, achievable, relevant, and time-bound (SMART) objectives for each E&S training program.
 - b. **Aligned with E&S Goals:** Ensure training objectives align with the company's overall E&S goals and objectives.
- 3) **Design Training Content:**
 - a. **Relevant and Engaging:** Develop training materials that cover essential E&S topics, such as risk assessment, stakeholder engagement, environmental monitoring, and incident reporting. Use a variety of formats (e.g., presentations, case studies, group discussions, site visits) to make the training engaging and interactive.
 - b. **Tailored to Audience:** Consider the different roles and responsibilities within the company when designing training content.
- 4) **Delivery Methods:**
 - a. **Internal Training:** Conduct training sessions led by internal experts or E&S managers.
 - b. **External Training:** Utilize external consultants or training providers for specialized E&S topics.
 - c. **Online Training:** Leverage online courses or webinars for flexible and accessible learning.
 - d. **Blended Learning:** Combine different delivery methods for a comprehensive and effective training experience.
- 5) **Assessment and Evaluation:**
 - a. **Knowledge Checks:** Incorporate quizzes, tests, or practical exercises to assess knowledge acquisition.
 - b. **Performance Evaluation:** Observe employees' application of E&S knowledge and skills in their daily work to gauge the effectiveness of the training.
 - c. **Feedback and Improvement:** Gather feedback from trainees and instructors to continuously improve

future training programs.

6) **Continuous Improvement:**

- a. **Regular Review:** Regularly review and update E&S training plans to reflect changes in regulations, industry standards, and company operations.
- b. **Ongoing Learning:** Encourage employees to participate in ongoing E&S training and professional development opportunities.

7. Environmental and Social Management Plans

Environmental and Social (E&S) Management Plans are a requirement as per **IFC PS 1 / WB ESS: Assessment and Management of Environmental and Social Risks and Impacts**. Environmental and Social Management Plans (ESMPs), which form a key part of an ESMS, outline measures to prevent, minimize, or address negative environmental, occupational, and social impacts of a project or company operations. ESMPs also detail how these measures will be implemented, monitored, and who's responsible (roles and responsibilities should be established for the ESMS as a whole, but also for each management plan). ESMPs typical include the following:

- **Impact Analysis:** Identification of types of potential impacts and ways to mitigate those impacts.
- **Mitigation Actions:** Specific steps to reduce or offset negative impacts.
- **Environmental and Social Standards:** Guidelines for products and operations, designed to ensure sustainable and responsible practices.
- **Monitoring Plan:** Outlines how environmental and social performance will be tracked and data to be collected.
- **Roles and Responsibilities:** Specification of who is responsible for tasks and activities in the plan
- **Communication Strategy:** How the company team will share information with affected suppliers, customers and communities and handle grievances.
- **Schedule and Costs:** Implementation schedule and estimated budget for implementing the ESMP.
- **Project/Program Integration:** Describes how the ESMP fits within the overall project or program structure and any relevant contract or operational requirements.
- **Community Engagement:** A plan for involving stakeholders throughout the project or program's entire lifecycle, addressing both social concerns and potential economic opportunities.

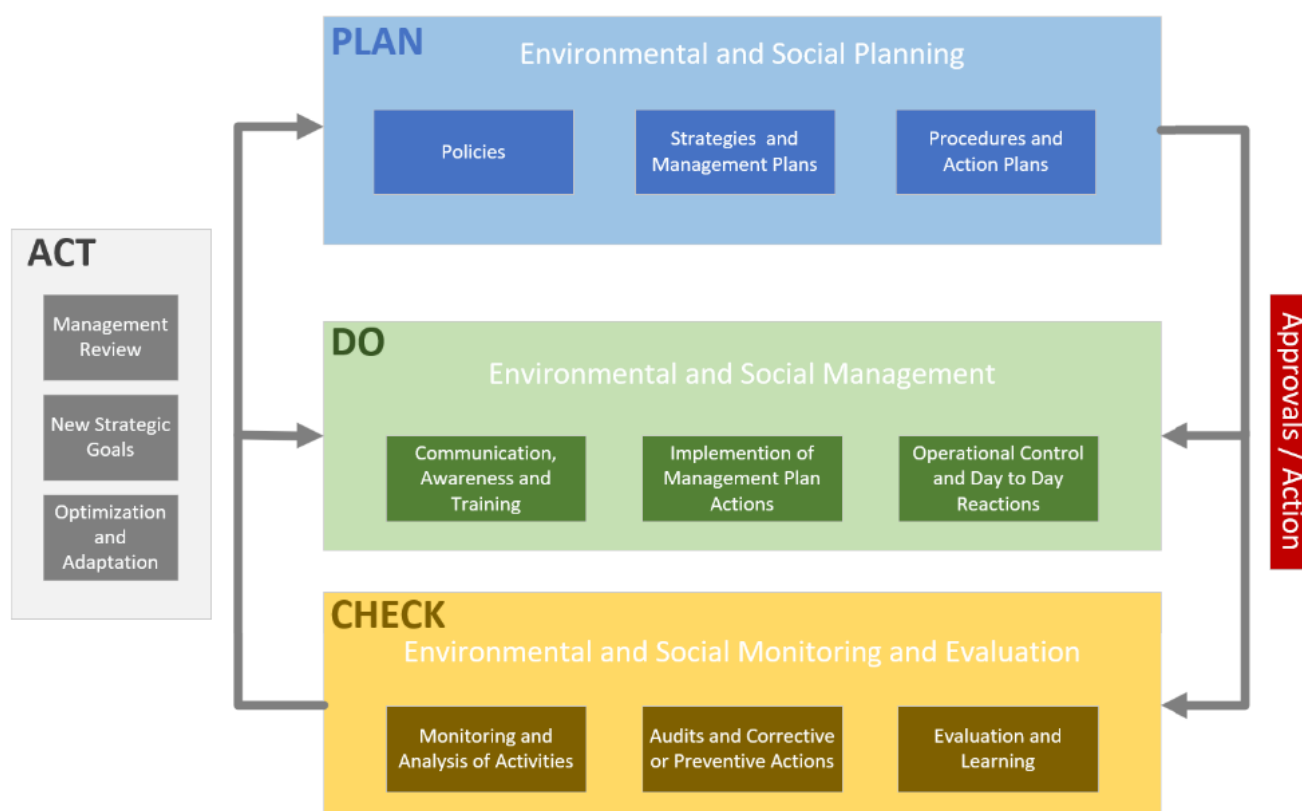


Figure 13: Environmental and social planning, management and review in the context of Plan-Do-Check-Act framework

As specified earlier in this document, an ESMS usually includes the following management plans:

- 1) **Occupational Health and Safety (OHS) Plan**
- 2) **Human Resource (HR) Plans**
- 3) **Waste Management Plans (WMP)**
- 4) **Emergency Preparedness and Response Plan (EPRP)**
- 5) **Stakeholder Engagement Plan (SEP)**
- 6) **Grievance Mechanism (GM)**
- 7) **Environmental and Social Action Plan (ESAP)**

ESMPs may include or be complemented by applicable standard operating procedures such as code of conduct, working at heights, working with electrical equipment, or storage of combustible products as required. The following sections

provide further details about each of these ESMPs.

8. Health and Safety Plan

This ESMS component is recommended as per **IFC PS 2 / WB ESS 2 Labor and Working Conditions**. An Occupational Health and Safety (OHS) plan is a crucial undertaking for companies involved with off-grid solar products and related services. Developing and maintaining an OHS plan is not merely a legal obligation; it's a strategic investment that provides value to an organization through providing the following benefits:

- **Protecting Employees:** Safeguarding employees from workplace hazards, preventing injuries, illnesses, and fatalities by identifying and assessing risks, implementing control measures, and providing training and resources.
- **Legal Compliance:** Ensuring the company complies with relevant occupational health and safety laws and regulations, avoiding legal penalties and reputational damage.
- **Creating a Safe Work Environment:** Fostering a culture of safety and implementing appropriate measures towards creating a safer workplace for everyone.
- **Reducing Costs:** Preventing accidents and injuries can significantly reduce costs associated with medical treatment, compensation, legal fees, lost productivity, and insurance premiums.
- **Improving Productivity and Morale:** Creating a safe workplace leading to higher employee morale, engagement, and productivity. When employees feel safe and valued, they are more likely to be committed to their work.
- **Improving Stakeholder Relations:** Demonstrating a commitment to OHS can strengthen relationships with stakeholders, including employees, investors, customers, and regulatory bodies.
- **Enhancing Reputation:** Enhancing a company's reputation as a responsible employer, attracting and retaining talent and improving customer trust.

An OHS Plan should address the following health and safety considerations: personal protective equipment, handling of hazardous materials and/or electric equipment, working from heights, and OHS related training. The plan's roles and responsibilities should be specified and should integrate the services of security personnel and third-party service providers. The OHS Plan should be harmonized with the company's emergency preparedness and response plan.

Key Elements of an Occupational Health and Safety (OHS) Plan

It is recommended to form a Safety Committee, consisting of management representatives, employees, and safety professionals, responsible for developing, implementing, and reviewing the company's OHS plan. The OHS plan is compiled into a document which includes the following elements:

- 1) **Company Commitment:** A clear statement from management highlighting the company's commitment to OHS, outlining goals, and emphasizing the importance of employee well-being.
- 2) **Hazard Identification and Risk Assessment:** A thorough assessment of potential hazards associated with solar product manufacturing, installation, maintenance, and transportation. Use checklists, inspections, and employee feedback to gather information. The OHS risks associated with off-grid solar products and services may include:
 - **Working at Heights:** Falls from ladders, roofs, or elevated platforms during installation or maintenance activities.
 - **Electrical Hazards:** Risk of electric shock, burns, or electrocution when working with solar panels, batteries, inverters, and wiring.
 - **Chemical Exposure / Battery Hazards:** Risk of acid burns, fire, or explosions from damaged or improperly handled batteries.
 - **Weather-Related Hazards:** Exposure to extreme heat, cold, or lightning strikes while working outdoors.
 - **Slips, Trips, and Falls:** Hazards related to uneven terrain, slippery surfaces, or cluttered work areas.
 - **Manual Handling Hazards:** Back injuries or strains from lifting, carrying, or moving heavy equipment or materials.
 - **Tool and Equipment Hazards:** Injuries caused by improper use of tools, equipment malfunctions, or lack of maintenance.
 - **Confined Space Hazards:** Risks associated with working in enclosed spaces, such as battery rooms or inverter compartments, including oxygen deficiency, toxic fumes, and entrapment.
 - **Vehicle Accidents:** Risk of collisions, rollovers, or other accidents while transporting solar equipment or materials.
 - **Loading and Unloading Hazards:** Injuries from lifting, carrying, or moving heavy equipment or materials during loading and unloading operations.

- **Cargo Securement:** Risks of falling objects or shifting loads during transportation.
- **Waste Disposal:** Environmental contamination and health hazards associated with improper disposal of hazardous waste materials, such as batteries or electronic components.
- **Remote Work Hazards:** Risks associated with working in remote locations, including limited access to medical care, communication difficulties, and exposure to wildlife.
- **Security Hazards:** Theft of equipment or materials from work sites or during transportation.
- **Gender based violence (GBV), Sexual Exploitation, Abuse, or Harassment (SEAH):** Risk of experiencing GBV/SEAH in the colleagues in the workplace from interactions with other stakeholders.
- **Environmental Hazards:** Risks associated with heat stress, working in remote areas, and exposure to wildlife.

3) **Risk Control Measures:** Specific protocols and procedures to mitigate identified risks. This includes using appropriate personal protective equipment (PPE), implementing safe work practices, providing training, conducting regular inspections, and establishing emergency response plans. Consider a hierarchy of controls, prioritizing elimination, substitution, engineering controls, administrative controls, and finally, PPE.

- **Working at Heights:** Fall Protection Systems such as guardrails, safety nets, or personal fall arrest systems (PFAS); Proper Ladder and Scaffold Use; Train employees on working at heights safely and ensure they are certified to use fall protection equipment.
- **Electrical Hazards:** Implement Lockout/Tagout (LOTO) procedures to de-energize electrical systems before working on them; Use insulated tools and wear appropriate PPE, including gloves, safety glasses, and arc flash protection; Train employees on electrical safety, including hazard recognition, safe work practices, and emergency procedures.
- **Chemical Exposure / Battery Hazards:** Use PPE when handling batteries, including gloves and eye protection. Store batteries in well-ventilated areas away from flammable materials.
- **Weather-Related Hazards:** Monitoring Weather Conditions before starting work and stop work if conditions become unsafe; Provide appropriate clothing for hot or cold weather, such as hats, sunscreen, and insulated gear; Encourage employees to stay hydrated and take breaks in shaded or heated areas; Seek shelter indoors during lightening/thunderstorms and avoid working near tall objects or open water;
- **Slips, Trips, and Falls:** Keep Work Areas Clean and Organized; Ensure employees wear non-slip footwear with good traction; Provide adequate lighting in work areas; Train employees on fall prevention techniques and hazard awareness.
- **Manual Handling Hazards:** Teach employees proper lifting techniques, emphasizing using legs, not back; Use lifting equipment, such as hoists, cranes, or dollies, for heavy objects; Encourage team lifting for heavy or awkward loads; Rotate tasks to prevent repetitive strain injuries.
- **Tool and Equipment Hazards:** Ensure employees are trained on the safe use and maintenance of tools and equipment; Conduct regular inspections of tools and equipment to identify defects or malfunctions; Maintain tools and equipment in good working order, following manufacturer recommendations; Provide and enforce the use of appropriate PPE, such as safety glasses, gloves, and steel-toed boots.
- **Confined Space Hazards:** Establish and follow confined space entry permits and procedures; Ensure adequate ventilation to prevent oxygen deficiency and the buildup of toxic fumes; Use gas detectors to monitor for hazardous atmospheres; Have rescue procedures in place and ensure employees are trained in their use.
- **Vehicle / Transportation Hazards:** Maintain vehicles in good working order, including regular inspections and servicing; Provide drivers with comprehensive training on safe driving practices and load securement; Use appropriate restraints, such as straps, chains, or binders, to secure loads properly; Plan routes to avoid hazardous road conditions or areas with high traffic.
- **Waste Disposal:** Follow local regulations for the disposal of hazardous waste materials.
- **Remote Work Hazards:** Provide employees working in remote locations with communication devices, first aid kits, and emergency response plans.
- **Security Hazards:** Implement security measures, such as fencing, alarms, and surveillance cameras, to protect worksites and equipment.
- **Gender-Based Violence (GBV), Sexual Exploitation, Abuse, or Harassment (SEAH):** Establish a clear, comprehensive, and zero-tolerance policy against GBV/SEAH. Communicate the policy to all employees and stakeholders; Conduct regular training sessions on GBV/SEAH prevention, raising awareness of the issue and providing employees with the tools to identify and report incidents; Establish confidential and easily accessible reporting mechanisms for employees to report incidents of GBV/SEAH without fear of retaliation; Ensure prompt and thorough investigation of all reported incidents of GBV/SEAH, and take appropriate disciplinary action against perpetrators; Provide access to counseling and support services for victims of GBV/SEAH; Engage with local communities and stakeholders to raise awareness of GBV/SEAH and promote a culture of respect and safety.
- **Environmental Hazards:**

- **Heat Stress:** Implement work-rest cycles to allow employees to cool down and rehydrate; Provide access to cool water and encourage employees to drink fluids regularly; Provide shaded areas or cooling stations for employees to rest and cool down; Gradually acclimatize employees to hot weather conditions.
 - **Remote Areas:** Ensure reliable communication systems are in place for emergencies; Develop and communicate emergency response plans for medical emergencies, natural disasters, or other unexpected events; Train employees in first aid and CPR.
 - **Wildlife Exposure:** Educate employees on local wildlife hazards and how to avoid encounters; Provide protective gear, such as snake-proof boots or insect repellent, as needed; Establish procedures for handling wildlife encounters, including bites or stings.
- 4) **Training and Education:** Comprehensive training programs for employees covering safe work practices, hazard awareness, emergency procedures, and the proper use of PPE.
 - 5) **Incident Reporting and Investigation:** A system for reporting and investigating workplace incidents, near misses, and injuries. This helps identify root causes and implement corrective actions.
 - 6) **Emergency Response:** Detailed plans for responding to various emergencies, such as electrical shocks, fires, chemical spills, or injuries. This includes evacuation procedures, first aid, and communication protocols.
 - 7) **Regulatory Compliance:** Ensuring compliance with relevant occupational health and safety laws and regulations in the company's operating regions.
 - 8) **Continuous Improvement:** Regular reviews of the OHS plan to assess its effectiveness, identify areas for improvement, and incorporate lessons learned from incidents and near misses. Conduct inspections, audits, and reviews to ensure compliance and identify areas for improvement.
 - 9) **Roles and Responsibilities:** The OHS plan should include a clear specification of the roles and responsibilities of company staff with respect to implementation of the OHS tasks and activities.

Successful OHS plan implementation will require effective communication about the OHS plan towards ensuring all employees are aware of the OHS plan, their responsibilities, and the specific safety protocols that apply to their roles. In case of any safety incidents, the company's staff responsible for monitoring OHS should make use of an incident/accident report form and keep a record of all reported incidents to enable evaluation and continual improvement in OHS practices and processes.

Table 11: Sample Incident / Accident Report Form

Employee Incident Report Form			
Employee Involved		Manager / Supervisor	
Name:		Name:	
Title/Position:		Title Position:	
ID Number:			
Company:			
Insurance Company (if relevant):			
Incident Details			
Date:		Time:	
Location:		Weather Condition:	
Description of Incident			
Employee Explanation			
Witnesses			
Treatment / Corrective Action Taken			
Actions To Be Taken			
Details of Person Filling the Incident Form			
Name:		Signature:	
Title/Position:			
ID Number:			
Company:		Date:	

Table 12: Sample OHS Plan Table of Risks and Management Measures

Hazard Category	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
Working at Heights	Falls from ladders, roofs, or elevated platforms during installation or maintenance activities.	- Implement fall protection systems (guardrails, safety nets, personal fall arrest systems). - Ensure proper ladder and scaffold use. - Train employees on working at heights safely and certify them in fall protection equipment usage.	- Immediately assess the injured person's condition and call for medical assistance if needed. - Secure the area to prevent further accidents. - Report the incident to the OHS officer/manager and relevant authorities. - Review and update safety procedures as needed.	OHS Officer/Manager: Develop and implement fall protection protocols, train employees, and conduct inspections. Employees: Use fall protection equipment correctly and report any hazards. Site Supervisor: Ensure safe work practices and enforce safety rules.
Electrical Hazards	Risk of electric shock, burns, or electrocution when working with solar panels, batteries, inverters, and wiring.	- Implement Lockout/Tagout (LOTO) procedures to de-energize electrical systems before work. - Use insulated tools and wear appropriate PPE (gloves, safety glasses, arc flash protection). - Train employees on electrical safety, hazard recognition, safe practices, and emergency procedures.	- Immediately disconnect the power source and ensure the injured person is not in contact with electrical current. - Provide first aid or CPR if needed and call for emergency medical services. - Investigate the incident and implement corrective actions.	OHS Officer/Manager: Develop and enforce LOTO procedures, provide electrical safety training, and conduct regular inspections. Electricians: Follow safe work practices and use appropriate PPE. Employees: Report any electrical hazards and do not attempt repairs without proper authorization and training.
Chemical Exposure/Battery Hazards	Risk of acid burns, fire, or explosions from damaged or improperly handled batteries.	- Use PPE when handling batteries (gloves, eye protection). - Store batteries in well-ventilated areas away from flammable materials. - Train employees on safe battery handling, storage, and disposal procedures. - Have spill kits and fire extinguishers readily available.	- In case of spills, contain and neutralize the spill using appropriate materials. - If a fire occurs, evacuate the area and use fire extinguishers or call the fire department. - Seek medical attention for any chemical burns or injuries. - Report the incident and implement corrective actions.	OHS Officer/Manager: Develop and enforce safe battery handling procedures, provide training, and ensure proper storage facilities. Employees: Follow safety procedures, use PPE, and report any leaks or damage. Emergency Response Team: Respond to spills and fires, provide first aid, and evacuate the area if necessary.
Weather-Related Hazards	Exposure to extreme heat, cold, or lightning strikes while working outdoors.	- Monitor weather conditions before and during work, stop work if unsafe. - Provide appropriate clothing for extreme weather (hats, sunscreen, insulated gear). - Encourage hydration and breaks in shaded/heated areas. - Seek indoor shelter during lightning/thunderstorms. - Avoid working near tall objects or open water during thunderstorms.	- In case of heat stroke or hypothermia, move the person to a cool/warm area, provide first aid, and call for medical help. - During lightning storms, seek shelter indoors immediately and avoid using electrical equipment. - Monitor weather forecasts and plan work accordingly to minimize exposure to extreme weather.	OHS Officer/Manager: Monitor weather conditions and establish work guidelines. Supervisors: Ensure employees follow weather-related safety protocols. Employees: Wear appropriate clothing and report any health issues related to weather exposure.

Hazard Category	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
Slips, Trips, and Falls	Hazards related to uneven terrain, slippery surfaces, or cluttered work areas.	- Keep work areas clean and organized. - Ensure employees wear non-slip footwear. - Provide adequate lighting in work areas. - Train employees on fall prevention and hazard awareness.	- If a fall occurs, assess the injured person's condition and provide first aid if needed. - Call for medical assistance if necessary. - Secure the area to prevent further accidents. - Investigate the incident and implement corrective actions.	OHS Officer/Manager: Conduct regular inspections of work areas and ensure proper housekeeping. Supervisors: Enforce clean work areas and ensure employees wear appropriate footwear. Employees: Report any slip, trip, or fall hazards and follow safety procedures.
Manual Handling Hazards	Back injuries or strains from lifting, carrying, or moving heavy equipment or materials.	- Teach proper lifting techniques (using legs, not back). - Use lifting equipment (hoists, cranes, dollies) for heavy objects. - Encourage team lifting for heavy or awkward loads. - Rotate tasks to prevent repetitive strain injuries.	- If an injury occurs, provide first aid and seek medical attention if necessary. - Report the incident and investigate the cause. - Review and adjust lifting procedures or equipment as needed.	OHS Officer/Manager: Provide training on safe lifting techniques and ensure proper lifting equipment is available. Supervisors: Monitor work practices and ensure safe lifting procedures are followed. Employees: Use proper lifting techniques and report any difficulties or injuries.
Tool and Equipment Hazards	Injuries caused by improper use of tools, equipment malfunctions, or lack of maintenance.	- Train employees on safe tool and equipment use and maintenance. - Conduct regular inspections and maintenance of tools and equipment. - Provide and enforce the use of appropriate PPE (safety glasses, gloves, steel-toed boots).	- If an injury occurs, provide first aid and seek medical attention if necessary. - Report the incident and investigate the cause. - Remove faulty equipment from service and repair or replace it. - Review and update training and safety procedures as needed.	OHS Officer/Manager: Ensure proper tool and equipment maintenance, provide training, and conduct inspections. Employees: Use tools and equipment correctly and report any defects or malfunctions.
Confined Space Hazards	Risks associated with working in enclosed spaces (battery rooms, inverter compartments) - oxygen deficiency, toxic fumes, entrapment.	- Establish and follow confined space entry permits and procedures. - Ensure adequate ventilation. - Use gas detectors to monitor the atmosphere. - Have rescue procedures in place and train employees in their use.	- In case of an emergency in a confined space, activate the rescue plan and call for help. - Do not enter the confined space without proper training and equipment. - Investigate the incident and implement corrective actions.	OHS Officer/Manager: Develop and enforce confined space entry procedures, provide training, and ensure rescue equipment availability. Authorized Entrants: Follow entry procedures and use appropriate equipment. Attendants: Monitor entrants and call for help if needed.
Vehicle Accidents	Risk of collisions, rollovers, or other accidents while transporting solar equipment or materials.	- Maintain vehicles in good working order (regular inspections and servicing). - Train drivers on safe driving practices and load securement. - Use appropriate restraints to secure loads. - Plan routes to avoid hazardous conditions or high-traffic areas.	- In case of an accident, assess the situation and provide first aid if necessary. - Call for emergency services (police, ambulance, fire department) as needed. - Secure the area and prevent	Fleet Manager/Supervisor: Oversee vehicle maintenance and driver training. Drivers: Follow safe driving practices and report any vehicle issues.

Hazard Category	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
			further accidents. - Report the incident to the company and relevant authorities.	Logistics Team: Plan safe transportation routes and ensure proper load
Loading and Unloading Hazards	Injuries from lifting, carrying, or moving heavy equipment/materials during loading/unloading.	- Use mechanical lifting aids (forklifts, cranes) for heavy items. - Train employees on safe lifting techniques and proper use of equipment. - Ensure adequate space and clear pathways for loading/unloading. - Provide adequate lighting in loading/unloading areas.	- If an injury occurs, provide first aid and seek medical attention if necessary. - Report the incident and investigate the cause. - Review and adjust lifting procedures or equipment as needed.	Logistics/Warehouse Manager: Ensure proper equipment and training for safe loading/unloading. Employees: Follow safe lifting practices and report hazards. Supervisors: Monitor loading/unloading operations and enforce safety protocols.
Cargo Securement	Risks of falling objects or shifting loads during transportation.	- Use appropriate restraints (straps, chains, binders) to secure loads. - Train drivers on proper load securement techniques. - Inspect and maintain restraints regularly. - Ensure proper weight distribution and stacking of cargo.	- In case of a shifting or falling load, stop the vehicle in a safe location. - Re-secure the load properly before continuing. - If objects have fallen and caused damage or injury, assess the situation and call for assistance if needed.	Logistics/Transportation Manager: Develop and enforce cargo securement protocols. Drivers: Follow load securement procedures and inspect restraints before each trip. Loaders: Ensure proper loading and securement of cargo.
Waste Disposal	Environmental contamination and health hazards from improper disposal of hazardous waste (batteries, electronic components).	- Follow local regulations for hazardous waste disposal. - Partner with certified e-waste recyclers for environmentally sound disposal. - Train employees on proper waste handling, labeling, and storage procedures. - Implement waste segregation and collection systems.	- In case of a spill or accidental release of hazardous waste, follow the emergency spill response plan. - Contain and clean up the spill using appropriate PPE and materials. - Report the incident to the OHS officer/manager and relevant authorities.	OHS Officer/Manager: Develop and implement waste disposal procedures, ensure compliance with regulations, and manage contracts with recyclers. Employees: Follow waste disposal procedures and report any spills or leaks. Waste Management Contractors: Ensure safe and compliant disposal of hazardous waste.
Remote Work Hazards	Risks associated with working in remote locations (limited medical care, communication difficulties, wildlife exposure).	- Provide employees with communication devices (satellite phones, radios) and first aid kits. - Develop and communicate emergency response plans for medical emergencies, natural disasters, or other unexpected events. - Train employees in first aid and CPR. - Educate employees on local wildlife hazards and avoidance. - Establish procedures for handling wildlife encounters.	- In case of a medical emergency, provide first aid and contact emergency services as soon as possible. - In case of wildlife encounters, follow established procedures and seek medical attention if needed. - Maintain communication with the main office or base camp and report any incidents or concerns.	OHS Officer/Manager: Develop emergency response plans and ensure employees have access to communication and medical resources. Employees: Follow safety protocols for remote work, carry communication devices and first aid kits, and report any incidents or concerns.
Security Hazards	Theft of equipment or materials from work sites or during transportation.	- Implement security measures at worksites (fencing, alarms, surveillance cameras, security	- In case of theft or vandalism, report the incident to the authorities	Security Manager/Officer: Implement and oversee security measures, conduct risk

Hazard Category	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
		personnel). - Track equipment and materials during transportation using GPS or other tracking systems. - Train employees on security protocols and reporting procedures. - Collaborate with local authorities and security providers to address security concerns.	and the company's security personnel. - Secure the area and preserve any evidence. - Cooperate with investigations and provide any relevant information.	assessments, and collaborate with authorities. Employees: Follow security protocols, report suspicious activities, and safeguard company assets. Logistics/Transportation Manager: Ensure secure transportation of equipment and materials.
GBV, SEAH	Risk of experiencing GBV/SEAH from colleagues or stakeholders.	- Establish a clear, comprehensive, zero-tolerance policy against GBV/SEAH. - Communicate the policy to all employees and stakeholders. - Conduct regular training on GBV/SEAH prevention and response. - Establish confidential reporting mechanisms. - Ensure prompt investigation and appropriate action against perpetrators. - Provide support services for victims. - Engage with local communities to promote respect and safety.	- Ensure the safety and well-being of the affected person(s). - Report the incident to the appropriate authorities and follow company reporting procedures. - Provide support and counseling to the victim(s). - Conduct a thorough investigation and take appropriate disciplinary action.	Human Resources Manager: Develop and implement GBV/SEAH policies and procedures, conduct training, and provide support services. Management: Enforce the policy, ensure a safe workplace, and support victims. Employees: Report incidents, respect colleagues, and create a safe and inclusive work environment.
Environmental Hazards	Risks associated with heat stress, working in remote areas, and exposure to wildlife.	- Implement work-rest cycles, provide cool water and shaded areas, and gradually acclimatize employees to heat. - Ensure reliable communication systems and emergency response plans for remote areas. - Educate employees on local wildlife hazards, provide protective gear (snake-proof boots, insect repellent), and establish procedures for handling wildlife encounters.	- In case of heat stroke or other heat-related illness, move the person to a cool area, provide fluids, and seek medical attention. - In case of wildlife encounters, follow established procedures, provide first aid if needed, and seek medical attention if necessary. - Report the incident to the OHS officer/manager.	OHS Officer/Manager: Conduct environmental risk assessments, develop mitigation measures, and provide training on environmental hazards. Employees: Follow safety protocols for remote work and report any environmental concerns. Supervisors: Ensure employees have access to necessary resources and support in remote areas.

Community Health and Safety

A key ESMS consideration closely related to OHS is Community Health and Safety, which is recommended under **IFC PS 4 / WB ESS 4: Community Health, Safety and Security**. The company is responsible for proactively identifying, preventing, and mitigating potential risks and negative impacts on the health, safety, and security of the community that may arise from products and services. Special consideration should be given to protecting vulnerable groups within the community. The following are some of the measures and actions to be considered for safeguarding community health and safety.

Product Safety:

- **Robustness and Durability:** Procure and sell products to withstand environmental conditions (heat, dust, rain) and potential user mishandling.
- **Electrical Safety:** Incorporate safety features like overcurrent protection, grounding, and insulation to prevent electric shock.
- **Labeling:** Provide clear instructions for safe use, installation, and maintenance, including hazard warnings and emergency procedures.
- **Material Safety:** Select components and materials that minimize the risk of fire, toxic fumes, or other hazards in case of product damage.
- **Child Safety:** If the product is accessible to children, consider additional safety features like tamper-proof enclosures or child-resistant locks.

Transport, Handling, and Installation:

- **Packaging:** Use sturdy packaging to protect products during transport and provide clear handling instructions to minimize the risk of damage.
- **Installation Safety:** Develop installation guidelines that prioritize safety, including secure mounting, proper wiring, and protection from environmental factors. This includes instructions on proper grounding, cable management, and avoiding contact with water.
- **Training:** Offer training or certification programs for installers to ensure safe and correct installation practices.

Product Usage:

- **Proper usage:** Clear instructions on how to operate the system, including how to turn it on/off, adjust settings (if applicable), and connect appliances.
- **Safe handling:** Consumers should be aware of potential electrical hazards and how to safely handle the devices.
- **Battery safety:** If the system includes batteries, consumers need to understand the risks associated with handling and charging them, such as fire hazards, chemical exposure, and proper disposal.
- **Energy efficiency:** Tips on how to maximize the system's efficiency, such as using energy-efficient appliances and optimizing the placement of solar panels.
- **Overloading:** Consumers should be informed about the maximum wattage capacity of the system and the importance of not overloading it with appliances beyond its limits, as this could damage the system or pose a safety risk.
- **Troubleshooting:** Common issues that consumers might encounter and basic troubleshooting steps to resolve them, or when to seek professional help.
- **Warranty terms:** Clear explanation of the warranty terms, including duration, coverage, and limitations.
- **Customer support:** Information on how to contact the company for technical support, repairs, or replacements.
- **Responsible disposal:** Instructions on how to dispose of the system and its components (especially batteries) at the end of their life in an environmentally responsible manner.

Maintenance and Decommissioning:

- **Maintenance Guidelines:** Provide clear instructions for routine maintenance, including cleaning, inspection, checking connections, monitoring battery health, and replacement of parts, emphasizing safety precautions.
- **Troubleshooting:** Common issues that consumers might encounter and basic troubleshooting steps to resolve them, or when to seek professional help.
- **Decommissioning Plan:** Develop a plan for safe decommissioning of products at the end of their lifespan, addressing the removal of hazardous components (e.g., batteries) and proper disposal or recycling.
- **Recycling and Waste Management:** Explore options for recycling or responsible disposal of components at the end of the product's life cycle to minimize environmental impact.

Community Engagement and Awareness:

- **Safety Education:** Conduct community awareness programs to educate users on the safe operation,

maintenance, and potential risks associated with off-grid solar products.

- **Feedback Mechanism:** Establish a system for collecting feedback from users and communities on safety issues or incidents related to the product.
- **Incident Response:** Develop a plan for responding to incidents involving product safety, including investigation, communication with affected parties, and corrective actions.
- **Vulnerable Groups:** Consider the specific needs and vulnerabilities of different community members (elderly, disabled) in the design and deployment of off-grid solar products.

Table 13: Examples of community health and safety concerns and management measures

Issue	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
Community Health	- Exposure to hazardous materials from improper disposal of e-waste - Fire hazards from faulty products or installations	- Partner with certified e-waste recyclers for environmentally sound disposal of end-of-life products. - Educate consumers on safe use and disposal of products. - Ensure proper installation and maintenance to prevent fires. - Conduct regular product quality checks and recalls for defective products.	- Immediately evacuate the area and call emergency services (fire department, ambulance) if needed. - Contain and isolate any hazardous materials spills or leaks, following established protocols. - Provide first aid to injured individuals. - Report the incident to relevant authorities and the company's OHS manager.	OHS Officer/Manager: Oversee e-waste disposal and recycling programs, ensure compliance with safety standards, and conduct product quality checks. Community Relations Officer: Educate communities on safe use and disposal of products, and report any safety concerns. Installation and Maintenance Teams: Ensure proper installation and maintenance of products to minimize risks. Customers: Follow instructions for safe use and disposal, report any product defects or malfunctions.
Community Safety	- Injuries from accidental contact with electrical components or equipment - Fires or explosions caused by faulty installations	- Ensure all electrical components and equipment are properly installed and maintained. - Conduct regular safety inspections of installations. - Educate community members on the potential risks and safety precautions associated with solar products. - Establish a clear reporting mechanism for community members to report any safety concerns or incidents.	- In case of an electrical shock or burn, disconnect the power source and provide first aid. - Evacuate the area in case of fire or explosion and call emergency services. - Investigate the incident and implement corrective actions.	OHS Officer/Manager: Conduct safety inspections, develop and implement safety protocols, and educate community members on safety. Installation and Maintenance Teams: Follow safety procedures during installation and maintenance and report any safety hazards. Community Representatives: Raise awareness of safety risks in the community and report any concerns or incidents.
Community Security	- Theft of solar products or components - Vandalism of installations	- Install security measures (e.g., fences, locks, alarms, surveillance cameras) at project sites and storage facilities. - Employ security personnel to patrol project sites and monitor installations. - Collaborate with local communities and law enforcement to enhance security measures and address any security concerns.	- Report any theft or vandalism to local authorities and the company's security personnel. - Increase security patrols and surveillance in affected areas. - Engage with community members to gather information and address security concerns.	Security Manager/Officer: Develop and implement security plans, oversee security operations, and coordinate with local authorities. Project Managers: Ensure security measures are in place at project sites. Community Relations Officer: Engage with communities to address security concerns and build trust.
Gender Equality and Inclusivity	Discrimination, harassment, exclusion based on gender.	- Establish and enforce a strong gender equality and inclusion policy. - Conduct regular training and awareness sessions on gender equality and inclusion. - Create a safe and inclusive workplace culture where all employees feel valued and respected.	- Investigate any reported incidents of gender discrimination or harassment promptly and thoroughly. - Take appropriate disciplinary action against perpetrators. - Provide support and counseling to affected employees.	Human Resources Manager: Develop and implement gender equality and inclusion policies, conduct training, and address grievances. Management: Promote a culture of respect and inclusivity, and ensure equal opportunities for all employees.

Issue	Potential Risks	Risk Control Measures	Emergency Response Measures	Roles and Responsibilities
		- Establish clear reporting mechanisms for addressing grievances related to gender discrimination or harassment. - Ensure equal opportunities for career development and advancement for all employees.	- Review and strengthen gender equality and inclusion policies and procedures as needed.	Employees: Treat all colleagues with respect and report any incidents of discrimination or harassment.
Vulnerable Populations Inclusivity	Marginalization, exclusion, and discrimination of vulnerable groups (e.g., people with disabilities, minorities, women).	- Conduct diversity and inclusion assessments to identify barriers and develop targeted strategies. - Provide reasonable accommodations for employees with disabilities. - Ensure equal access to training, resources, and opportunities for career advancement. - Foster a culture of respect and sensitivity towards diverse backgrounds and needs. - Engage with local communities to understand and address the needs of vulnerable groups.	- Provide immediate assistance and support to affected individuals. - Address any concerns or complaints raised by vulnerable groups promptly and effectively. - Review and adjust policies and procedures to ensure inclusivity and address any discriminatory practices.	Human Resources Manager/Diversity and Inclusion Officer: Develop and implement diversity and inclusion policies, conduct assessments, and provide training. Management: Promote an inclusive work environment and address any issues related to discrimination or exclusion. Employees: Respect the diversity of colleagues and report any concerns or incidents of discrimination.
Child and Forced Labor	Employment of children below the legal working age or any form of forced labor.	- Strictly adhere to national and international laws prohibiting child and forced labor. - Conduct due diligence on suppliers and contractors to ensure they comply with labor laws. - Implement a robust child labor monitoring and remediation system. - Provide training to employees and suppliers on child and forced labor risks and prevention.	- Immediately report any suspected cases of child or forced labor to relevant authorities. - Remove any child or forced labor from the workplace and provide necessary support and assistance. - Conduct a thorough investigation and take appropriate action against any individuals or entities involved. - Review and strengthen policies and procedures to prevent future occurrences.	Human Resources Manager: Ensure compliance with labor laws, conduct due diligence on suppliers, and implement monitoring and remediation systems. Management: Establish a zero-tolerance policy for child and forced labor and enforce it rigorously. Employees: Report any suspicions of child or forced labor and participate in training programs.
Corruption	Bribery, extortion, fraud, or any unethical practices in business operations.	- Develop and implement a comprehensive anti-corruption policy. - Conduct regular anti-corruption training for employees and partners. - Establish clear reporting mechanisms for employees to report suspected corruption without fear of retaliation. - Implement internal controls and audits to detect and prevent corrupt practices. - Cooperate with authorities in investigations of corruption allegations.	- Immediately report any suspected or known cases of corruption to relevant authorities. - Conduct internal investigations and take appropriate disciplinary action against any employees involved. - Review and strengthen anti-corruption policies and procedures to prevent future occurrences.	Compliance Officer/Ethics Committee: Develop and implement anti-corruption policies, conduct training, and oversee investigations. Management: Create a culture of integrity and ethical behavior, and lead by example. Employees: Adhere to the company's code of conduct and report any suspected corruption.

Consumer Protection

An important aspect of Health and Safety critical for an Off-Grid Solar Business is ensuring consumer protection, with respect to the OGS products and services provided by the business. Protecting consumers fosters trust, builds a loyal customer base, and strengthens the reputation of the company, ultimately leading to a thriving market. It involves fair treatment of customers, transparent communication, responsible lending practices, and the provision of high-quality products and services.

One of the established ways an OGS company can demonstrate commitment to consumer protection is by adopting the GOGLA Consumer Protection Code. This code serves as the industry's benchmark for consumer protection. The GOGLA Consumer Protection Code is built upon six principles, which establish the baseline standards that every consumer should expect from an off-grid solar company and includes a comprehensive framework for assessment and improvement. Key Components of the Code include:

- **Consumer Protection Principles:** These represent the minimum standards of practice, covering areas like fair treatment, transparent communication, and responsible product quality.
- **Self-Assessment Tool:** This tool allows companies to measure and monitor their performance against the Consumer Protection Principles, facilitating ongoing improvement.
- **Consumer Protection Indicators:** These provide a quantitative framework for measuring performance against each of the six Principles, plus overarching governance and management.

GOGLA's Consumer Protection Code Principles, which serve as the cornerstone of the Code, guiding companies in their interactions with customers and ensuring ethical and responsible business practices, include the following¹⁸:

1) Transparency

- a) The company shares clear and sufficient information on the product, service, payment plan and personal data privacy practices to enable consumers to make informed decisions.*
- b) The company shares relevant and timely information before, during and after sales.*
- c) The company communicates in a language and manner consumers can understand.*

2) Responsible sales and pricing

- a) The company takes adequate care to ensure consumers can afford to pay for the product and / or service without becoming overburdened.*
- b) The company ensures consumers' characteristics are taken into account in the price, payment structure and fees of the product and / or service.*

3) Good consumer service

- a) The company ensures availability of technical and after-sales service support, including warranty and post-warranty service for a reasonable period of time.*
- b) An accessible, effective and timely mechanism for complaints and problem resolution is in place.*
- c) The company instructs consumers on proper use and care, and any health and safety risks, related to the product usage or disposal.*
- d) The company considers measures to enable continued operability of the product in the event of the failure of the company.*

4) Good product quality

- a) The company ensures the product / system is appropriate, good quality, safe and performs as advertised.*
- b) The company ensures the user interface / payment platform is appropriate for consumers.*
- c) The company takes reasonable measures to ensure product longevity (including ease of operation and maintenance, repairability and durability).*

¹⁸ Mahoney, T. (2022, November 10). The Consumer Protection Code. Gogla.Org. <https://www.gogla.org/what-we-do/business-services-and-standards/consumer-protection-code/what-is-the-consumer-protection-code/>

5) Data privacy

- a) *The company applies good practices and comply with relevant laws and regulations governing consumer data privacy.*
- b) *The company only collects, use, retain and share personal information that is necessary for the stated consumer service and legitimate interests of the business.*
- c) *The company ensures consumer data is kept secure and confidential.*

6) Fair and respectful treatment

- a) *The company ensures fair and respectful treatment of current and prospective consumers, with adequate safeguards in place against corruption and abusive treatment.*
- b) *The company promotes inclusive practices and do not discriminate, for example, by gender, religion or ethnicity.*
- c) *The company seeks consumers' views and feedback on the design and delivery of the product, service and payment plan.*

By committing to the Code, OGS companies pledge to uphold these standards, using the Self-Assessment Tool to track and improve their practices. This not only benefits consumers, but also enhances the company's reputation and its dialogue with investors, as it provides clear evidence of their commitment to responsible business practices. The box below provides further details on how to commit to GOGLA's consumer project code.

Box: How to Commit to GOGLA's Consumer Protection Code¹⁹

- 1) *If you wish to make a commitment, we strongly recommend an internal **review of your company's alignment** with the spirit of the Principles*
- 2) *Send a **letter of commitment** (using this template²⁰) signed by a Director of your company to GOGLA (please send via email to consumerprotection@gogla.org)*
- 3) *Within three months of sending the letter of commitment, measure your performance against the CP indicators using the **Self-Assessment** tool.*
- 4) *Once the assessment is complete, send a **letter of declaration** (using this template²¹) signed by a Director to GOGLA (via consumerprotection@gogla.org). GOGLA does not request any visibility of the self-assessment results.*
- 5) *To **maintain commitment** to the CP Code, companies must repeat the self-assessment and declaration to GOGLA every 12 months thereafter.*



¹⁹ Mahoney, T. (2022, November 14). How to Commit & Endorse to the Consumer Protection Code. Gogla.Org. <https://www.gogla.org/what-we-do/business-services-and-standards/how-to-commit-endorse-to-the-consumer-protection-code/>

²⁰ <https://www.gogla.org/wp-content/uploads/2024/05/CP-Code-Endorsement-letter-template-May2024.docx>

²¹ https://www.gogla.org/wp-content/uploads/2023/03/cp_self_assessment_declaration_template_-_draft.docx

9. Human Resources Policy and Plan

Human Resources Policy and/or Plan are recommended ESMS components as per **IFC PS 2 / WB ESS 2 Labor and Working Conditions**. A Human Resources (HR) Policy and/or Plan serves as a framework for fostering a fair, safe, and productive workplace. It ensures legal compliance, guides effective workforce management, promotes employee well-being, and aligns HR practices with the company's broader environmental and social goals. HR Policies and Plans help mitigate HR-related risks, enhance the company's reputation, and contribute to sustainable business success by building a motivated and responsible workforce.

Components of a Human Resources Policy and/or Plan include:

- **Purpose and Scope:** Clearly state the policy's purpose, aligning it with the company's mission and values, while emphasizing its role in achieving the ESMS's environmental and social objectives. Define the scope, explaining who the policy applies to (employees, contractors, etc.) and the areas of HR it covers.
- **Legal and Regulatory Compliance:** Outline the company's commitment to complying with relevant national labor laws, regulations, and international standards (e.g., ILO conventions). Identify specific laws and regulations that the policy will address (e.g., minimum wage, working hours, anti-discrimination).
- **Recruitment and Hiring:** Describe the company's recruitment and selection process, emphasizing fair and transparent practices. Detail how the company will ensure equal opportunity for all candidates, regardless of gender, ethnicity, age, disability, etc. Outline any specific recruitment goals related to diversity and inclusion.
- **Employee Rights and Responsibilities:** Clearly define the rights and responsibilities of employees. Roles and Responsibilities, which are a key aspect of Human Resource Plans, are addressed in the section on Organizational Capacity and Competency.
- **Performance Management and Development:** Explain the company's performance appraisal process, including how employee performance will be evaluated and feedback provided. Detail how the company will support employee development through training, mentoring, or other programs. Outline procedures for addressing performance issues or disciplinary actions.
- **Compensation and Benefits:** Specify the company's compensation philosophy and how wages and salaries will be determined. List the types of benefits offered to employees (e.g., health insurance, paid leave, retirement savings).
- **Health and Safety:** Describe the company's commitment to providing a safe and healthy workplace. Outline procedures for identifying and mitigating workplace hazards, how the company will comply with relevant health and safety regulations, and how the company will handle workplace accidents and injuries. Health and Safety considerations are addressed in more detail in the OHS Plan section.
- **Grievance Mechanism:** Establish a clear and accessible process for employees to raise concerns or complaints related to their employment or working conditions. Grievance Mechanism is addressed in the Grievance Mechanism section of this document.
- **Training and Awareness:** Outline a plan for training employees. Describe how the company will raise awareness of environmental and social issues among employees. Training is addressed in the section on Organizational Capacity and Competency.
- **Monitoring and Review:** Establish a process for monitoring the implementation and effectiveness of the HR policy and plan. Specify how often the policy will be reviewed and updated to ensure it remains relevant and effective.

The human resource policy and plan should address the following recommended policies and practices:

- **Fairness and Equality:** Ensure fair treatment, gender equality, non-discrimination, and equal opportunity for all employees.
- **Equitable Working Conditions:** Adopt and implement HR policies that ensure fair working conditions, aligning with national laws and regulations, as well as any applicable corporate standards.
- **Non-Discriminatory Practices:** Employment decisions (hiring, promotion, termination, etc.), professional development opportunities, and retrenchment plans should be based on merit, skills, and experience, free from any form of discrimination.
- **Diversity and Inclusion:** Demonstrate a commitment to gender mainstreaming and closing of gender gaps that may exist in the workplace. Promote inclusiveness and diversity at all levels of the company, particularly in management roles.
- **Integration with Other Plans:** HR policies should be coordinated with the organisational capacity and competency, emergency preparedness and response plan and the grievance mechanism.
- **Positive Workplace Relations:** Foster healthy relationships between workers and management.
- **Legal Compliance:** Adhere to national employment and labor laws and international standards.

- **Worker Protection:** Prioritize the protection of workers, especially those in vulnerable groups.
- **Women's Health and Safety:** Address the specific safety, health, and hygiene needs of women in the workplace.
- **Living Wage:** Commit to paying a living wage that covers basic needs and allows for some discretionary income, particularly in regions with low minimum wages. Reference can be made to the Global Living Wage Coalition for guidance.
- **Grievance Mechanism:** Establish a clear and accessible grievance mechanism for employees and contractors.
- **No Forced or Child Labor:** Zero tolerance for use of forced or child labor.

The box below presents a generic sample Human Resources Policy statement that can be adapted to the context of your business.

Box 1: Sample Human Resource Policy
Human Resource Policy for [Company Name] I. Introduction [Company Name] is committed to fostering a diverse, inclusive, and equitable workplace. This Human Resource (HR) Policy outlines the company's commitment to fair and equitable treatment of all employees, regardless of their background, ensuring compliance with all applicable laws and regulations, and promoting a positive and productive work environment. II. Equal Employment Opportunity [Company Name] provides equal employment opportunities (EEO) to all employees and applicants for employment without regard to race, color, religion, sex (including pregnancy or childbirth), gender, national origin, age, disability, or any other characteristic protected by applicable national, regional, or local laws. This policy applies to all terms and conditions of employment, including recruiting, hiring, placement, promotion, termination, layoff, recall, transfer, leaves of absence, compensation, and training. III. Non-Discrimination and Harassment [Company Name] strictly prohibits discrimination and harassment of any kind. This includes, but is not limited to, harassment based on any of the protected characteristics listed above. All employees are expected to treat each other with respect and dignity. The company will not tolerate any form of harassment, whether it is verbal, physical, or visual. IV. Compensation and Benefits [Company Name] is committed to providing fair and competitive compensation and benefits to all employees. Compensation will be based on factors such as skills, experience, performance, and market rates. The company will regularly review compensation and benefits packages to ensure they remain competitive and equitable. V. Performance Management [Company Name] will conduct regular performance appraisals for all employees. These appraisals will be based on objective criteria and will provide feedback on performance, identify areas for improvement, and set goals for future development. Performance will be the primary factor in determining promotions, salary increases, and other rewards. VI. Training and Development [Company Name] will provide opportunities for employees to enhance their skills and knowledge through training and development programs. These programs will be designed to improve job performance, promote career advancement, and ensure that employees have the necessary skills to succeed in their roles. VII. Employee Rights and Responsibilities Employees have the right to a safe and healthy work environment, fair treatment, and respect for their privacy and dignity. They also have the responsibility to perform their duties to the best of their abilities, follow company policies and procedures, and treat others with respect. VIII. Complaint and Grievance Procedure

[Company Name] has established a formal complaint and grievance procedure to address any concerns or complaints raised by employees, including those related to Sexual Exploitation, Abuse, and Harassment (SEAH). This procedure ensures that all complaints are investigated fairly and impartially and that appropriate action is taken to resolve them.

IX. Retaliation Prohibited

[Company Name] prohibits any form of retaliation against employees who report discrimination, harassment, or other violations of company policy. The company will protect the confidentiality of those who report such issues and will take appropriate action against anyone who engages in retaliation.

X. Code of Conduct

All employees are expected to adhere to the company's Code of Conduct, which outlines acceptable behavior in the workplace. The Code of Conduct prohibits any form of harassment, discrimination, violence, or other unethical behavior. It emphasizes respect for diversity, inclusivity, and the prevention of gender-based violence.

XI. Compliance with Laws and Regulations

[Company Name] will comply with all applicable laws and regulations relating to employment, including those relating to equal opportunity, non-discrimination, minimum age of employees, compensation, social security, leave entitlements, benefits, and safety. The company will also comply with any applicable industry standards or codes of conduct.

XII. Prohibition of Forced and Child Labor

[Company Name] maintains a zero-tolerance policy against forced labor and child labor. We comply with all national laws regarding minimum age requirements and hazardous work restrictions for children. We conduct due diligence on our supply chains to ensure that our suppliers also adhere to these standards.

XIII. Amendments

This HR Policy may be amended from time to time by [Company Name]. Any amendments will be communicated to employees in writing.

XIV. Acknowledgement

By signing below, you acknowledge that you have read and understood this HR Policy and agree to abide by its terms.

Employee Signature:

Date:

Company Representative Signature:

Date:

Code of Conduct

A key element of an Human Resources Policy or Plan should be development of an employee code of conduct statement, which serves as a guide for expected behavior, outlining the company's values and ethical principles. It fosters a positive work environment, promotes professionalism and integrity, and clarifies standards for decision-making. By setting clear expectations, the code helps prevent misunderstandings, minimizes risk, and protects the company's reputation. Ultimately, it creates a shared understanding of appropriate conduct, contributing to a more cohesive and successful organization.

Box 2: Sample Code of Conduct statement that a company can adapt to its context.

This Code of Conduct is part of our measures to deal with environmental, social, health and safety (ESHS) risks related to our operations and the products and services we provide. The Code of Conduct specifies behaviour that we expect of all our employees and subcontractors providing the Services for the project. Our workplace is an environment where

unsafe, offensive, abusive or violent behaviour will not be tolerated and where all persons should feel comfortable raising issues or concerns without fear of retaliation.

Code of Conduct

Each company staff is required to:

- 1) carry out his/her duties competently and diligently;
- 2) comply with this Code of Conduct and all applicable laws, regulations and other requirements, including requirements to protect the health, safety and well-being of other persons;
- 3) maintain a safe working environment including by:
 - a. ensuring that workplaces, equipment and processes under each person's control are safe and without risk to health;
 - b. wearing required personal protective equipment; and
 - c. following applicable emergency operating procedures.
- 4) report work situations that he/she believes are not safe or healthy and remove himself/herself from a work situation which he/she reasonably believes presents an imminent and serious danger to his/her life or health;
- 5) treat other people with respect, regardless of gender, race, ethnicity, nationality, language, religion, political opinion, social origin, or disability;
- 6) not discriminate against specific groups such as women, people with disabilities, migrant workers or children;
- 7) not engage in Sexual Harassment, which means unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct of a sexual nature with other Experts, Contractor's Personnel or Client's Personnel;
- 8) not engage in Sexual Exploitation, which means any actual or attempted abuse of position of vulnerability, differential power or trust, for sexual purposes, including, but not limited to, profiting monetarily, socially or politically from the sexual exploitation of another;
- 9) not engage in Sexual Abuse, which means the actual or threatened physical intrusion of a sexual nature, whether by force or under unequal or coercive conditions;
- 10) not engage in any form of sexual activity with individuals under the age of 18, except in case of pre-existing marriage;
- 11) complete relevant training courses that will be provided related to the environmental and social aspects of the Contract, including on health and safety matters, Sexual Exploitation and Abuse (SEA), and Sexual Harassment (SH);
- 12) report violations of this Code of Conduct; and not retaliate against any person who reports violations of this Code of Conduct, whether to us or the Client, or who makes use of grievance mechanism for Experts or the project's Grievance Redress Mechanism.

Reporting Code of Conduct Violations

If any person observes behaviour that he/she believes may represent a violation of this Code of Conduct, or that otherwise concerns him/her, he/she should raise the issue promptly, for example through the company's grievance mechanism. The person's identity will be kept confidential, unless reporting of allegations is mandated by the country law. Anonymous complaints or allegations may also be submitted and will be given all due and appropriate consideration. There will be no retaliation against any person who raises a concern in good faith about any behaviour prohibited by this Code of Conduct. Such retaliation would be a violation of this Code of Conduct.

Consequences of Violating the Code of Conduct

Any violation of this Code of Conduct by Experts may result in serious consequences, up to and including suspension of employment (without payment of salary), termination of employment and possible referral to legal authorities, depending on the nature and seriousness of the violation.

Signing of Code of Conduct by Each Staff

I have received a copy of the Code of Conduct written in a language that I comprehend. I have read the Code of Conduct and understand the behaviour and conduct expected of me as specified in the Code of Conduct. I understand the consequences of behaviour inconsistent with the Code of Conduct and I commit to fully comply with the Code of Conduct. I understand that it is my responsibility to report any behaviour by others working on the project, that I have witnessed, which is in violation of the Code of Conduct.

Full Name: _____ **Position:** _____

Signature: _____

Date: (day month year): _____

Table 14: HR related risks and management measures

HR Issue	Potential Risks	Management Measures	Incident Response Measures	Roles and Responsibilities
Fairness and Equality	- Discrimination (gender, age, ethnicity, etc.) - Unequal pay for equal work - Lack of career development opportunities	- Establish clear anti-discrimination and equal opportunity policies. - Conduct regular diversity and inclusion training. - Implement fair and transparent compensation and promotion practices based on merit and performance. - Provide mentorship and professional development opportunities for all employees. - Establish a confidential reporting mechanism for grievances.	- Investigate and address any complaints of discrimination or unfair treatment promptly. - Conduct mediation or conflict resolution sessions if necessary. - Revise policies and procedures to ensure fairness and equality.	HR Manager: Develop and enforce anti-discrimination and equal opportunity policies, conduct training, and address grievances. Management: Promote a fair and inclusive workplace culture and ensure equal opportunities for all. Employees: Report any instances of discrimination or unfair treatment and participate in diversity and inclusion training.
Equitable Working Conditions	- Unsafe or unhealthy working conditions - Excessive working hours - Lack of rest breaks	- Comply with all relevant labor laws and regulations regarding working hours, rest breaks, and safety standards. - Conduct regular workplace inspections to identify and address hazards. - Provide necessary safety equipment and training to employees. - Encourage a healthy work-life balance and provide flexible work arrangements where possible.	- Provide immediate medical attention for any work-related injuries or illnesses. - Report incidents to relevant authorities. - Review and improve working conditions to prevent similar incidents in the future.	OHS Manager: Ensure compliance with safety regulations, conduct workplace inspections, and provide safety training. HR Manager: Monitor working hours and ensure compliance with labor laws. Employees: Report any unsafe or unhealthy working conditions and follow safety procedures.
Non-Discriminatory Practices	- Discrimination in hiring, promotion, termination, or other employment decisions	- Base all employment decisions on merit, skills, and experience. - Use standardized and objective evaluation criteria for recruitment, performance appraisals, and promotions. - Document all employment decisions and ensure transparency. - Implement blind recruitment practices to eliminate bias in the hiring process.	- Investigate any complaints of discrimination thoroughly and impartially. - Take corrective action if discrimination is found. - Review and update policies and procedures to prevent future occurrences.	HR Manager: Ensure fair and transparent employment practices and investigate complaints. Managers: Make employment decisions based on objective criteria and avoid personal biases. Employees: Report any instances of discrimination.
Diversity and Inclusion	- Lack of diversity in the workforce - Exclusion of certain groups from decision-making	- Set diversity and inclusion goals and targets. - Actively recruit candidates from diverse backgrounds. - Provide training on unconscious bias and cultural competence. - Create employee resource groups for underrepresented groups. - Ensure representation of diverse groups in leadership and decision-making positions.	- Review recruitment and promotion practices to identify and address any barriers to diversity and inclusion. - Implement measures to increase representation of underrepresented groups in leadership and decision-making positions.	HR Manager: Develop and implement diversity and inclusion strategies and initiatives. Management: Foster a culture of diversity and inclusion and lead by example. Employees: Participate in diversity and inclusion initiatives and embrace differences.

Integration with Other Plans	- Lack of coordination between HR policies and other company plans (e.g., emergency response, grievance mechanism)	- Ensure that HR policies are aligned with and integrated into other company plans. - Regularly review and update policies to reflect changes in other plans. - Communicate the integration of plans to all employees.	- Review and update HR policies in response to emergencies or grievances. - Ensure that all employees are aware of the integrated nature of the plans and their roles and responsibilities.	HR Manager: Coordinate with other departments to ensure integration of plans. Management: Oversee the integration process and ensure consistency between plans. Employees: Be aware of all company plans and follow procedures accordingly.
Positive Workplace Relations	- Conflicts between employees and management - Low employee morale and engagement	- Foster a positive workplace culture based on open communication, respect, and trust. - Encourage employee feedback and involvement in decision-making. - Provide regular training on conflict resolution and communication skills. - Implement employee recognition and reward programs.	- Address conflicts promptly and fairly through mediation or other dispute resolution mechanisms. - Conduct employee surveys to assess morale and engagement levels and identify areas for improvement.	HR Manager: Foster positive workplace relations, conduct training, and address conflicts. Managers: Communicate openly with employees, listen to feedback, and address concerns promptly. Employees: Communicate concerns respectfully and participate in conflict resolution processes.
Legal Compliance	- Violations of labor laws and regulations - Legal disputes and lawsuits	- Stay informed about all relevant labor laws and regulations. - Conduct regular audits to ensure compliance. - Provide legal training to HR staff and managers. - Consult with legal counsel on complex issues.	- Address any legal issues promptly and seek legal counsel if necessary. - Review and update policies and procedures to ensure compliance with changing regulations.	HR Manager/Legal Counsel: Ensure compliance with labor laws, conduct audits, and provide legal guidance. Management: Prioritize legal compliance and provide resources for legal support. Employees: Follow company policies and procedures and report any legal concerns.
Worker Protection	- Unsafe working conditions - Lack of social protection (e.g., insurance, pensions) - Exploitation of workers	- Provide safe and healthy working conditions in compliance with regulations. - Offer adequate social protection benefits (e.g., health insurance, workers' compensation, pension plans). - Establish mechanisms to prevent and address worker exploitation, such as whistleblower hotlines and grievance procedures.	- Investigate any reports of unsafe working conditions or worker exploitation. - Take corrective action to address any violations or deficiencies. - Provide support and assistance to affected workers.	OHS Manager: Ensure safe working conditions and provide training on safety procedures. HR Manager: Provide social protection benefits and address worker exploitation issues. Employees: Report any unsafe conditions or exploitation and participate in safety training.
Women's Health and Safety	- Lack of access to sanitary facilities - Inadequate protection against gender-based violence	- Provide clean and accessible sanitary facilities for female employees. - Implement policies and procedures to prevent and address gender-based violence in the workplace. - Offer training on gender sensitivity and harassment prevention.	- Respond promptly to any reports of gender-based violence or harassment. - Provide support and counseling to affected employees. - Take disciplinary action against perpetrators.	HR Manager: Ensure provision of sanitary facilities, implement policies to prevent gender-based violence, and provide training on gender sensitivity. Management: Create a safe and respectful workplace for all employees. Employees: Report any incidents of gender-

				based violence or harassment and participate in training.
Living Wage	- Insufficient wages to meet basic needs	- Conduct regular wage reviews to ensure that wages are aligned with the cost of living. - Benchmark wages against industry standards and the Global Living Wage Coalition recommendations. - Provide additional benefits or allowances to supplement wages where necessary. - Consider profit-sharing or bonus schemes to incentivize productivity and improve employee well-being.	- Review and adjust wages in response to changes in the cost of living. - Negotiate with employee representatives to ensure fair compensation.	HR Manager: Conduct wage reviews and ensure compliance with minimum wage laws. Finance Department: Monitor wage costs and ensure financial sustainability. Management: Make informed decisions about wage adjustments and benefit packages.
Grievance Mechanism	- Lack of a fair and accessible grievance mechanism	- Establish a clear and transparent grievance mechanism that is accessible to all employees. - Ensure confidentiality and protection against retaliation for those who report grievances. - Train managers and supervisors on how to handle grievances fairly and effectively. - Regularly review and evaluate the effectiveness of the grievance mechanism.	- Ensure that all grievances are addressed promptly and fairly. - Conduct investigations into serious grievances and take appropriate action. - Communicate the outcome of grievances to all parties involved.	HR Manager: Establish and manage the grievance mechanism, investigate complaints, and provide support to employees and contractors. Managers and Supervisors: Handle grievances at the initial stage, follow procedures, and escalate issues when necessary. Employees and Contractors: Report grievances promptly and participate in the resolution process.
No Forced or Child Labor	- Employment of children below the legal working age or any form of forced labor.	- Strictly adhere to national and international laws prohibiting child and forced labor. - Conduct due diligence on suppliers and contractors to ensure they comply with labor laws. - Implement a robust child labor monitoring and remediation system. - Provide training to employees and suppliers on child and forced labor risks and prevention.	- Immediately report any suspected cases of child or forced labor to relevant authorities. - Remove any child or forced labor from the workplace and provide necessary support and assistance. - Conduct a thorough investigation and take appropriate action against any individuals or entities involved. - Review and strengthen policies and procedures to prevent future occurrences.	HR Manager: Ensure compliance with labor laws, conduct due diligence on suppliers, and implement monitoring and remediation systems. Management: Establish a zero-tolerance policy for child and forced labor and enforce it rigorously. Employees: Report any suspicions of child or forced labor and participate in training programs.

10. Waste Management Plan

Waste Management is an important ESMS component as per **IFC PS 3 / WB ESS 3: Resource Efficiency and Pollution Prevention and Management**.

All off-grid solar (OGS) products companies should develop a waste management plan towards addressing e-waste, which is considered the source of the most significant environmental and social impacts associated with the OGS sector. Proactive waste management in the OGS sector is a multi-faceted approach that yields significant benefits.

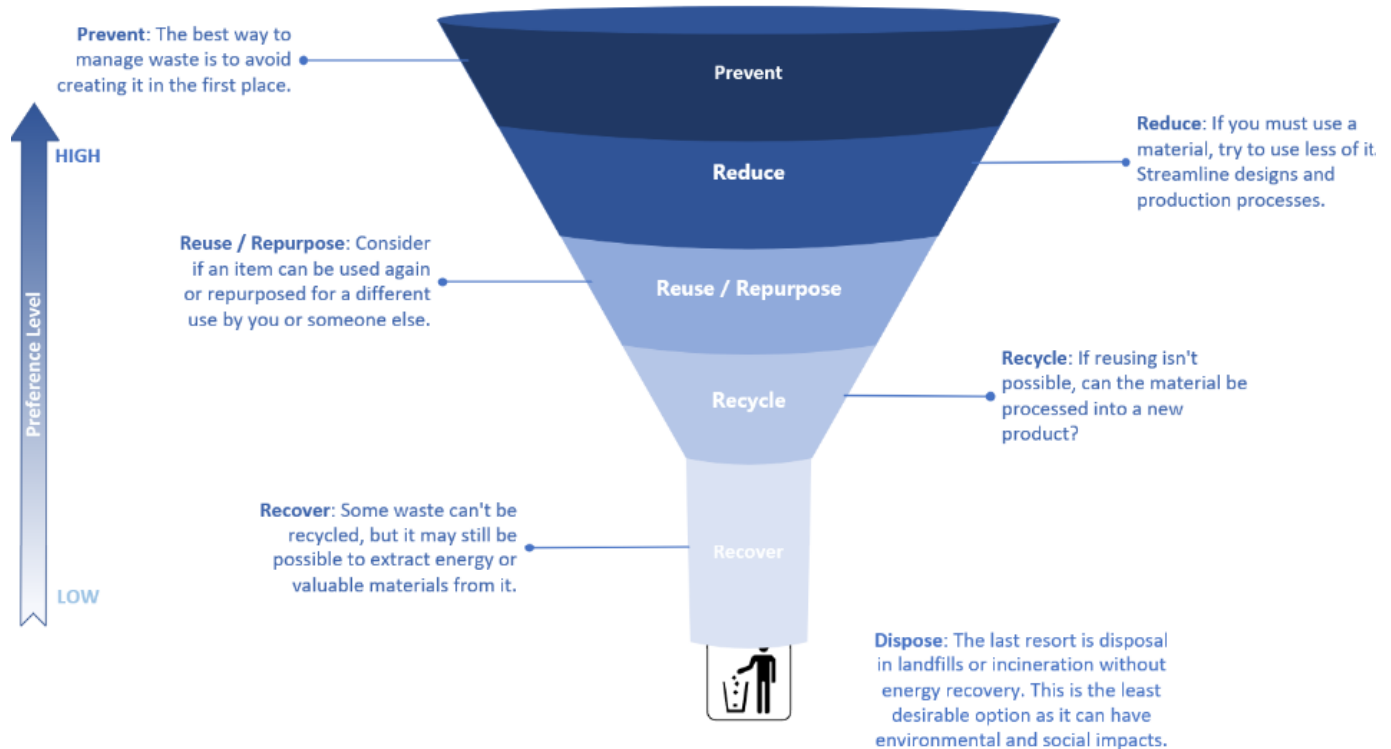


Figure 14: Waste management hierarchy

By minimizing waste sent to landfills (or other dumping sites), conserving valuable resources, and preventing pollution, companies demonstrate a strong commitment to environmental responsibility. Adherence to environmental regulations and extended producer responsibility obligations ensures legal compliance and avoids potential penalties. Embracing resource efficiency through recovery and reuse not only contributes to a circular economy but also unlocks cost savings by reducing disposal expenses and generating revenue from recycled materials. Responsible waste management can enhance brand reputation, showcasing the company's dedication to sustainable and responsible business practices. Engaging local communities, where feasible, in waste management initiatives fosters positive relationships and creates shared social benefits. In preparing a waste management plan the company should taking into consideration the following:

- **Waste Generation and Practices Assessment:**
 - Conduct a thorough waste audit to identify the types and quantities of waste generated throughout the product lifecycle (manufacturing, distribution, use, end-of-life).
 - Assess existing waste management practices within the company and identify gaps or areas for improvement.
 - Evaluate the regulatory landscape and identify relevant environmental and waste management regulations.
 - Study and map the local and regional e-waste management ecosystem (e.g. prevalence and distribution of OGS products, locations of e-waste collection points, and info on existing repair and recycling facilities) to inform your waste management strategies.
- **Waste Management Strategies:**
 - **Extended Producer Responsibility (EPR) Systems:** Establish a system for taking back and managing end-of-life products, including collection, transportation, and processing.
 - **Formalization of Recycling:** Partner with certified recyclers to ensure safe and responsible recycling

of lithium-ion batteries (LIBs) and other components. Consider investing in local recycling infrastructure if feasible.

- **Circular Design:** Collaborate with product designers to incorporate principles of easy disassembly, modularity, and use of recyclable materials.
- **5R Principles:** Integrate the 5R principles (Reduce, Reuse, Repair, Refurbish, Recycle) into the company's operations and product lifecycle.

- **Implementation:**

- Develop and implement collection mechanisms for end-of-life products, potentially through partnerships with local organizations or retailers, while leveraging existing infrastructure for take-back and collection of e-waste.
- Establish procedures for safe transportation and storage of e-waste.
- Provide incentives to staff and agents to participate in e-waste collection process.
- Building awareness of consumers and installers on the environmental and human health risks of improper e-waste disposal and educate them on proper disposal and recycling practices.
- Convince and incentivize consumers to give up their e-waste products for reuse/repurposing/recycling.
- Follow the 5R principles, prioritizing repair and refurbishment of OGS products over recycling and disposal, where feasible.
- Track and monitor waste flows to measure the effectiveness of the plan.

- **Review and Improvement:**

- Regularly review and evaluate the waste management plan to identify areas for improvement.
- Adapt to changing regulations and technologies.
- Seek feedback from stakeholders and incorporate their input into the plan.

In regions where the informal recycling sector is prevalent, it is crucial for off-grid solar companies to actively engage with this sector. This engagement includes partnering with local government agencies and regulatory bodies to develop and implement waste management policies that address the unique challenges posed by the informal sector. Collaborating with informal recyclers, community organizations, and formal recycling stakeholders to establish safe and efficient waste collection and recycling channels; this may include setting up collection points, providing fair compensation for collected materials, and promoting awareness about safe handling practices. Promote training and capacity building programs for informal recyclers to equip them with the knowledge and skills necessary for safe and environmentally sound e-waste and solar product waste recycling. Explore opportunities to introduce appropriate technologies and tools to the informal sector to enhance their efficiency and safety.

Table 15: Waste disposal management template.

Waste Type	Waste Classification	Disposal Method	5R Principles Target	Responsibility

Source: adapted from CAMO Clean Energy. N.d. REPP Environmental and Social Management System Toolkit.

Table 16: Waste management options for common off-grid solar products

Product	Component	Material	Environmental/Health Concerns	Reuse Options	Repurposing Options	Recycling Options	Recycling Practices	Best
Stand-Alone Home Solar System	Solar panel	Entire Solar Panel	Discarded or stored panels can have safety (e.g. fall risk) and visual aesthetics impacts; See concerns for individual materials below.	Use in less demanding applications (e.g., powering small devices).	Integrate into DIY projects like solar phone chargers.	Specialized PV panel recycling facilities	Separate components for recycling	
		Silicon	Resource depletion	N/A	N/A	Specialized PV panel recycling facilities	Separate from other components, recover high-purity silicon for reuse in new panels. Test for functionality before reuse or repurposing.	
		Glass	Energy-intensive production, potential for breakage	Reuse for greenhouses or building materials.	Create decorative items or mosaics.	Glass recycling facilities	Crush and melt glass to create new glass products.	
		Aluminum	Energy-intensive production	Repurpose as structural material for other projects.	Create crafts or art pieces.	Aluminum recycling facilities	Melt and recast aluminum to create new products.	
		Copper	Resource depletion	Reuse in electrical wiring or plumbing.	Create sculptures or jewelry.	Copper recycling facilities	Separate copper wiring and components for recycling.	
		Silver	Resource depletion	Recover and reuse in new electrical contacts.	Use in jewelry making or other crafts.	Specialized precious metal recovery facilities	Recover silver from electrical contacts and solder.	
		Lead (in solder)	Toxic to humans and the environment	N/A	N/A	Lead recycling facilities	Carefully handle and recycle lead to prevent contamination.	

Product	Component	Material	Environmental/Health Concerns	Reuse Options	Repurposing Options	Recycling Options	Recycling Practices	Best
	Batteries	Lead-acid Battery	Lead contamination, sulfuric acid leaks	Use for backup power in off-grid systems.	Repurpose as weights or anchors.	Battery recycling centers	Collect and recycle lead-acid batteries separately from other types.	
		Lead (Lead-acid)	Lead contamination, toxic to humans and environment	N/A	N/A.	Battery recycling centers	Collect and recycle lead-acid batteries separately from other types.	
		Sulfuric acid (Lead-acid)	Corrosive, can cause burns and environmental damage	N/A	N/A	Specialized hazardous waste facilities	Neutralize and dispose of properly according to regulations.	
		Lithium-ion Battery	Fire hazard, potential for thermal runaway	Repurpose for lower-power applications (e.g., portable lights).	Use in DIY power banks or battery packs.	Specialized lithium-ion battery recycling facilities	Follow manufacturer guidelines for safe disposal and recycling.	
		Lithium (Lithium-ion)	Flammable, can cause burns	N/A	N/A	Specialized lithium-ion battery recycling facilities	Follow manufacturer guidelines for safe disposal and recycling.	
		Cobalt (Lithium-ion)	Toxic to humans and the environment	N/A	N/A	Specialized battery recycling facilities	Recover cobalt for reuse in new batteries.	
		Nickel (Lithium-ion)	Toxic to humans and the environment	N/A	N/A	Specialized battery recycling facilities	Recover nickel for reuse in new batteries.	
	Charge controller	Plastic	Non-biodegradable, can release toxins when burned	Reuse for storage or other purposes.	Repurpose into planters or art projects.	Plastic recycling facilities	Separate different types of plastics for proper recycling.	
		Copper, electronic components	Resource depletion, potential release of hazardous substances	N/A	Repurpose circuit boards for educational or artistic projects.	E-waste recyclers	Separate electronic components from other materials for specialized recycling.	
	Inverter	Aluminum	Energy-intensive production	Repurpose as heat sink for other electronics.	Create decorative items or sculptures.	Aluminum recycling facilities	Melt and recast aluminum to create new products.	
		Copper	Resource depletion	Reuse in electrical wiring or plumbing.	Create sculptures or jewelry.	Copper recycling facilities	Separate copper wiring and components for recycling.	
		Electronic components	Contain hazardous substances (e.g., lead, mercury)	N/A	Repurpose circuit boards for educational or artistic projects.	E-waste recyclers	Separate electronic components from other	

Product	Component	Material	Environmental/Health Concerns	Reuse Options	Repurposing Options	Recycling Options	Recycling Practices	Best
							materials for specialized recycling.	
Solar Pumps	Motor	Copper	Resource depletion	Reuse in other applications with similar voltage and power needs.	Repurpose for DIY projects such as water pumps.	Copper recycling facilities	Separate copper wiring and components for recycling.	
		Iron	Resource depletion	N/A	Repurpose as scrap metal for art or construction projects.	Scrap metal recycling facilities	Separate iron components for recycling.	
		Plastic	Non-biodegradable, can release toxins when burned	Reuse for storage containers or other purposes.	Repurpose into planters or art projects.	Plastic recycling facilities	Separate different types of plastics for proper recycling.	
	Pump	Steel	Resource depletion	N/A	Repurpose for DIY projects or as scrap metal for art.	Scrap metal recycling facilities	Separate steel components for recycling.	
	Solar panel	(See solar panel materials above)						
Solar Lanterns	Casing	Plastic	Non-biodegradable, can release toxins when burned	Reuse for storage containers or other purposes.	Repurpose into planters or art projects.	Plastic recycling facilities	Separate different types of plastics for proper recycling.	
	LED lights	Electronic components	Contain hazardous substances (e.g., lead, mercury)	N/A	Repurpose circuit boards for educational or artistic projects.	E-waste recyclers	Separate electronic components from other materials for specialized recycling.	
		Plastic	Non-biodegradable, can release toxins when burned	Reuse for storage containers or other purposes.	Repurpose into planters or art projects.	Plastic recycling facilities	Separate different types of plastics for proper recycling.	
	Battery	Lithium-ion	Fire hazard, potential for thermal runaway	Repurpose for lower-power applications	Use in DIY power banks or battery packs.	Specialized lithium-ion battery recycling facilities	Follow manufacturer guidelines for safe disposal and recycling.	
		Lead (Lead-acid)	Lead contamination, toxic to humans and environment	N/A	Repurpose as weights or anchors.	Dispose of at authorized battery recycling centers only.	Collect and recycle lead-acid batteries separately from other types.	

Product	Component	Material	Environmental/Health Concerns	Reuse Options	Repurposing Options	Recycling Options	Recycling Practices	Best
		Sulfuric acid (Lead-acid)	Corrosive, can cause burns and environmental damage	N/A	N/A	Specialized hazardous waste facilities	Neutralize and dispose of properly according to regulations.	
		Lithium Iron Phosphate (LiFePO4)	Fire hazard, potential for thermal runaway	N/A	N/A	Specialized lithium iron phosphate battery recycling facilities	Follow manufacturer guidelines for safe disposal and recycling.	
	Wiring	Copper	Resource depletion	Reuse in electrical wiring or other applications.	Use for crafts or jewelry making.	Copper recycling facilities	Separate copper wiring from other components for recycling.	
	Solar cell	Silicon	Resource depletion	Use in less demanding applications (e.g., powering calculators).	Integrate into DIY projects or educational kits.	Specialized PV cell recycling facilities	Separate from other components, recover silicon for reuse.	
Solar Fan	Motor	Copper, iron, electronic components	Resource depletion, potential release of hazardous substances (e.g., lead, mercury)	Reuse in similar applications with appropriate voltage and power ratings.	Repurpose for DIY projects like electric bikes or other small appliances.	Copper recycling facilities, e-waste recyclers	Separate copper and electronic components from other materials for specialized recycling.	
	Blades	Plastic, aluminum	Non-biodegradable (plastic), energy-intensive production (aluminum)	N/A	Repurpose as decorative items or use in art projects.	Plastic recycling facilities, aluminum recycling facilities	Separate plastic and aluminum components for recycling.	
	Housing	Plastic	Non-biodegradable, can release toxins when burned	Reuse for storage containers or other purposes.	Repurpose into planters or art projects.	Plastic recycling facilities	Separate different types of plastics for proper recycling.	
	Solar cell	Silicon	Resource depletion	Use in less demanding applications (e.g., powering calculators).	Integrate into DIY projects or educational kits.	Specialized PV cell recycling facilities	Separate from other components, recover silicon for reuse.	
	Battery	Lithium-ion	Fire hazard, potential for thermal runaway	Repurpose for lower-power applications	Use in DIY power banks or battery packs.	Specialized lithium-ion battery recycling facilities	Follow manufacturer guidelines for safe disposal and recycling.	

Product	Component	Material	Environmental/Health Concerns	Reuse Options	Repurposing Options	Recycling Options	Recycling Practices	Best
		(note lead-acid batteries may be used in fans, but are less common)						



Source: ACE 2019 citing source as IFC 2018

Figure 15: E-waste management value chain²²

²² Africa Clean Energy (ACE) 2019. E-Waste Policy Handbook. Citing source as IFC 2018. E-waste Dismantling: An Entrepreneur's Guide

Resource Efficiency Plan

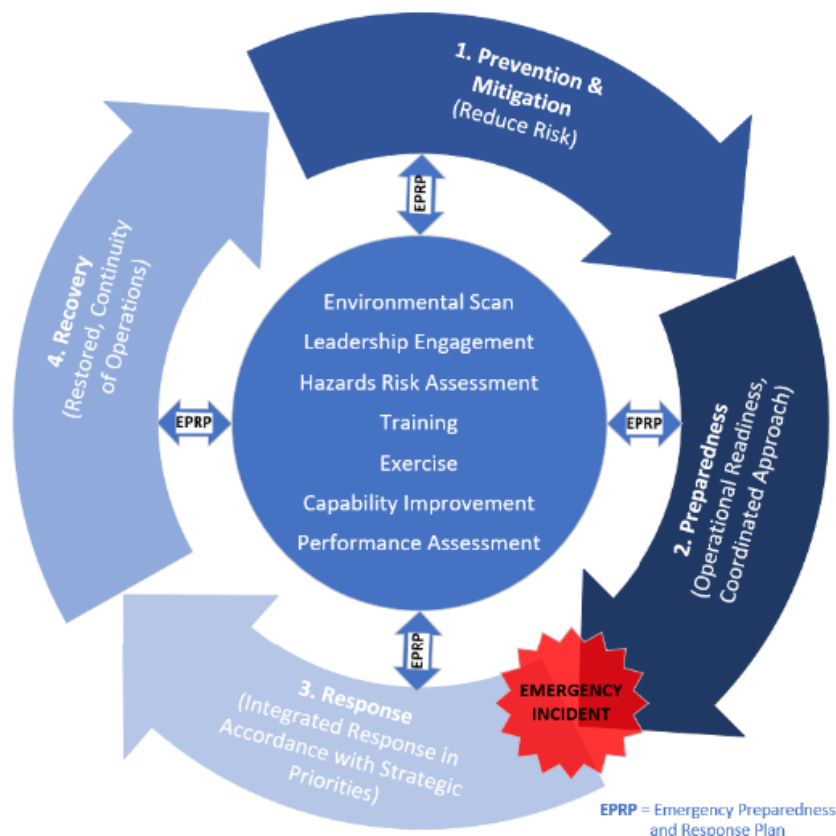
The company should also develop a Resource Efficiency Plan to promote conservation of natural resources and more environmentally sustainable operations. The developing of the Resource Efficiency Plans includes the following:

- 1) **Baseline Assessment:**
 - **Data Collection:** Gather historical data on energy consumption (electricity, fuel), greenhouse gas emissions (from operations, transportation, etc.), noise levels (at the workplace, during installations), and water usage. Utilize utility bills, fuel records, noise monitoring equipment, and water meters.
 - **Benchmarking:** Compare data with industry standards or similar companies to understand the company's relative performance.
 - **Identify Opportunities:** Analyze data to pinpoint areas of high consumption, inefficiency, and potential for improvement.
- 2) **Goal Setting:**
 - **Targets:** Establish measurable and ambitious, yet achievable, reduction targets for energy consumption, emissions, noise, and water usage. Consider aligning with science-based targets or industry initiatives.
 - **Timeframe:** Define a timeline for achieving these targets, considering short-term and long-term goals.
- 3) **Action Plan Development:**
 - **Energy:**
 - Implement energy-efficient lighting and appliances in offices and warehouses.
 - Optimize heating, ventilation and air-conditioning (HVAC) systems for maximum efficiency.
 - Encourage the use of energy-saving modes on computers and electronics.
 - Explore renewable energy options for powering company facilities.
 - **Emissions:**
 - Transition company vehicles to electric or hybrid models.
 - Optimize transportation routes and logistics to reduce fuel consumption.
 - Implement carbon offset projects to compensate for unavoidable emissions.
 - **Noise:**
 - Use quieter equipment and machinery during installations.
 - Schedule noisy activities during less sensitive hours.
 - Implement noise barriers or insulation to reduce sound transmission.
 - **Water:**
 - Install water-efficient fixtures in bathrooms and kitchens.
 - Collect and reuse rainwater for irrigation or cleaning purposes.
 - Implement water recycling systems where feasible.
- 4) **Stakeholder Engagement:**
 - **Employee Engagement:** Encourage employee participation in resource conservation efforts through awareness campaigns, incentives, and training programs.
 - **Stakeholder Communication:** Share progress and achievements with stakeholders to demonstrate the company's commitment to environmental responsibility.
- 5) **Monitoring and Reporting:**
 - **Tracking:** Establish a system for tracking energy consumption, emissions, noise levels, and water usage on a regular basis.
 - **Reporting:** Create clear and concise reports summarizing the data, comparing it to targets, and identifying areas for improvement.
 - **Frequency:** Determine the appropriate frequency for monitoring and reporting (e.g., monthly, quarterly, annually) based on the nature of the operations and the rate of change in resource usage.
- 6) **Review and Improvement:**
 - **Regular Evaluation:** Periodically review the Resource Efficiency Plan to assess progress towards targets and identify any obstacles or challenges.
 - **Corrective Actions:** Implement corrective actions to address any deficiencies or areas where targets are not being met.
 - **Adaptation:** Update the plan as needed to reflect changes in operations, technologies, or regulations.
 - **Continuous Improvement:** Foster a culture of continuous improvement by seeking out new and innovative ways to reduce resource consumption and minimize environmental impact.

11. Emergency Preparedness and Response Plan

Emergency Preparedness and Response Plans are key ESMS component as per **IFC PS 4 / WB ESS 4: Community Health Safety and Security**.

An Emergency Preparedness and Response Plan (EPRP) is developed to enable a company to respond effectively to emergencies towards ensuring the safety of employees, communities, and the environment in the face of potential hazards. An emergency preparedness and response plan prioritizes the safety of all involved parties while ensuring regulatory compliance and mitigating potential risks. By proactively identifying and addressing hazards, a company is better equipped to handle emergencies, minimizing injuries, property damage, and environmental harm. This preparedness not only fosters trust with stakeholders but also ensures business continuity by minimizing disruptions and enabling swift recovery in the event of an incident.



Source: Adapted from Public Safety Canada

Figure 16: General emergency preparedness and response management process

The following steps outline the process and highlight the value of such an EPRP plan:

- **Hazard Identification and Assessment:** Conduct a comprehensive assessment of potential hazards associated with off-grid solar operations, including those listed above (moving equipment, electrical hazards, chemical exposure, etc.). Consider the specific geographic location and environmental factors that could exacerbate risks (e.g., severe weather, natural disasters, civil unrest).
- **Emergency Response Procedures:** Develop detailed procedures for responding to each identified hazard, including:
 - a. Immediate actions to take (e.g., shutting down equipment, evacuating personnel).
 - b. Communication protocols for notifying relevant parties (employees, emergency services, community members).
 - c. Roles and responsibilities for emergency response teams.
- **Shutdown Procedures:** Create clear instructions for safely shutting down powered equipment and systems in an emergency. Ensure all employees are trained on these procedures.
- **Equipment and Facilities:** Maintain a list of emergency equipment (fire extinguishers, first aid kits, etc.) and

their locations. Establish protocols for the proper use of emergency equipment and inspection and maintenance schedules for the equipment and facilities.

- **Emergency Contacts:** Compile a list of emergency contacts, including local emergency services, medical providers, and relevant authorities, and ensure the list is readily available to all employees.
- **Evacuation Plans:** Identify clear evacuation routes and designate safe meeting points. Train employees on evacuation procedures.
- **Training and Drills:** Conduct regular training sessions to ensure employees understand the emergency response plan and their roles in an emergency. Perform drills to practice and evaluate the effectiveness of the plan. Consider coordinating training and drills with local emergency response services.
- **Review and Update:** Review and update the plan regularly to incorporate lessons learned from drills, incidents, or changes in the operating environment.

The emergency preparedness and response plan should take into account the following possible risks and hazards:

- **Workplace Hazards:**
 - a. Electrical: Electric shock, arc flash, fires from faulty wiring or equipment, battery-related hazards (e.g., thermal runaway, leaks).
 - b. Mechanical: Injuries from moving parts, falls from heights (during installation or maintenance), cuts or lacerations from tools.
 - c. Chemical: Exposure to hazardous substances (e.g., battery electrolytes, cleaning solvents), spills, fires.
 - d. Physical: Extreme heat or cold, noise, vibration, ergonomic hazards from repetitive tasks.
- **Environmental Hazards:**
 - a. Fires: Risk of fires from electrical malfunctions, battery incidents, or improper waste disposal.
 - b. Chemical Spills: Accidental release of hazardous chemicals (e.g., battery acid) into the environment.
 - c. Waste Management: Improper handling or disposal of hazardous waste, including batteries and electronic components.
- **Transportation and Logistics Hazards:**
 - a. Vehicle Accidents: Road accidents during transportation of solar products or equipment.
 - b. Cargo Damage: Damage to products during transit, potentially leading to leaks or other hazards.
- **Community-Related Hazards:**
 - a. Public Safety: Potential hazards to the public during installation or maintenance activities (e.g., falling objects, electrical hazards).
 - b. Community Health: Risks of exposure to hazardous materials or waste improperly handled by the company.
 - c. Pandemics: Disruption of operations, potential health risks to employees.
 - d. Social Impacts: Disruptions to community life or livelihoods due to company operations or accidents.
- **Natural Disasters and Extreme Weather Events:**
 - a. Floods: Damage to equipment or facilities, disruption of operations.
 - b. Storms: High winds, lightning strikes, power outages.
 - c. Earthquakes: Structural damage to buildings or equipment, disruption of operations.
- **Security Hazards:**
 - a. Theft or Vandalism: Theft or deliberate damage to products and equipment.
 - b. Civil Unrest, Kidnapping or Terrorism: Threats to personnel or facilities.
 - c. Cybersecurity: Data breaches, ransomware attacks, disruption of operations.

Table 17: Examples of common types of emergencies, preparedness measures and response procedures

Emergency Type	Potential Scenarios	Preparedness Measures	Response Procedures	Roles and Responsibilities
Natural Disasters	- Earthquakes, floods, storms, etc. impacting work sites or facilities.	- Develop a natural disaster preparedness plan, including evacuation routes, communication protocols, and emergency shelters. - Conduct drills and exercises to practice emergency procedures. - Monitor weather forecasts and warnings. - Secure equipment and materials in case of a storm.	- Activate the natural disaster preparedness plan. - Evacuate employees and community members to safe locations. - Secure equipment and facilities. - Assess damage and communicate with relevant authorities.	Emergency Response Team: Coordinate evacuation, communication, and damage assessment. Employees: Follow evacuation procedures and report any hazards. OHS Officer/Manager: Develop and maintain natural disaster preparedness plans, conduct drills, and coordinate with authorities.
Security Incidents	- Theft or vandalism of equipment or materials. - Threats or violence against employees.	- Implement security measures (e.g., fencing, alarms, surveillance cameras, security personnel). - Train employees on security protocols and reporting procedures. - Establish a security incident response plan. - Collaborate with local law enforcement agencies.	- Activate the security incident response plan. - Call for law enforcement assistance if needed. - Secure the area and preserve any evidence. - Report the incident to the OHS officer/manager and relevant authorities.	Security Manager/Officer: Develop and implement security plans, conduct risk assessments, and collaborate with authorities. Employees: Follow security protocols, report suspicious activities, and safeguard company assets.
Community Unrest	- Protests or demonstrations near work sites. - Civil unrest or conflict in the region.	- Monitor social and political developments in the operating area. - Develop a community unrest response plan, including communication protocols and evacuation procedures. - Engage with local communities to build relationships and address grievances.	- Activate the community unrest response plan. - Evacuate employees and secure the work site if necessary. - Communicate with local authorities and community leaders. - Report the incident to the company's management.	Community Relations Officer: Monitor social and political developments, engage with communities, and develop unrest response plans. Management: Make informed decisions based on risk assessments and prioritize employee safety. Security Personnel: Implement security measures and coordinate with authorities.
Fire	- Electrical fires due to faulty wiring or equipment. - Battery fires due to overheating or damage. - Fires caused by external factors (e.g., lightning strikes, wildfires).	- Install fire detection and suppression systems (smoke detectors, fire alarms, sprinklers). - Maintain fire extinguishers and ensure employees are trained in their use. - Conduct regular fire drills and evacuation exercises. - Develop and communicate a fire emergency plan.	- Activate fire alarms and evacuate the building. - Call the fire department. - Use fire extinguishers to attempt to control the fire (if safe to do so). - Account for all personnel and report any injuries or missing persons.	Emergency Response Team: Coordinate evacuation, fire suppression, and communication with emergency services. Employees: Follow evacuation procedures and report any fires. OHS Officer/Manager: Develop and maintain fire emergency plans, conduct drills, and ensure proper equipment is available.

Emergency Type	Potential Scenarios	Preparedness Measures	Response Procedures	Roles and Responsibilities
Chemical Spills/Leaks	- Acid spills from batteries. - Leaks of hazardous chemicals from other components.	- Store hazardous materials in designated areas with proper containment and labeling. - Provide spill kits and train employees on their use. - Develop and communicate a chemical spill response plan. - Ensure proper ventilation in storage areas.	- Isolate the spill area and prevent access. - Put on appropriate PPE. - Contain and clean up the spill using spill kits and procedures. - Report the incident to the OHS officer/manager and relevant authorities.	OHS Officer/Manager: Develop and maintain chemical spill response plans, provide training, and ensure proper storage and handling of hazardous materials. Employees: Follow spill response procedures and use PPE. Emergency Response Team: Respond to spills, contain and clean them up, and report to authorities.
Waste Management	- Improper handling or disposal of hazardous waste, including batteries and electronic components.	- Follow local regulations for hazardous waste disposal. - Partner with certified e-waste recyclers. - Train employees on proper waste handling, labeling, and storage procedures. - Implement waste segregation and collection systems.	- Isolate and contain any hazardous waste. - Follow proper disposal procedures. - Report the incident to the OHS officer/manager and relevant authorities.	OHS Officer: Develop and implement waste disposal procedures, ensure compliance with regulations, and manage contracts with recyclers. Employees: Follow waste disposal procedures and report any spills or leaks.
Vehicle Accidents	- Road accidents during transportation of solar products or equipment.	- Regular vehicle maintenance and inspections. - Driver training on safe driving practices. - Route planning to avoid hazardous conditions. - Compliance with traffic laws.	- Assess the situation and provide first aid if necessary. - Call for emergency services. - Secure the area and prevent further accidents. - Report the incident to the company and relevant authorities.	Fleet Manager/Supervisor: Oversee vehicle maintenance and driver training. Drivers: Follow safe driving practices and report any vehicle issues. Logistics Team: Plan safe transportation routes and ensure proper load securement.
Cargo Damage	- Damage to products during transit, potentially leading to leaks or other hazards.	- Proper packaging and load securement. - Use of appropriate transportation containers. - Training for loaders and unloaders on safe handling procedures.	- Assess the damage and isolate any hazardous materials. - Follow spill response procedures if necessary. - Report the incident to the logistics/transportation manager.	Logistics/Transportation Manager: Develop and enforce cargo handling and securement protocols. Loaders/Unloaders: Follow safe handling procedures and report any damage.
Public Safety	- Potential hazards to the public during installation or maintenance activities (e.g., falling objects, electrical hazards).	- Secure work areas during installation and maintenance. - Use warning signs and barriers. - Communicate with the community about potential hazards and safety precautions.	- If an incident occurs, provide first aid and call for medical assistance if needed. - Secure the area and prevent further accidents. - Report the incident to the company and relevant authorities.	Site Supervisor: Ensure safe work practices and communicate with the community. Community Relations Officer: Engage with the community to address concerns and provide information on safety measures.
Community Health	- Risks of exposure to hazardous materials or waste improperly handled by the company.	- Proper storage, handling, and disposal of hazardous materials. - Regular inspections and maintenance of equipment. - Community education on the risks and safe practices.	- If an exposure occurs, provide first aid and seek medical attention if needed. - Contain and clean up any spills or leaks. - Report the incident to the OHS	OHS Officer: Develop and enforce hazardous material management protocols. Community Relations Officer: Educate the community on the risks and safe practices.

Emergency Type	Potential Scenarios	Preparedness Measures	Response Procedures	Roles and Responsibilities
			officer/manager and relevant authorities.	
Medical Emergencies	- Injuries from falls, electrical shocks, or other accidents. - Sudden illnesses (e.g., heat stroke, heart attack).	- Provide first aid kits and ensure employees are trained in first aid and CPR. - Establish a designated first-aid station. - Develop and communicate a medical emergency response plan. - Maintain a list of emergency contact numbers.	- Assess the situation and provide first aid. - Call for emergency medical services if needed. - Escort the injured person to the designated first-aid station or a medical facility. - Report the incident to the OHS officer/manager.	OHS Officer/Manager: Develop and maintain medical emergency response plans, ensure first aid kits and training are available, and coordinate with medical providers. Employees: Provide first aid if trained and report any medical emergencies. First Aid Responders: Provide first aid and assist with medical evacuations if necessary.
Electrical	- Electric shock, arc flash - Fires from faulty wiring or equipment - Battery-related hazards (thermal runaway, leaks)	- Regular inspections and maintenance of electrical systems and equipment. - Lockout/Tagout (LOTO) procedures. - Insulated tools and PPE (gloves, safety glasses, arc flash suits). - Employee training on electrical safety. - Battery management systems for safe charging and discharging. - Proper ventilation and fire suppression systems in battery storage areas.	- Disconnect power source. - Administer first aid or CPR if needed. - Call emergency services. - Investigate and report incidents. - Review and update electrical safety procedures.	OHS Officer: Develop and enforce electrical safety protocols, conduct training and inspections. Electricians: Ensure safe work practices and use appropriate PPE. Employees: Report electrical hazards and follow safety procedures.
Mechanical	- Injuries from moving parts - Falls from heights (during installation/maintenance) - Cuts or lacerations from tools	- Machine guarding and regular maintenance. - Fall protection systems (guardrails, harnesses, safety nets). - Proper training on tool use and maintenance. - Use of appropriate PPE (gloves, safety shoes, hard hats).	- Provide first aid and seek medical attention if needed. - Secure the area and prevent further injuries. - Report the incident. - Investigate and implement corrective actions.	OHS Officer: Develop and enforce safety protocols for mechanical work, conduct training and inspections. Supervisors: Ensure employees use proper tools and techniques. Employees: Follow safety procedures and report hazards or injuries.
Chemical	- Exposure to hazardous substances (battery electrolytes, cleaning solvents) - Spills - Fires	- Use of PPE (gloves, respirators, goggles). - Proper storage and handling of chemicals. - Spill kits and procedures. - Training on chemical safety and emergency response.	- Evacuate the affected area. - Contain and clean up spills following proper procedures. - Provide first aid or decontamination if needed. - Call emergency services if necessary. - Investigate and report incidents.	OHS Officer: Develop and enforce chemical safety protocols, provide training, and manage chemical inventory and disposal. Employees: Follow safety procedures, use PPE, and report spills or leaks.
Physical	- Extreme heat or cold - Noise - Vibration	- Provide appropriate clothing and protective gear for weather conditions. - Hearing protection in noisy environments. - Anti-vibration tools and gloves. - Ergonomic workstations and regular breaks for repetitive tasks.	- Move to a safe location in case of extreme weather. - Provide first aid for heat or cold-related illnesses. - Rotate tasks or provide ergonomic tools to reduce	OHS Officer: Monitor environmental conditions, provide training on ergonomic practices, and ensure availability of protective gear. Supervisors: Enforce safe work practices and monitor employee well-being

Emergency Type	Potential Scenarios	Preparedness Measures	Response Procedures	Roles and Responsibilities
	- Ergonomic hazards from repetitive tasks		repetitive strain injuries. - Report incidents and seek medical attention if needed.	Employees: Report health concerns and follow safety protocols.

12. Stakeholder Engagement Plan (SEP)

A Stakeholder Engagement Plan is an essential ESMS component as per **WB ESS 10: Stakeholder Engagement and Information Disclosure**. Stakeholder Engagement is emphasized in IFC PS 1: Assessment and Management of Environmental and Social Risks and Impacts as crucial for identifying and managing the environmental and social risks and impacts of a project and requires identification of stakeholders, engagement of stakeholders and provision of a grievance mechanism.

A well-designed stakeholder engagement plan is invaluable for off-grid solar companies. By gathering diverse perspectives, companies can make informed decisions, mitigate risks, and earn the trust necessary for long-term success. Meaningful engagement fosters collaboration, leading to project designs that better meet community needs and minimize negative impacts. Ultimately, this collaborative approach builds a sense of shared responsibility, enhancing project sustainability and contributing to positive social and environmental outcomes.



Figure 17: Stakeholder map depicting the level engagement with stakeholders depending on their level of interest and influence

Key steps for stakeholder engagement planning:

1) Stakeholder Identification and Analysis:

- Comprehensive Identification:** Utilize various methods to identify all relevant stakeholders, both internal (employees, management) and external (customers, suppliers, local communities, government agencies, financiers, media, competitors).
- In-depth Analysis:**
 - Interests:** Determine each stakeholder's specific interests in the business (e.g., economic benefits, environmental concerns, social impacts).
 - Influence:** Assess the level of power or influence each stakeholder has over the business' outcome.
 - Impact:** Evaluate the potential positive or negative impacts the business could have on each stakeholder.
 - Attitude:** Gauge the current attitude of each stakeholder towards the business (supportive, neutral, opposed).
- Stakeholder Mapping:** Categorize stakeholders based on their level of interest and level of influence with respect to the business, its operations and projects.

2) Planning Engagement:

- Strategic Objectives:** Define clear, measurable objectives for stakeholder engagement at each stage of business operations and projects. These might include raising awareness, gathering feedback, building consensus, or resolving conflicts.
- Tailored Strategies:** Develop customized engagement strategies for each stakeholder group based on

their analysis. This could involve:

- i. **Communication channels:** Choose appropriate channels based on stakeholder preferences (e.g., face-to-face meetings, newsletters, social media, workshops, online forums).
 - ii. **Communication frequency:** Determine how often to engage with each stakeholder group (e.g., regular updates for highly impacted groups, occasional updates for less affected groups).
 - iii. **Engagement activities:** Design activities that foster meaningful dialogue and participation (e.g., consultations, public hearings, collaborative workshops, site visits).
- c. **Timeline and Budget:** Create a detailed timeline for engagement activities, allocating sufficient time and resources for each stage.

3) **Implementation:**

- a. **Transparent Communication:** Ensure all communication is clear, concise, and accessible to the target audience. Determine the most effective and appropriate channels for disseminating information to different stakeholder groups. Use plain language, avoid jargon, and consider translating materials into local languages if necessary.
- b. **Proactive Disclosure:** Share relevant project information early and often, including potential risks and impacts, mitigation measures, and grievance mechanisms.
- c. **Responsive Feedback:** Actively seek feedback from stakeholders and demonstrate responsiveness to their concerns. Facilitate two-way communication by providing channels for feedback, questions, and concerns.
- d. **Grievance Mechanism:** Establish a well-defined, accessible, and culturally appropriate grievance mechanism for stakeholders to raise concerns or complaints. Ensure that grievances are addressed promptly and fairly.

4) **Monitoring and Evaluation:**

- a. **Key Performance Indicators (KPIs):** Develop KPIs to track the effectiveness of stakeholder engagement efforts (e.g., participation rates in meetings, satisfaction surveys, number of grievances resolved).
- b. **Regular Reviews:** Conduct regular reviews of the engagement plan to assess progress, identify challenges, and make necessary adjustments.
- c. **Lessons Learned:** Document lessons learned for future projects and share them with relevant stakeholders.

Stakeholder engagement should adhere to the principles of free, prior, and informed consent (FPIC). This means ensuring engagement is free from external pressures, providing clear and accessible information beforehand, and tailoring communication to local languages and cultural contexts. The process should be responsive to the needs of both men and women, potentially requiring separate forums. Additionally, it should empower impacted individuals and communities, especially vulnerable groups, through targeted capacity building or other assistance, enabling their full and effective participation.

IFC good practice recommends that stakeholder engagement and consultation should be:

- **Targeted:** Focusing engagement efforts on specific stakeholder groups with distinct interests and concerns ensures that communication is tailored to their needs and that their voices are heard. This avoids a one-size-fits-all approach and maximizes the relevance of the engagement process.
- **Early:** Initiating engagement early in the project lifecycle allows stakeholders to provide input before critical decisions are made. This builds trust and fosters a sense of ownership among stakeholders, increasing their support for the business.
- **Informed:** Providing stakeholders with clear, accurate, and accessible information about the project, its potential impacts (both positive and negative), and the company's mitigation plans enables them to make informed decisions and participate meaningfully in the decision-making process.
- **Two-way:** Facilitating open and ongoing dialogue with stakeholders allows for feedback, concerns, and suggestions to be shared and addressed. This iterative process builds trust, strengthens relationships, and ensures that the business is responsive to community needs.
- **Gender Inclusive:** Actively seeking the participation of women and other marginalized groups ensures that their perspectives and concerns are considered, leading to more equitable and inclusive business operations and projects outcomes.
- **Localized:** Adapting engagement strategies to the local context and cultural norms ensures that communication is effective and respectful. This involves using local languages, working with community leaders, and recognizing traditional decision-making processes.
- **Documented:** Maintaining thorough records of all stakeholder engagement activities (meetings, consultations, feedback, grievances) provides transparency, accountability, and a basis for learning and improvement.
- **Reported:** Regularly reporting on stakeholder engagement activities to relevant stakeholders (e.g., investors, regulators, communities) demonstrates the company's commitment to transparency and accountability. This reporting can include summaries of feedback received, actions taken in response to concerns, and lessons learned.

Table 18: Examples of typical stakeholders and engagement methods

Stakeholder Group	Interest/Concerns	Engagement Level	Engagement Methods	Frequency	Responsible Party	Resources Required
Customers	- Product affordability, reliability, and efficiency. - Access to information and customer support. - Safe disposal of old products.	High	- Marketing - Before- and after-sales services - Consumer outreach programs and education - Surveys and feedback forms. - Community meetings and workshops. - Customer service hotlines and email support. - Educational materials (brochures, videos, social media).	Ongoing	Sales and Marketing Team	- Customer feedback collection tools (surveys, online forms, suggestion boxes) - Educational materials (brochures, videos, infographics, social media content) - Communication channels (hotlines, email, social media platforms) - Event organization resources (venues, materials)
Employees	- Fair wages and working conditions. - Safety training and equipment. - Opportunities for growth and development.	High	- Regular staff meetings and town halls. - Internal communication channels (newsletters, intranet). - Training programs and workshops. - Performance reviews and feedback sessions.	Ongoing	HR Department, OHS Manager	- Meeting facilities and communication tools (projectors, microphones, online platforms) - Training materials and resources (manuals, presentations, videos) - Employee feedback mechanisms (surveys, suggestion boxes, anonymous reporting channels)
Regulators, Import and Tax Agencies	- Regulatory compliance. - Environmental and social impact of operations. - Economic development and job creation.	High	- Regular meetings and consultations. - Formal reports and presentations. - Participation in government-led initiatives and working groups. - Collaboration on policy development and implementation.	Quarterly	Government Relations Officer, CEO	- Registration and compliance procedures and documentation - Meeting preparation materials (presentations, reports) - Legal expertise and consultation (if needed)
Local Communities	- Environmental and social impact of projects (e.g., land use, waste disposal, job opportunities). - Community engagement and participation.	High	- Community consultations and meetings. - Participatory planning and decision-making processes. - Community development initiatives (e.g., education, healthcare, infrastructure). - Social impact assessments and monitoring.	Ongoing	Community Relations Officer, Project Managers	- Community engagement tools (facilitation guides, consultation materials) - Social impact assessment and monitoring tools - Project communication materials (brochures, newsletters, websites) - Resources for community development projects
Manufacturers and Products Suppliers	- Fair business practices - Timely deliveries and payments - Product quality guarantees - Collaboration on product development and sustainability initiatives.	Medium	- Supplier meetings and workshops. - Contractual agreements and performance reviews. - Joint training and capacity building programs. - Collaboration on research and development projects.	Annually	Procurement Team, Sustainability Manager	- Supplier code of conduct - Contract templates and negotiation resources - Training materials and resources

Stakeholder Group	Interest/Concerns	Engagement Level	Engagement Methods	Frequency	Responsible Party	Resources Required
Investors, Financiers, and Donors	- Financial performance and return on investment. - Environmental and social responsibility. - Compliance with industry standards.	Medium	- Financial reports and presentations. - Environmental, Social, and Governance (ESG) reporting. - Investor meetings and conference calls. - Participation in industry associations and events. - Participation in grant support programmes like ROGEAP	Annually	Finance Department, Investor Relations	- Financial reporting software and tools - ESG reporting framework and guidelines - Investor communication materials (presentations, fact sheets)
Industry Associations	- Industry advocacy and representation. - Networking and knowledge sharing opportunities. - Development of industry standards.	Medium	- Participation in industry associations and events. - Collaboration on research and advocacy projects. - Development of industry guidelines and best practices.	Ongoing	CEO, Industry Relations Manager	- Membership fees for industry associations - Participation fees for industry events - Contribution to research and advocacy projects
e-Waste Management Service Providers	- Proper collection, recycling, and disposal of e-waste. - Compliance with environmental regulations.	High	- Regular meetings and consultations. - Training and capacity-building workshops. - Collaboration on collection and recycling infrastructure. - Joint advocacy for stronger e-waste regulations.	Quarterly	Sustainability Manager, OHS Manager	- Training materials and resources - Funding for infrastructure development - Legal expertise for regulatory compliance - Advocacy materials and resources

13. Grievance Mechanism

The Grievance Mechanism (GM) is a key ESMS component as per **WB ESS 10: Stakeholder Engagement and Information Disclosure**. Grievance Mechanism is emphasized in **IFC PS 1: Assessment and Management of Environmental and Social Risks and Impacts** which requires provision of a grievance mechanism. The GM is also relevant for **IFC PS 2 / WB ESS2: Labor and Working Conditions**, as a GM serves as an important mechanism for addressing concerns and complaints related to labor practices and working conditions.

A grievance mechanism is a system, process or procedure that allows employees or other stakeholders to raise concerns or complaints about a company's products, operations, policies, or activities. It is a crucial tool for conflict resolution, ensuring accountability, and building trust between the company and its stakeholders. By offering a platform for early issue identification, it enables proactive problem-solving and prevents minor concerns from escalating into major conflicts. Through fair and transparent dispute resolution, trust and goodwill are fostered between the company and its stakeholders. Addressing stakeholder concerns leads to valuable insights, driving operational improvements and better project outcomes. Ultimately, a well-functioning grievance mechanism strengthens the company's social license to operate, demonstrating a commitment to accountability and responsible practices, while also ensuring compliance with relevant environmental and social standards.

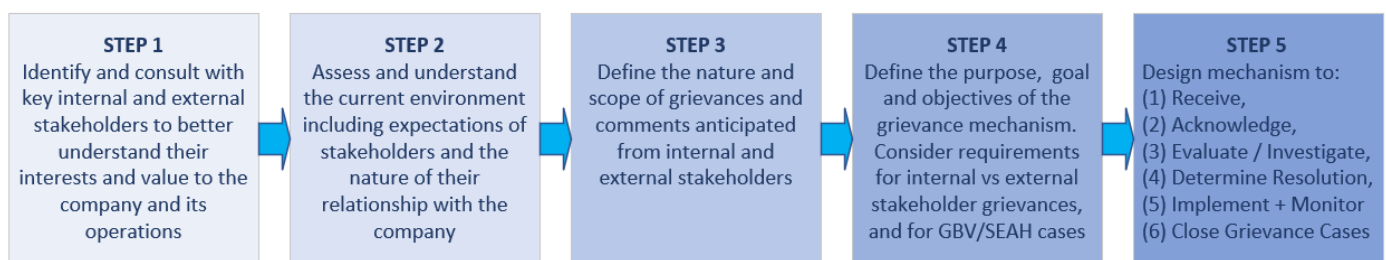
The basic requirements of a GM include:

- 1) A system/communication channel/mechanism to receive inquiries (comments, questions and complaints) and respond to the public and stakeholders (workers, customers and community members).
- 2) Collection, processing, responding to, and resolving raised issues were conducted promptly.
- 3) Informing stakeholders of the existence of the GM and how to submit complaints and suggestions.
- 4) Need to have a unique GM process in place to accommodate sensitive SEA/SH-related complaints.

Process for establishing a grievance mechanism: Developing a grievance mechanism can be approached as a three-phase process: design, implementation, and monitoring/evaluation. The design phase focuses on defining the mechanism's objectives, ensuring accessibility for all stakeholders, establishing transparent procedures, and prioritizing fairness and adaptability. During implementation, the mechanism is widely publicized, staff are trained to handle grievances, and records are meticulously maintained. The monitoring/evaluation phase involves regular reviews, data analysis, stakeholder feedback, and continuous improvement to ensure the mechanism remains effective and responsive over time.

1) Grievance Mechanism Design:

The following flowchart describes key steps in developing a grievance mechanism.



Source: Adapted from cao-grm.org

Figure 18: Grievance mechanism development steps

Some key aspects to consider while designing the grievance mechanism include:

- **Legitimate and Scalable:** The mechanism should be perceived as fair, credible, and proportionate to the project's potential impacts. It should encourage dialogue and shared responsibility for resolving issues.
- **Accessible and Anonymous:** Ensure that the mechanism is easily accessible to all stakeholders, regardless of their literacy level or background. Allow for anonymous complaints where feasible to encourage reporting.

- **Transparent and Fair:** Clearly communicate the steps involved in the grievance process, including timelines and responsible parties. Maintain transparency throughout the process and ensure fair treatment of all parties involved.
- **Adaptive and Dialog-Oriented:** Tailor the mechanism to accommodate different communication preferences (e.g., in-person meetings, written submissions, hotlines). Encourage dialogue and mediation to resolve grievances amicably.
- **Predictable and Timely:** Establish clear timelines for each step of the grievance process, assign responsibilities, and track progress. Ensure prompt resolution of complaints.

2) Grievance Mechanism Implementation:

The steps to consider building into the grievance mechanism process are depicted in the diagram below.

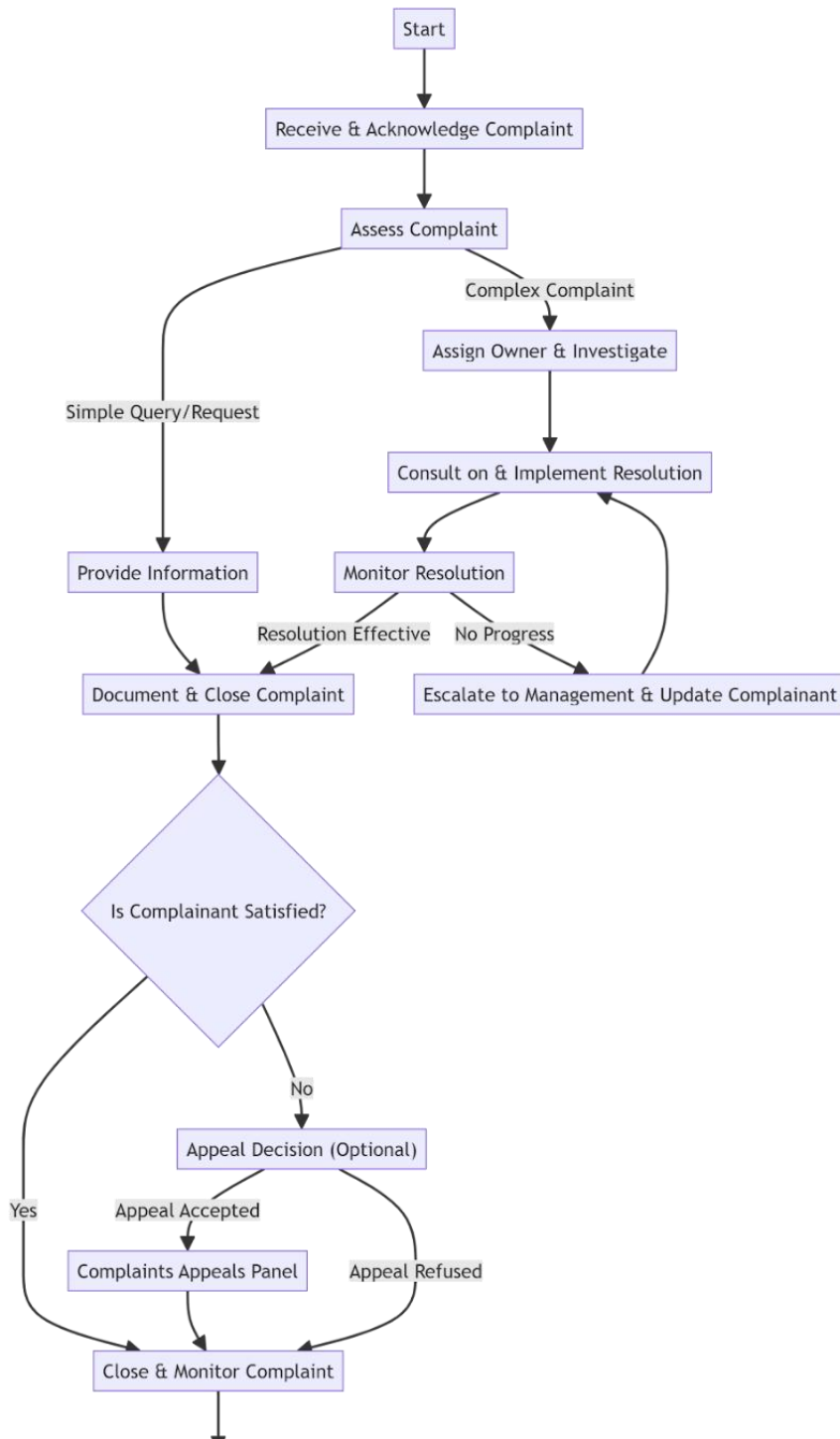


Figure 19: Grievance mechanism process suggested by the World Bank's Compliance Advisory Ombudsman (CAO)

Key considerations for grievance mechanism implementation include:

- **Communication:** Widely publicize the grievance mechanism through various channels (e.g., website, posters, community meetings) and ensure stakeholders understand how to use it.

- **Training:** Train staff on how to handle grievances effectively, including active listening, conflict resolution, and cultural sensitivity.

- **Record-Keeping:** Maintain detailed records of all grievances, including the nature of the complaint, the steps taken to address it, and the outcome.

3) Monitoring and Evaluation:

- Regularly review the effectiveness of the grievance mechanism by analyzing grievance data, seeking feedback from stakeholders, and identifying areas for improvement.

- Update the mechanism as needed to ensure it remains relevant and effective.

Complaints and resolution information should be collected and accurately recorded in a GRM Grievance Log (e.g., in the format of a table in a spreadsheet file) for non-GBV/SEAH complaints.

Table 19: Grievance mechanism process suggested in ROGEAP Operational Manual

Process	Description	Time Frame	Other Information
Identification of grievance	Face to face; telephone; letter; mail; e-mail; website; recorded during public/ community interaction; others The grievance can also be passed through other parties, such as the chief office because the public are more conversant with this office.	1 Day	Email address; hotline number
Grievance assessed and logged	Significance assessed and grievance recorded or logged (i.e. in a log book) It will be prudent to have a grievance record book where the grievances are recorded for follow up. Grievances concerning sexual exploitation and abuse/gender-based violence should be treated as confidential. Only the nature of the complaint and the processing outcome should be recorded.	3-6 Days	Significance criteria: Level 1 –one off event; Level 2 – complaint is widespread or repeated; Level 3- any complaint (one off or repeated) that indicates breach of law/ policy
Grievance is acknowledged	Acknowledgement of grievance through appropriate medium	3 Days	
Development of response	Grievance assigned to appropriate party for resolution Response development with input from management/ relevant stakeholders	4-8 Days	
Response signed off	Redress action approved at appropriate	8-15 Days	
Implementation /communication of response	Redress action implemented and update of progress on resolution communicated to complainant	5-9 Days	

Source: ROGEAP Operational Manual

The following information should be included in the Grievance Mechanism log/record:

- **Case Number:** A unique identifier for each grievance, allowing for easy tracking and reference (e.g., GRM-2023-001).
- **Date Claim Was Received:** The exact date when the grievance was initially submitted.
- **Name of Person Receiving Complaint:** The full name and title of the staff member who first received the complaint.
- **Where/How Complaint Was Received:** The specific channel used to submit the grievance (e.g., online form, hotline, complaint box, in-person meeting).
- **Name and Contact Details of Complainant (if known and anonymity is not requested):** If the complainant provides their information, record their full name, phone number, email address, and any other relevant contact information.
- **Content of the Claim:** A detailed summary of the grievance, including all issues raised, suggestions made, and inquiries posed by the complainant.
- **Did the company acknowledge the receipt of the complaint from the Complainant?** A simple "Yes" or "No" answer.
- **If yes, include the date and method of communicating Receipt Confirmation:** Record the date the company acknowledged receipt and the specific method used (e.g., email, phone call, letter).
- **Expected Date of Decision:** The estimated date by which a decision on the grievance is expected to be reached.
- **Decision Outcome:** A concise summary of the final decision reached on the grievance.
- **Include names of participants and date of decision:** List the names and titles of all individuals involved in the decision-making process and the date the decision was made.
- **Was the decision communicated to the complainant?** A "Yes" or "No" response.
- **If yes, state when, by whom, and via what method of communication:** Record the date, the name of the person who communicated the decision, and the method used (e.g., email, phone call, letter).
- **Was the complainant satisfied with the decision?** A "Yes" or "No" response based on the complainant's feedback (if available).
- **State the Decision:** Briefly summarize the complainant's feedback on the decision (e.g., "Complainant expressed satisfaction" or "Complainant requested further clarification").
- **Any Follow-Up Action:** Describe any further actions taken to address the grievance or implement the decision.
- **(and by whom, by what date?)** Specify who is responsible for the follow-up action and the deadline for completion.

Specific GM procedures and protocols should be developed for GBV/SEAH-related complaints adhering to the following guiding principles:

- **Ethical:** Ensure all actions respect the dignity, rights, and autonomy of the survivor. This involves informed consent, respecting confidentiality, and avoiding victim-blaming.
- **Confidential:** Protect the privacy of all parties involved. Information should only be shared on a need-to-know basis with individuals directly involved in the investigation and resolution process.
- **Non-Biased:** Guarantee impartiality and fairness throughout the process. Avoid conflicts of interest and ensure investigators are trained to recognize and address their own biases.
- **Rapid:** Prioritize a timely response to complaints to minimize the potential for further harm. Establish clear timelines for each step of the process and communicate them to the survivor.
- **Survivor-Centered:** Place the needs and well-being of the survivor at the center of the process. Offer support services such as counseling, medical care, and legal assistance. Empower the survivor to make informed decisions about their participation in the process.

Examples of Grievance Mechanism Procedures and Protocols adapted to GBV/SEAH-related complaints:

1. **Reporting and Intake:**
 - Provide multiple, confidential channels for reporting complaints (e.g., dedicated hotline, online form, in-person reporting to a trusted staff member).
 - Ensure intake staff are trained to handle sensitive disclosures with empathy and professionalism.
 - Document all complaints thoroughly, including date, time, details of incident, and supporting evidence.
2. **Investigation:**
 - Conduct prompt and thorough investigations, following established protocols.
 - Assign trained and impartial investigators who are knowledgeable about GBV/SEAH issues.
 - Interview the survivor, the accused, and any witnesses separately and confidentially.
 - Gather and preserve relevant evidence, such as medical records, photographs, or digital communications.
 - Maintain clear and detailed records of the investigation process.
3. **Support and Protection:**
 - Provide immediate support to the survivor, such as counseling, medical care, and legal assistance.
 - Implement measures to protect the survivor from retaliation or further harm. This may include temporary reassignment, security measures, or restraining orders.
4. **Decision-Making and Action:**
 - Based on the investigation findings, make fair and impartial decisions regarding appropriate action. This may include disciplinary measures, termination of employment, or referral to law enforcement.
 - Communicate the decision to the survivor and the accused, explaining the reasoning behind it.
 - Implement any necessary corrective actions to prevent future incidents.
5. **Monitoring and Evaluation:**
 - Track the progress of all GBV/SEA/SH complaints and their resolution.
 - Regularly evaluate the effectiveness of the grievance mechanism and identify areas for improvement.
 - Conduct periodic training and awareness-raising activities for employees on GBV/SEAH issues.

This approach not only protects individuals from harm but also fosters a culture of accountability and respect.

Table 20: Sample grievance form to be completed by a Complainant. Similar form can be made accessible on the company's website for online submissions.

Comments and Grievance Submission Form
Name:
Address:
Telephone:
E-mail:
Description of complaint: Who, what, where, when, how
Date and/or duration of incident that led to the complaint:
Suggestions for how the complaint could be resolved:
Signature:
Date:
Company Reference No. (for office use only):

Source: World Bank's Compliance Advisory Ombudsman (CAO)

Table 21: Example of a Grievance Mechanism Log where received grievances are documented along with the response process undertaken

Date Received	Grievance ID	Name and Contact of the Complainant (if known)	Stakeholder	Grievance Category	Description of Grievance	Mode of Receipt	Assigned To	Status (Open, In Progress, Resolved, Closed)	Resolution/ Action Taken	Follow-up Actions	Closing Date
2024-04-15	G001	Aissatou Mary	Employee	Working Conditions	Employee reported unsafe working conditions in the warehouse due to inadequate lighting and clutter.	In-person	EHS Manager	Closed	Additional lighting installed, warehouse reorganized, and safety training conducted for all employees.	-Conduct regular safety inspections of the warehouse. -Implement a checklist for warehouse organization and maintenance.	2024-04-22
2024-05-03	G002	Ben Musa	Customer	Product Quality	Customer reported solar lantern malfunctioning after a few weeks of use.	Email	Customer Service	Resolved	Lantern replaced with a new one and a discount offered on the next purchase as compensation for inconvenience.	-Review quality control procedures for solar lanterns. -Investigate the root cause of the malfunction.	2024-05-10
2024-05-20	G003	N/A	Community Member	Environmental Impact	Community member raised concerns about the disposal of old solar panels in a nearby landfill.	Phone	EHS Manager	In Progress	Investigating the issue and gathering information on the company's waste disposal practices.	-Conduct a site visit to the landfill to assess the situation. -Review the company's e-waste management plan and update if necessary. -Organize a community meeting to address concerns and discuss sustainable disposal solutions.	N/A
2024-05-28	G004	N/A	Employee	Gender Discrimination	Female employee reported discriminatory behavior from a supervisor during a performance review.	In-person	HR Manager	Open	Conducting an internal investigation into the complaint and gathering statements from both parties.	-Schedule mediation sessions between the employee and supervisor. -Provide sensitivity training to all employees and managers.	N/A

14. Monitoring and Review

A Monitoring and Review Plan serves as a roadmap for tracking, assessing, and improving the company's environmental and social performance across all aspects of its operations. By systematically collecting data on key environmental and social indicators, the plan enables the company to evaluate its progress against predefined objectives and targets, ensuring alignment with its ESMS commitments and relevant standards like the IFC Performance Standards. Through this data-driven approach, the company can pinpoint areas where performance is falling short or risks are emerging, allowing for timely implementation of corrective actions. The plan also emphasizes transparent communication of monitoring results and lessons learned with stakeholders, fostering accountability and building trust. This continuous cycle of monitoring, evaluation, and improvement not only ensures regulatory compliance and effective risk management but also drives performance optimization, leading to a reduced environmental and social footprint.



Source: Adapted from United Nations Development Program, 2009

Figure 20: Planning, monitoring, and evaluation process for results-based management

Process for developing a monitoring and review plan:

- 1) **Establish Monitoring Objectives:**
 - a. Clearly define the specific goals and objectives of the monitoring and review process, aligning them with the overall ESMS objectives and relevant performance standards.
 - b. Identify key performance indicators (KPIs) to measure progress towards the objectives.
- 2) **Develop Monitoring Methodology:**
 - a. Determine the appropriate methods for data collection (e.g., inspections, audits, surveys, stakeholder consultations).
 - b. Set clear responsibilities for data collection, analysis, and reporting.
 - c. Establish a schedule for monitoring activities, ensuring adequate frequency and coverage of relevant environmental and social aspects.
- 3) **Data Analysis and Reporting:**
 - a. Analyze monitoring data to identify trends, patterns, and areas where performance is not meeting expectations.
 - b. Prepare regular reports summarizing monitoring results, including an assessment of performance against established KPIs and any corrective actions taken.
- 4) **Review and Improvement:**
 - a. Conduct periodic reviews of the monitoring and review plan to assess its effectiveness and identify areas for improvement.
 - b. Use lessons learned from monitoring activities to refine the ESMS and adapt to changes.

Stakeholder engagement should be prioritized during monitoring, ensuring stakeholder concerns are heard and addressed throughout the monitoring and review process. Transparency is also key, with public disclosure of monitoring results and corrective actions demonstrating accountability and a commitment to improvement. Independent reviews, such as audits, can provide an objective assessment of the ESMS and its implementation. Capacity building should be ongoing for staff involved in monitoring and review to ensure their competence and effectiveness in carrying out the

monitoring tasks. Below is an example of the environmental and social risk monitoring form for solar companies.

Table 22: Environmental and social risk monitoring form for solar companies (source: ROGEAP Operational Manual)

ENVIRONMENTAL AND SOCIAL RISK MONITORING FORM FOR SOLAR COMPANIES

Please provide responses to the questions below. Please include additional sheets or attachments as required to provide details on questions that have been answered Yes.

Name of the company			
Completed by (name):			
Position in the company:		Date:	
Reporting period	From:	To:	

Environmental & Social Management System (ESMS)

General	Yes/No	
Has the company developed and implemented an ESMS?		If yes, please attach a copy of the ESMS to this report.
Please provide the name and contact information of the main responsible who has the overall responsibility for the implementation of the company ESMS		
Please provide current staffing of other core ESMS persons in the organization involved with ESMS implementation		
Policies & Processes	Yes/No	
Has the company recently updated: - E&S policy - HR policy - OHS procedures - Stakeholder engagement plan - Grievance mechanism (including SEA/SH related GM)		
Has management signed off on the required policies/procedures (E&S policy, HR policy, OHS procedures)?		If yes, please provide the date and internal communication indicating the same.
Is the Environmental and Social Action Plan (ESAP) being carried out?		If yes, please provide information on the evaluation against the performance and processes indicators including deadlines.

Please state any difficulties and/or constraints related to the implementation of the ESAP		
If there is an ESMS already in place, have there been any updates to the ESMS or policy and procedures adopted by your company during the reporting period?		<i>If yes, please provide a copy of the updates including dates and reasons for the same.</i>
Please state any difficulties and/or constraints related to the implementation of the Environmental and Social procedures.		
Capacity	Yes/No	
Are the roles and responsibilities of the staff well defined and understood as per the implementation of the ESAP and/or ESMS?		<i>Please provide details about the definition of the roles and responsibilities.</i> <i>Please describe the training (if any) provided to the ESMS persons and other team members during year.</i>
Monitoring	Yes/No	
Do you have an internal monitoring and review plan?		<i>If yes, please describe the plan and its processes, also considering it against the performance and processes indicators</i>
Have you implemented an external communication and grievance mechanism?		<i>If yes, please describe the grievance mechanism</i>
Reporting	Yes/No	
Is there an internal process to report internally on Environmental and Social issues?		<i>If yes, please explain the process, reporting format and frequency and actions taken if any.</i>

15. Environmental and Social Action Plan (ESAP)

An environmental and social action plan (ESAP) consists of measures an organisation (business) implements or plans to address environmental, social, health and safety risks and impacts associated with their products, services and operations. ESAPs are like a roadmap to guide and ensure an organisation (business) complies with E&S safeguard requirements and guidelines. The focus of the ESAP should be based on the level of development of the company's ESMS and the nature of environmental and social risks faced by the company. An ESAP facilitates developing and implementing any missing or outstanding E&S risk management measures (e.g., elements of an ESMS) in a timely and appropriate manner.

Before implementing an ESAP, an organisation should identify and understand the E&S issues relevant to the business operations and projects, E&S guidelines and requirements applicable to the organisation, the status of the organisation's E&S management efforts, and any gaps to be filled towards having a fully compliant ESMS aligned with achieving the organisation's E&S sustainability vision and goals.

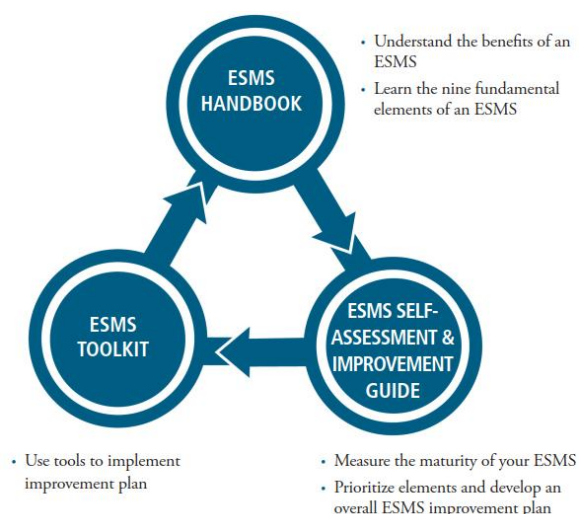
The key elements of an ESAP, which can be prepared in a paragraph/bullet or table form, include Objectives, Outputs, Activities/Actions, Key Performance Indicators, Baseline Status, Target, Timeline, Responsibilities, and Resources.



Figure 21: Components of an Environmental and Social Action Plan (ESAP)

An ESAP is required of solar businesses that still need to have all elements of an ESMS to help these businesses develop and implement a full ESMS. An ESAP is required of solar businesses classified as having medium to high E&S risks to ensure the businesses have measures, E&S covenants, and E&S monitoring to avoid and mitigate adverse impacts of the E&S risks. The ESAP should be developed within a year and included in the financing agreement between financial institutions and the solar business.

The IFC has developed an ESMS Handbook accompanied by an ESMS toolkit and ESMS self-assessment & improvement guide, which can be helpful complementary resources to this current document, to assist in assessing your business' ESMS²³.



²³ IFC. 2015. Environmental and Social Management System (ESMS) Implementation Handbook - GENERAL. Retrieved June 11, 2024, from <https://www.ifc.org/en/insights-reports/2015/publications-handbook-esms-general>

The following table is a template for a simple ESAP.

Source: IFC 2015

Figure 22: IFC ESMS Handbook and Supplemental Resources

Table 23: Template for Environmental and Social Action Plan (ESAP)

No.	Action Required	Deliverable	Completion Indicator	Deadline
Environmental and Social Assessment and Management System				
Labor and Working Conditions				
Occupational Health and Safety				
E-Waste Management				
End User's Health and Safety				
Stakeholder Engagement and Grievance Mechanisms (including SEA/SH related GM)				
Land-related Issues - Solar Business to verify that end-users have adequate mitigation measures in place				
Other Issues				

Source: ESRM Sector Guide on Off-Grid Solar for CFIs

Sample ESAP for an Off Grid Solar Business

The table below presents an example of what an ESAP may look like for an off grid solar company focused on putting in place missing elements of its ESMS.

Table 24: Example of an ESAP for a solar business

Outputs	Activity/Actions	KPI	Baseline	Targets	Timeline	Responsibilities	Budget (USD)
Company E&S policy	- Draft E&S policy - Review and approve E&S policy	Approved E&S Policy	No E&S Policy	- E&S policy document prepared - E&S policy approved	- Feb 2025 - Mar 2025	- Management drafts and implements - The Board of Directors reviews and approves	500
E&S risks and impacts identified	E&S risk and impact assessment	Number of E&S risks and impacts identified	E&S risks unknown	List of identified E&S risks and impacts	Feb 2025	Hired consultant assesses and reports on E&S risks	1000
E&S management plan	- Develop an environmental and social management plan (ESMP) to address E&S risks and impacts - Review and approve the ESMP	Approved ESMP	Partial or no ESMP	- ESMP document prepared and approved - ESMP implemented	- Mar 2025 - Mar 2025	- Hired consultant drafts ESMP - Management reviewed and approved ESMP and ensured the implementation	1000
Organisational awareness and capacity development	- Assess organisational E&S awareness and capacity for managing E&S risks and impacts; - Train staff on E&S risks and management - Hire an E&S Specialist to strengthen E&S competency	- Number and % of staff trained on E&S risks and impacts - Number of E&S Specialists hired	- Limited awareness and capacity - No E&S training - No E&S Specialist in the company	- E&S awareness and capacity development plan document - All staff participated in E&S training - At least one (1) E&S Specialist hired	- Apr 2025 - May-Jun 2025 - Jan 2026	- Hired Consultant to assess and report on E&S awareness and capacity and conduct training - Management review report, ensure training is conducted, and recruit an E&S specialist	1000
Emergency preparedness and response plan	- Prepare an emergency preparedness and response plan (EPRP) - Review and approve the EPRP - Train staff on EPRP procedures	- Approved EPRP - Number and % of staff trained on EPRP	- No EPRP 0 staff trained on EPRP	- EPRP document prepared and approved - All staff trained on EPRP procedures	- Mar 2025 - Mar 2025 - May-Jun 2026	- Hired a Consultant to draft EPRP and conducted training - Management reviews EPRP and ensures readiness to implement SEP	500
Stakeholder engagement plan	- Prepare a stakeholder engagement plan (SEP) - Review and approve the SEP	Approved SEP	No SEP	- SEP document prepared and approved - SEP implemented	- Mar 2025 - Apr 2025	- Management drafts and oversees the implementation - The Board of Directors reviews and approves	500
Grievance mechanism	- Prepare a grievance mechanism (GM) - Review and approve the company GM	- Approved GM - Number of grievances received of - Number of grievances responded to	- No GM 0 grievances documented	- GM prepared and approved - GM operational	- Mar 2025 - Apr 2025	- Management prepares GM and ensures that GM is operational - The Board of Directors reviews and approves	500
Monitoring and evaluation plan	- Prepare the E&S monitoring and evaluation (M&E) plan - Review and approve the E&S M&E Plan	Approved E&S M&E Plan	- No E&S M&E plan 0 E&S M&E engagement activities	- E&S M&E Plan prepared and approved - E&S M&E plan implemented	- Mar 2025 - Apr 2025	- Hired Consultant to prepare M&E plan - Management reviews and ensures implementation	500

	Implement the E&S M&E activities as per plan	Number of E&S M&E engagement activities				Designated staff conduct M&E	
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